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Systematic Benchmarking of Biological Information Sources for Viral Mutation Hotspot Detection

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Abstract

This dissertation presents **MutBench**, a region-level hotspot-detection benchmark for 11 RNA-virus surface glycoproteins under single-position scoring (gap-aware), with substitutions only as ground-truth labels, that compares 10 biological information types (20 scoring formulas) $\times$ 14 detection families (39 variants) $\times$ 11 RNA viruses = 8,580 evaluations. Within this regime, the optimal information source is pathogen-dependent, and HIV-1 vaccine-escape enrichment provides the primary retrospective external anchor.

Detecting mutation hotspots in viral genomes is essential for variant surveillance and vaccine design, yet existing approaches have relied almost exclusively on mutation frequency and Shannon entropy, leaving comparatively unexplored a rich landscape of complementary information sources (phylogenetic selection signals, protein language models, structural features). Concurrent fitness-regression benchmarks (ViroGym, EVEREST v3, ProteinGym v1.3) target per-variant prediction; to our knowledge, MutBench is the broadest *region-level, single-position hotspot-detection* benchmark to date for RNA viruses, with 8,580 (scoring$\times$detector$\times$pathogen) evaluation cells. MutBench comprises three core components: (1) a three-layer ground truth system integrating literature-curated positive markers (Layer A: a pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types, with immune-escape positions dominating the curated set), conserved structural regions under purifying selection as negative controls (Layer B), and Deep Mutational Scanning (DMS) experimental fitness data as functional validation (Layer C);

(2) a composite evaluation metric defined as hotspot-score $= (\text{recall}+\text{precision}+\text{stability})/3$; and (3) a two-stage main experimental design plus a Stage 3 information-integration layer, in which Stage 1 conducts in-depth analysis on SARS-CoV-2, Stage 2 performs the large-scale cross-pathogen comparison, and Stage 3 evaluates multi-source feature integration.

The main evidence is summarized below.

Feature-ablation and negative sensitivity analyses bound the practical use: on the 12-pathogen Stage 3 panel, the historical 4-feature core (homoplasy, pLDDT, entropy, frequency) has paired delta $+0.011$ MCC (95% CI [-0.018, $+0.042$], permutation $p = 0.61$) against the full 10-feature EqualWeight ensemble, ranks $87/1023$ in the follow-up full-lattice audit, and is paired with a pre-registered callability rule that abstains on $0/12$ folds; UniProt-derived Layer A$'$ sensitivity is negative (mean MCC $= -0.055$; precision@20 $= 0.005$), indicating label-provenance non-equivalence rather than a panel extension. H3N2 ($9.36\times$, 69% overlap) is a self-consistency check, SARS-CoV-2 ($7.63\times$) is exploratory after Bonferroni, and EqualWeight integration also achieves H3N2 escape enrichment $4.012\times$ ($p = 0.0023$).

A systematic analysis of the 20 scoring types spanning six information categories (frequency, MSA-derived, phylogenetic, structural, AI-based, and composite) reveals 9 distinct optimal information types across 11 pathogens (Tranception/ESM-2 channel collinearity is not separated at headline granularity; see Chapter 2): FUBAR phylogenetic selection scores achieve the highest MCC for H3N2 (0.534), protein language model scores dominate for SARS-CoV-2 and RSV, and homoplasy scores excel for Norovirus. The external practical anchor for downstream search-space reduction remains HIV-1, where the Layer-A-disjoint

Table 0.1: Main evidence ledger for MutBench

Claim	**Statistic**	**Boundary**	**Defense role**
Framework sanity check	Stage 1 SARS-CoV-2 MutClust-Hybrid hotspot-score 0.778	Controlled single-pathogen check	Validates the benchmark mechanics before cross-pathogen analysis
Pathogen-dependent information utility	Stage 2 $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$ (11-pathogen cluster-bootstrap 95% interval [0.201, 0.346]); 6-category aggregation $\omega^2 = 0.103$; within-cell residual $\omega^2 \approx 0.39$	Layer A labels partly confound pathogen biology with curation protocol	Primary statistical evidence for pathogen-specific information choice
No universal best combination	Friedman $\chi^2 = 7.69$, $p = 0.990$; LOPO 0/11 exact matches; 0.265 oracle-vs-generalized MCC gap	LOPO 0/11 alone is null-consistent under permutation	Corroborates the interaction effect without treating LOPO as independent proof
Practical external anchor	HIV-1 full-set Fisher enrichment $7.19\times$ ($p_{\text{adj}} = 2.5\times10^{-16}$, 82% Layer-A-disjoint); novel-only 7.16–8.24$\times$ on 37 Layer-A-disjoint positions (worst-case $p = 5 \times 10^{-12}$; Bonferroni $p_{\text{adj}} \approx 3.9 \times 10^{-9}$ across 780 combinations)	Retrospective; no prospective time-forward validation; validation restricted to 3/11 pathogens by curated antibody-escape annotation availability	Shows search-space reduction from $\sim$1,000 to $\sim$10–50 positions per protein with 7–9$\times$ enrichment for escape sites

escape subset supports 7.16–$8.24\times$ enrichment under the novel-only audit.

Keywords: mutation hotspot detection, MutBench, biological information types, viral genomics, multi-source information analysis, pathogen-dependent optimality, SARS-CoV-2, Deep Mutational Scanning

Chapter 1

Introduction

1.1 Research Background

Viral mutation hotspots are genomic positions where mutations concentrate and can mark changes in transmissibility, immune escape, or drug resistance; they therefore help prioritize variant monitoring, vaccine updates, and antiviral-risk assessment [8, 9]. Despite the scale of SARS-CoV-2 genomic surveillance [8], most hotspot analyses still rely on mutation frequency or Shannon entropy, which cannot by themselves distinguish functional signal from founder effects or compare against phylogenetic, structural, protein-language-model, or DMS-derived information.

Adjacent benchmarks evaluate per-variant fitness or escape rather than region-level hotspot selection: ViroGym [10], EVEREST v3 [11], ProteinGym [12, 13], EVEscape [14], Livesey & Marsh [15], and partial cancer-genomics precedents [16]. This gap also limits operational platforms such as Nextstrain [17] and PANGO [18], which track lineages but do not benchmark which information sources best identify mutation-prone positions. The author's prior MOSD [19] and MutClust [20] work motivated MutBench as a standardized evaluation framework for the computational-analysis step between sequence collection and experimental validation. MutBench evaluates how well computational rankings concentrate biologically meaningful positions for wet-lab triage within the RNA-virus single-position scope of Chapter 2; extension to DNA viruses, eukaryotic pathogens, and prokaryotic systems remains future work (Chapter 4, Section 4.6).

1.2 Related Research

This section reviews prior work in viral mutation hotspot detection and identifies the benchmarking gap that motivated MutBench.

1.2.1 Mutation Hotspot Detection Methods

Mutation hotspots are regions within a genome where mutations are concentrated in a non-uniform manner, likely reflecting functionally important sites under structural or immunological selective pressure. The concept originates from the observation that mutations are not distributed uniformly across viral genomes: certain positions or regions accumulate mutations at rates far exceeding the background, driven by positive selection for immune evasion, receptor binding optimization, or adaptation to new host environments. For RNA viruses, which exhibit mutation rates of 10^{-3} to 10^{-5} substitutions per nucleotide per replication cycle [21, 22], the interplay between high mutation supply and strong selective pressure creates pronounced mutational clustering in functionally critical regions. Identifying hotspots in viral genomes is critical for three practical applications: (1) predicting variant emergence by identifying positions predisposed to future mutations, (2) selecting vaccine targets by avoiding highly mutable epitopes or targeting conserved adjacent regions, and (3) developing broadly effective therapeutics that target structurally constrained positions unlikely to mutate without fitness cost. Existing hotspot detection approaches can be categorized into frequency-based, entropy-based, density-based, and selection-based methods.

Frequency-based approaches calculate mutation frequency at the amino acid or nucleotide level and designate positions with high frequency as hotspots. While intuitive and easy to implement, they have two fundamental limitations. First, **lineage bias**: when a specific lineage is overrepresented in the database, its defining mutations receive disproportionately high frequency scores. Mutations such as D614G, observed in nearly all modern SARS-CoV-2 sequences, reflect genuine fitness advantages, but the rate of near-fixation is amplified by founder effects during the early pandemic, making it difficult to distinguish the relative contributions of selection and genetic drift from frequency data alone. Second, **neglect of rare variants**: mutations that occur at low frequency but exhibit diverse forms—potentially indicating immune evasion or viral adaptation—are systematically overlooked.

Entropy-based approaches use Shannon entropy to measure nucleotide diversity at a given position. Mullick et al. [23] combined Shannon entropy with K-means clustering to detect hotspots in the SARS-CoV-2 Spike protein. However, conventional nucleotide entropy reflects the distribution of all nucleotides regardless of whether mutation has occurred, failing to capture effective diversity conditioned on the occurrence of mutation. Furthermore, K-means clustering requires the number of clusters to be specified in advance and does not account for density structure. Large-scale frequency-based analyses have been applied to SARS-CoV-2 [24, 25], but these studies used single information types (mutation frequency or fitness proxies) without systematic comparison against alternative scoring approaches.

Cancer genomics methods. In cancer genomics, a rich ecosystem of methods has been developed to detect somatic mutation hotspots. MutSigCV [26] identifies significantly mutated genes by modeling background mutation rates that account for gene expression, replication timing, and chromatin state. OncodriveCLUST [27] detects genes with significant spatial clustering of mutations in protein sequence space, under the assumption that driver mutations cluster in functional domains. HotMAPS [28] maps mutations onto protein three-dimensional structures to identify spatial hotspots in 3D, and MutSpot [29] models position-specific background mutation rates using epigenomic features. While these tools represent sophisticated approaches to hotspot detection, they target somatic mutations in tumor genomes, which are more than 1,000 times larger than viral genomes and involve fundamentally different mutation dynamics and selective pressures (neutral somatic mutation followed by clonal selection vs. population-level viral evolution), making direct transfer infeasible.

An important limitation of frequency-based mutation analysis is **phylogenetic non-independence**: mutations observed in closely related sequences are not independent events [30]. Phylogeny-aware methods such as TreeTime [31], BEAST [32], and HyPhy [33] can address this but require high-quality phylogenetic trees and face scaling and placement challenges in densely sampled datasets, motivating ultra-fast tree-placement frameworks such as UShER [34]. Several tools detect homoplasic (independently recurrent) mutations on phylogenies. HomoplasyFinder [35] identifies positions where the same mutation arises independently on multiple branches of a phylogenetic tree, which provides evidence for convergent evolution driven by positive selection rather than shared ancestry. ConDor [36] extends this approach to detect convergent amino acid substitutions across large protein alignments, using statistical tests to distinguish genuine convergence from chance recurrence. Homoplasy-based approaches are valuable because they explicitly account for phylogenetic non-independence, but they identify individual recurrent sites rather than performing the spatial clustering and multi-information-type comparison that MutBench provides. In MutBench, homoplasy is incorporated as one of the 10 underlying biological information families, which are expanded into 20 scoring channels (with normalization or compositing variants) and evaluated across pathogens (Section 3.5).

Simpler approaches to hotspot identification include sliding window methods, which scan fixed-length windows across the genome and flag those exceeding a mutation count threshold. While computationally straightforward, sliding window methods suffer from two inherent limitations: the window size must be specified in advance (too small misses extended hotspots; too large merges distinct hotspots), and they impose a fixed spatial resolution that cannot adapt to the variable density of mutations across different genomic regions. Signal processing techniques such as wavelet transforms offer a multi-scale alternative, decomposing mutation frequency profiles to identify localized peaks at multiple spatial scales simultaneously. However, these methods treat the genome as a one-dimensional signal and do not incorporate biological information about mutation importance or selective pressure.

Selection-based methods represent a fundamentally different approach. Rather than detecting spatial clusters of mutations, codon-based methods estimate the ratio of non-synonymous to synonymous substitution rates ($\omega = dN/dS$) at each codon position. The HyPhy phylogenetic platform [33] packages a family of such tests: MEME (Mixed Effects Model of Evolution) [37] detects episodic positive selection at sites where $\omega > 1$ on a subset of phylogenetic branches, while FUBAR (Fast Unconstrained Bayesian AppRoximation) [38] provides a Bayesian estimate of per-site selection pressure pooled across the whole tree. Closely related tests in the same family include FEL (Fixed Effects Likelihood) and SLAC (Single Likelihood Ancestor Counting) [39], which provide site-level ω estimates under a maximum-likelihood and a counting framework respectively, and BUSTED (Branch-Site Unrestricted Statistical Test for Episodic Diversification) [40], which tests for episodic diversifying selection along a designated set of branches. These methods provide a principled evolutionary framework for identifying positions under positive selection, but they detect individual selected sites rather than spatial clusters of hotspot regions, and they require reliable phylogenetic trees and codon-aligned sequences—prerequisites that are not always available for all pathogens. MutBench incorporates four phylogenetic selection-pressure scoring channels—three FUBAR-derived posterior summaries (FUBAR posterior probability, FUBAR positive-selection indicator, FUBAR Bayes factor) plus a Nei–Gojobori dN/dS proxy (Chapter 2, Section 2.2.4)—enabling direct comparison of selection-based features against frequency-based and PLM-based alternatives within a unified framework. FEL, SLAC, and BUSTED are reviewed here for completeness of the HyPhy family but are not benchmarked as scoring channels in the present panel; coverage of the broader selection-test family is left to future work (Chapter 4, Section 4.6).

1.2.2 Density-Based Clustering

DBSCAN [41] identifies density-connected clusters and noise from ε-neighborhoods and core points, allowing arbitrary cluster shapes without a pre-set cluster count. Its fixed global epsilon is poorly matched to viral genomes, where mutation density can differ sharply between immunodominant surface regions and conserved domains; in SARS-CoV-2, default DBSCAN merged 99.76% of points into one cluster, and tuned parameters ($\varepsilon = 0.15$, MinPts $= 5$) still yielded only 4 broad clusters [20]. HDBSCAN [42] and OPTICS [43] address variable density through hierarchical stability or reachability plots, but remain generic spatial algorithms. This limitation motivated MutClust [20], which replaces a global epsilon with an H-score-scaled, position-specific radius so biologically important positions receive wider neighborhoods.

1.2.3 Deep Mutational Scanning as Ground Truth

Deep Mutational Scanning (DMS) is an experimental technique that systematically measures the functional effects of all possible single amino acid substitutions within a protein [44]. In a typical DMS experiment, a library of mutant variants—each carrying a single amino acid substitution at one position—is generated through saturation mutagenesis. These variants are subjected to a selection assay (e.g., receptor binding, viral entry, or replication), and the relative fitness effect of each mutation is quantified by comparing variant frequencies before and after selection using deep sequencing. The result is a comprehensive *fitness landscape*: a matrix of fitness scores for every possible substitution at every position in the protein.

DMS is the gold standard for characterizing mutational effects because it is **unbiased** (every position and substitution is tested, not just those observed in nature), **comprehensive** (thousands of mutations are measured simultaneously in a single experiment), and **experimental** (fitness effects are directly measured rather than computationally predicted). Unlike natural sequence variation, which reflects the combined effects of selection, drift, and sampling bias, DMS directly measures the intrinsic functional consequence of each mutation.

Several landmark DMS studies have shaped our understanding of viral protein fitness landscapes. Starr et al. [45] measured ACE2 binding affinity and protein expression for all single amino acid substitutions in the SARS-CoV-2 receptor-binding domain (RBD), revealing that most RBD positions are highly constrained for binding. Dadonaite et al. [46] extended this to the full-length Spike protein by measuring pseudovirus cell entry fitness, identifying positions critical for viral entry beyond the RBD. Lee et al. [3] performed DMS of influenza H3N2 hemagglutinin (HA), demonstrating that DMS fitness measurements can predict the evolutionary fates of naturally circulating variants. Bloom and Neher [47] extended this further by inferring per-protein fitness effects across the entire SARS-CoV-2 proteome from public sequence phylogenies—rather than from direct DMS assays, since comprehensive DMS had then been performed on only two SARS-CoV-2 proteins—establishing a sequence-based reference complementary to direct DMS measurements.

A parallel lineage of **antibody-escape DMS** maps not viral fitness in isolation but the per-position escape from neutralization by specific antibodies or convalescent/vaccinee sera. Greaney et al. [48] pioneered this design for the SARS-CoV-2 receptor-binding domain by measuring escape from a panel of monoclonal antibodies, and Greaney et al. [49] extended it to the full Spike with polyclonal sera (both SARS-CoV-2-specific). These SARS-CoV-2 antibody-escape DMS datasets, together with the Bloom-lab broader escape-DMS catalog, are the direct experimental antecedent of the **vaccine-escape cross-validation analysis** reported in Chapter 4 (Section 4.4.2; results tabulated in Chapter 3 Table 3.20), which audits per-pathogen MutBench top-k output against curated antibody-/vaccine-escape position lists for HIV-1 (Moore–Williamson 2015), H3N2 (Lee et al. 2018) and SARS-CoV-2. Layer C ground truth in MutBench remains *fitness*-DMS based and is sourced from Dadonaite et al. 2024 (SARS-CoV-2 cell entry), Lee et al. 2018 (H3N2 replicative fitness), Haddox et al. 2018 (HIV-1 viral growth in TZM-bl), Simonich et al. (RSV), Aditham et al. (Rabies) and Bakhache et al. (EV-A71); see Section 1.2.3 of this chapter and Table 2.4 of Chapter 2. The two ground-truth tracks are therefore orthogonal: antibody-escape DMS feeds the vaccine-escape audit, fitness-DMS feeds Layer C, and each contributes a distinct per-position evaluation signal that is largely independent of literature-curated adaptive positives (Layer A) and of purifying-selection negatives (Layer B).

Despite its power, DMS has important limitations. First, it is **expensive and labor-intensive**: each DMS experiment requires constructing a complete mutant library and performing selection assays, limiting its application to a small number of viral proteins. Second, many surface-protein DMS datasets (especially for SARS-CoV-2 Spike and RSV F) use **pseudovirus systems** rather than authentic replication-competent virus, potentially missing fitness effects that depend on the complete viral replication cycle, although authentic replicative-fitness assays are also used in this dissertation's panel (e.g., MDCK culture for H3N2, RD culture for EV-A71). Third, DMS measures fitness **in vitro**, which may not fully capture in vivo selective pressures such as immune evasion in the context of population-level immunity. Finally, DMS coverage across the viral world remains sparse: the EVEREST v3 catalog [11] covers only 45 viral DMS datasets even after explicitly targeting WHO priority viruses, and the catalog spans far fewer distinct viral proteins than the 40-virus WHO panel itself, leaving most pathogens (and most non-surface proteins within covered pathogens) without DMS measurements. This availability gap means that DMS alone cannot serve as a universal ground truth for all pathogens—motivating MutBench's three-layer ground truth design (Chapter 2), where DMS serves as Layer C for the subset of pathogens with available data.

DMS data provides a natural ground truth for hotspot benchmarking: positions where mutations exhibit large fitness effects correspond to biologically functional sites that detection methods should identify [15]. Specifically, positions where the average DMS fitness effect of mutations is strongly negative (indicating functional constraint) or where specific mutations show large positive effects (indicating adaptive potential) are expected to overlap with biologically meaningful hotspots. Livesey and Marsh [15] demonstrated that DMS datasets can serve as unbiased benchmarks for computational variant effect predictors, establishing a precedent for the use of DMS as ground truth in method evaluation.

As a complementary approach, deep generative models have been shown to predict the effects of mutations from sequence variation alone [50], learning the statistical patterns of amino acid co-occurrence in evolutionarily related sequences to infer functional constraints without experimental data. This approach was extended by EVE [51], a Bayesian variational autoencoder (VAE) that learns evolutionary constraints from multiple sequence alignments (MSAs) without requiring experimental labels. EVE achieves state-of-the-art performance on clinical variant classification for human proteins and provides uncertainty estimates through its Bayesian formulation. The success of EVE and related generative models suggests that evolutionary information encoded in natural sequence variation captures much of the same functional signal as DMS experiments, albeit with lower resolution for individual mutations.

1.2.4 Protein Language Models for Variant Effect Prediction

Protein language models score mutations from evolutionary plausibility learned over large sequence corpora, offering structure- and assay-free features for variant-effect prediction. ESM-2 [52, 53] uses masked log-likelihood ratios, whereas Tranception [54] is autoregressive and can retrieve homologous context; both are relevant to MutBench because viral proteins may be underrepresented in cellular-protein-heavy training corpora. Broader PLM and structure-aware systems include ProtTrans [55], ESM-3 [56], ESM-IF [57], SaProt [58], and ProteinMPNN [59]; ESM-3 is excluded from the headline 8,580-evaluation grid because its Cambrian non-commercial license conflicts with the MIT-redistribution requirement (Chapter 2, Section 2.2.7), and structure-aware channels are deferred as future work. EVEscape [14] shows that generative fitness, accessibility, and residue dissimilarity can be productively combined; MutBench adopts only an EVEscape-style composite that substitutes ESM-2 mutation surprise for the EVE term (Section 2.2.4). AlphaMissense [60] is important but human-variant-centered, while EVEREST [11, 61] reports PLM underperformance on viral proteins as a cross-study parallel, not independent confirmation of MutBench's pathogen-dependence result (Section 1.2.8). ProteinGym [12] ranks predictors for per-mutation fitness regression over DMS assays; MutBench instead asks which information types, including ESM-2 masked-marginal and semantic-change scores, identify mutation-enriched regions across pathogens (Section 3.5).

1.2.5 Viral Genomic Surveillance Platforms

Real-time viral genomic surveillance platforms provide the operational context in which hotspot detection methods are deployed, but serve a fundamentally different purpose.

Nextstrain [17] integrates phylogenetic analysis with geographic and temporal metadata to track pathogen evolution in real time. Its primary function is to answer "which lineages are currently spreading and where?"—for example, tracking the geographic expansion of SARS-CoV-2 Omicron sublineages or seasonal influenza clades. Nextstrain visualizes the evolutionary relationships among circulating variants but does not evaluate or compare methods for detecting mutation-prone positions.

PANGO [18] provides a standardized lineage classification system for SARS-CoV-2, assigning labels (e.g., BA.2.86, JN.1) to sequences based on their phylogenetic placement. Its purpose is taxonomic organization—labeling "what is this variant?"—rather than identifying which genomic positions are likely to accumulate future mutations of functional significance.

These platforms cannot substitute for hotspot detection benchmarking for three reasons: (1) they operate primarily at the *lineage/variant level* (which variant is spreading), not at the *position level* (which genomic position is mutation-prone); (2) they are predominantly *retrospective* (tracking variants that have already emerged), although the Bedford-lab forecasting line (Łuksza & Lässig [62]; Huddleston et al. [63]) extends Nextstrain toward short-horizon clade-level prediction at the *lineage* rather than per-position level; and (3) they do not compare or evaluate detection algorithms—they are end-user tools, not method evaluation frameworks. The distinction is therefore one of *primary function* rather than absolute exclusion: surveillance answers "what has happened, and which clades are growing?" while hotspot detection aims to answer "where, at the residue level, might important mutations occur next?" MutBench addresses the latter question by providing a systematic framework for evaluating which detection methods most effectively identify biologically meaningful mutation-prone positions.

Other surveillance-related resources include **GISAID** [64], the primary repository for pathogen genome sequences providing the raw data from which mutation frequencies are computed, and **outbreak.info**, which aggregates variant prevalence data from GISAID. These databases are essential data *sources* for hotspot detection studies (including the present dissertation), but they do not themselves provide detection or benchmarking capabilities. The relationship between surveillance platforms and hotspot detection is thus complementary: surveillance platforms provide the data and operational context, while benchmarking frameworks like MutBench provide the methodological infrastructure for evaluating detection methods applied to that data.

1.2.6 Feature Importance Analysis in Viral Mutation Prediction

Prior feature-importance studies motivate MutBench's multi-source analysis but remain mostly single-pathogen: Maher et al. [65] found epidemiological features dominated sequence features for SARS-CoV-2 variant emergence, Rodriguez-Rivas et al. [66] showed epistatic DCA models improved escape prediction (AUC 0.76 vs. 0.63), Hie et al. [67] linked PLM semantic change to antigenic novelty, and EVEscape [14] combined fitness, accessibility, and residue dissimilarity. Coupling-aware methods such as DCA, EVcouplings, and GREMLIN remain outside MutBench's 20-scoring panel (Section 2.2.4), so the scoring $\times$ pathogen interaction in Chapter 3 is an information-type contrast within a single-position regime, not an upper bound on epistasis-aware methods. MutBench addresses the remaining gap by evaluating 20 scoring channels from 10 biological information families across 11 pathogens and reporting pathogen-specific feature recommendations (Section 3.5).

1.2.7 Algorithm Selection and Benchmarking Methodology

The finding that the best-performing detection method depends on the pathogen (Chapter 3) is an instance of Rice's algorithm selection problem [68]. Rice formalized this as a mapping from a *problem space* (characterized by measurable features of each problem instance) through a *feature space* to an *algorithm space*, with the goal of selecting the algorithm that maximizes a given *performance measure* for each instance. In the context of hotspot detection, each pathogen constitutes a problem instance characterized by features such as genome size, evolutionary rate, and mutation density distribution; the algorithm space comprises the set of scoring–detection combinations; and the performance measure is benchmark-stage-specific—hotspot-score (recall+precision+stability) for Stage 1 SARS-CoV-2 validation and MCC for the Stage 2 11-pathogen comparison (see Section 2.2.5).

This perspective is closely related to the **No Free Lunch (NFL) theorem** [69], which states that no single optimization algorithm outperforms all others across all possible problem instances. Importantly, NFL does *not* imply that all algorithms perform equally on any given problem—rather, it implies that superior performance on one class of problems necessarily comes at the cost of inferior performance on others. For hotspot detection, this means that a scoring type optimized for one pathogen's evolutionary characteristics may be suboptimal for another, motivating the systematic cross-pathogen comparison that MutBench provides.

This framework has been formalized in the machine learning community through meta-learning [70, 71] and automated algorithm selection (AutoML) [72, 73], where the goal is to automatically select the best-performing model or pipeline configuration for a given dataset. The parallel to viral hotspot detection is direct in principle: pathogen characteristics (genome size, mutation rate, population structure, selective pressure regime) constitute the problem-instance features, and scoring–detection pipelines constitute the algorithm space. The concrete pathogen-to-scoring recommendation produced by MutBench under this framing is presented in Chapter 4 as an application of the prior-work framework, not as part of the background itself. However, the algorithm selection framework has not previously been applied to viral genomics in this form, where the "problem instances" are pathogens with diverse evolutionary characteristics and the "algorithms" are scoring–detection pipelines.

The benchmarking methodology adopted in this dissertation follows established principles from computational biology. Weber et al. [74] proposed guidelines for neutral benchmarking, articulating three core requirements: (1) *independent evaluation*, where the benchmark designer is distinct from the method developers; (2) *multiple metrics*, to capture different aspects of performance and avoid optimization for a single criterion; and (3) *controlled experimental design*, ensuring that all methods are evaluated on identical data under identical conditions. Mangul et al. [75] extended these principles to a community-wide audit of omics computational tools, documenting systematic disparities in tool reliability that motivated the present study's emphasis on multi-pathogen rather than single-pathogen evaluation. Boulesteix et al. [76] further emphasized the importance of avoiding "self-assessment bias" where method developers evaluate their own methods on self-selected datasets, which systematically inflates reported performance. MutBench adheres to these principles—with the partial exception that MutClust [20], developed in the same lab as the present dissertation, is one of the 14 evaluated detection families and therefore violates Weber's strict "benchmark designer $\neq$ method developer" criterion. This self-assessment-bias risk (in the sense of Boulesteix et al. [76]) is mitigated, but not eliminated, by three design choices: (i) all 14 detection families are evaluated under identical preprocessing, scoring, and metric pipelines; (ii) the three-layer ground truth (literature-curated adaptive positives, purifying-selection negatives, DMS-anchored Layer C) is constructed independently of any method's predictions; and (iii) MutClust's hyperparameters were frozen prior to Stage 2 sweep execution. The remaining design influence is acknowledged as a limitation in Chapter 4. With this caveat, MutBench evaluates all methods using a composite metric (Stage 1 hotspot-score: recall+precision+stability) for in-depth SARS-CoV-2 validation and MCC for the cross-pathogen Stage 2 comparison (see Section 2.2.5 for the rationale).

1.2.8 Benchmarking Gap in Viral Hotspot Detection

The absence of benchmark studies for hotspot detection methods in viral genomics stands in contrast to cancer genomics, where Bailey et al. [16] conducted a comprehensive Pan-Cancer analysis applying 26 computational tools to 9,423 tumor exomes under uniform conditions, producing a widely used catalog of cancer driver genes. No analogous *detection-task* benchmark exists for viral mutation hotspots: while several recent fitness-regression benchmarks now span comparable virus panels, the region-level hotspot-detection task itself remains unstandardized—detection methods are evaluated using ad hoc validation against small sets of known functional mutations, without standardized datasets, metrics, or ground truth definitions. Related efforts in adjacent domains include: (1) ProteinGym [12, 13], which benchmarks variant effect predictors using 217 DMS assays and 90+ baseline models (as of v1.3) but addresses a per-substitution fitness-regression task rather than hotspot region detection; (1b) **ViroGym** [10] (Mar 2026), which benchmarks PLM and alignment-based scorers across 13 viruses, 79 DMS assays, 552,937 sequences, and 7 phenotypic categories including immune escape—larger virus count than MutBench, but focused on per-variant fitness rather than per-region detection; (2) ConDor [36], which detects convergent mutations across phylogenies but is a single tool rather than a benchmark framework; (3) Pons et al. [77], who reviewed 40+ cancer hotspot detection methods but produced a descriptive survey without a unified benchmark, and cancer-specific models do not transfer to viral evolutionary dynamics; and (4) individual SARS-CoV-2 studies using entropy [8] or clustering that validate against small variant sets without standardized ground truth. Among the 35+ mutation clustering tools catalogued in the computational oncology literature [77], none was designed for or validated on viral population-level mutation data. Cancer hotspot tools (OncodriveCLUST [27], HotMAPS, MutSpot) require tumor cohort somatic mutation calls and human-genome-specific background models (trinucleotide signatures, replication timing). Phylogenetic selection tools (MEME, FUBAR, FEL, SLAC, BUSTED) detect per-codon dN/dS signals but do not perform spatial clustering of hotspot regions. The only virus-specific hotspot clustering tool is MutClust [20], validated on a single pathogen (SARS-CoV-2). This gap—the absence of systematic multi-pathogen hotspot detection benchmarking—is what MutBench addresses. Thus, a systematic benchmark with multi-source ground truth, composite evaluation metrics, and cross-pathogen validation remains absent for viral hotspot detection.

Variant Effect Prediction Benchmarks

Recent years have seen significant progress in benchmarking variant effect prediction (VEP) methods on proteins. ProteinGym [12] established a large-scale benchmark comprising 217 DMS assays and 2.7 million substitution variants, primarily for human and model organism proteins. EVE [51], a Bayesian VAE trained on evolutionary sequences, demonstrated that unsupervised generative models can predict disease-associated variants without clinical labels. EVEscape [14] extended this framework to viral immune escape prediction by combining EVE fitness scores with structural accessibility and physicochemical dissimilarity, successfully forecasting SARS-CoV-2 Omicron escape mutations from pre-pandemic data.

Most recently, EVEREST [11, 61] specifically benchmarked VEP methods across WHO priority viruses; the v3 expansion (Jan 2026) covers 45 viral DMS datasets, 31 forecasting clades across 4 viruses, and reliability assessment across 40 WHO priority RNA viruses spanning 16 viral families. EVEREST finds that protein language models substantially underperform alignment-based approaches on viral proteins, that reliability fails for over half of the WHO-40 panel (especially surface glycoproteins), and that prediction reliability varies dramatically across pathogens. Because MutBench's LOPO 0/11 match rate is itself null-consistent under permutation (Chapter 3), we describe this as a cross-study *parallel* on pathogen-dependence rather than as independent corroboration of non-transferability per se. The PLANT framework [78] and Shah et al. [79] provide PLM-based and ML-based H3N2 antigenic cartography that explicitly competes with WHO vaccine-strain selection, building on the foundational antigenic-cartography paradigm of Smith et al. [80]; MutBench's H3N2 self-consistency check (Section 4.4.2) is positioned as a detection-side complement to these prediction-side cartographies, not as a competitor. A particularly relevant finding from EVEREST is that the performance gap between methods varies by viral family: alignment-based methods such as EVE consistently outperform PLMs across most viral proteins, but the magnitude of this advantage differs substantially between, for example, influenza HA and SARS-CoV-2 Spike. This pathogen-dependent performance variation parallels the central finding of MutBench (Chapter 3). AlphaMissense [60] demonstrated proteome-wide variant classification using an AlphaFold-derived architecture, though its focus on human proteins limits direct viral applicability.

MutBench's unique contribution is the *scoring × pathogen interaction analysis*: while EVEREST establishes that method performance varies across viruses, MutBench quantifies this interaction through three-way ANOVA ($\omega^2 = 0.296$) and demonstrates that the choice of *biological information source* (not just the computational method) is the dominant performance factor. This finding—that the optimal information source is pathogen-dependent within the 20-type/39-variant design, with the interaction as the largest modeled variance component—has no counterpart in EVEREST or ProteinGym, which treat the scoring/feature set as fixed. Specifically, MutBench provides three capabilities absent in EVEREST and other VEP benchmarks: (1) quantification of the scoring–pathogen interaction through three-way ANOVA, revealing that the scoring × pathogen interaction ($\omega^2 = 0.296$; 11-pathogen cluster-bootstrap 95% interval [0.201, 0.346]) is the largest modeled variance component (within-cell residual $\omega^2 \approx 0.39$); (2) cross-validation via vaccine escape enrichment for the 3 of 11 pathogens with curated escape annotations, with HIV-1 (7.19×, Bonferroni-corrected $p = 2.5 \times 10^{-16}$; escape set 82% disjoint from Layer A) as the primary external anchor and H3N2 9.36× reported as a self-consistency check given 69% Layer A overlap (circularity audit in Table 3.20); and (3) a concrete pathogen-to-scoring recommendation table enabling application to new pathogens without requiring DMS experimental data. Furthermore, MutBench introduces the vaccine escape cross-validation paradigm with a per-pathogen circularity audit: HIV-1 provides the cleanest external validation, while H3N2 and SARS-CoV-2 contribute self-consistency and exploratory evidence respectively—a distinction absent from EVEREST or ProteinGym. Table 1.1 summarizes the key differences between MutBench and these related benchmarks.

Table 1.1: Cross-benchmark comparison: task, scope, ground-truth, methods compared. The ViroGym row reflects the March 2026 arXiv preprint (pre-peer-review) and is included for completeness; EVEREST v3 (Jan 2026) is a bioRxiv preprint and ProteinGym v1.3 is a versioned release/update—both are well-established community benchmarks but the v3 / v1.3 versions specifically have not been confirmed peer-reviewed in this dissertation's bibliography.

	MutBench	**ViroGym**	**EVEREST v3**	**ProteinGym v1.3**	**Bailey 2018**
Domain	Viral hotspot	Viral VEP	Viral VEP	General VEP	Cancer driver
Task	Detection	Prediction	Prediction	Prediction	Detection
Pathogens/proteins	11 viruses	13 viruses (79 DMS)	4–40 viruses (45 DMS)	217 proteins	Cancer genes
Ground truth	3-layer	DMS+escape	DMS	DMS	Patient cohorts
Methods compared	780 combos	$\sim$10 PLM/MSA	$\sim$20 VEP	90+ VEP	26 tools
Cross-pathogen ANOVA	Yes (ω^2)	No	No	No	No

Direct performance comparison between MutBench and ProteinGym, EVEREST, or ViroGym is not possible because the tasks differ fundamentally: VEP benchmarks (including ViroGym's per-variant fitness/escape regression and forecasting tasks) evaluate Spearman correlation or AUROC between predicted and measured per-variant outcomes, while MutBench evaluates MCC for binary per-region hotspot classification. The benchmarks are complementary rather than competing.

This gap—systematic multi-pathogen benchmarking for hotspot detection—is the fundamental motivation for MutBench (Chapter 2), the core study of this dissertation.

Related research summary. This section has reviewed the two bodies of prior work that directly inform MutBench. Section 1.2 surveyed viral mutation hotspot detection methods—from frequency-based and entropy-based scoring through density-based clustering to protein language models and selection-based approaches—and identified a critical benchmarking gap: no standardized framework exists for evaluating and comparing these methods across multiple pathogens with independent ground truth. The only virus-specific hotspot detection tool, MutClust [20], relies on a single pathogen, single information type (H-score), and self-referential validation, leaving open the question of how different scoring approaches and detection algorithms compare across diverse viral pathogens. Together, these gaps motivate the design of MutBench as a systematic, multi-pathogen benchmark with three-layer ground truth, composite evaluation metrics, and statistical analysis of scoring–pathogen interactions. Coupling-aware methods (DCA, EVcouplings, GREMLIN, and similar pairwise/epistatic models) remain explicitly out of scope of the 20-scoring panel; the scoring $\times$ pathogen interaction $\omega^2 = 0.296$ and all downstream Stage 2/3 conclusions should therefore be read within a single-position scoring regime, with epistasis-driven hotspot detection deferred as future work (Chapter 4, Section 4.6). Chapter 2 presents the MutBench framework in detail, including the construction of multi-layer ground truth datasets for 11 pathogens, the benchmark design comprising 20 scoring types across 6 categories and 14 detection method families, and the factorial ANOVA methodology for quantifying scoring–pathogen interactions.

1.3 Research Motivation and Problem Definition

The absence of systematic evaluation for viral mutation hotspot detection gives rise to the following three specific problems.

First, there is a **lack of a standardized evaluation framework**. Existing hotspot detection methods, including MutClust, report their performance using their own datasets and evaluation criteria, making fair comparison among methods impossible. The bootstrap test, MutClust's sole validation method, is a self-referential approach that can only confirm the statistical significance of detected clusters; it cannot verify whether those clusters correspond to biologically meaningful functional regions.

Second, there is an **absence of validation based on independent ground truth**. Objectively evaluating hotspot detection results requires a ground-truth standard that is independent of the algorithm, yet no such ground truth framework has been defined for viral mutation hotspot detection. Although independently obtainable evidence sources exist—such as convergent evolution sites, sites under purifying selection (natural selection that removes deleterious mutations), and Deep Mutational Scanning (DMS; a technique that experimentally measures the functional impact of every possible single amino-acid substitution in a protein) data—they have not been systematically integrated. Convergent evolution sites are mutations that independently arise at the same genomic position in multiple lineages within the same viral species; for example, mutations at Spike position 484 emerged independently across multiple SARS-CoV-2 lineages: E484K (glutamic acid to lysine) in Beta, Gamma, and Mu, and E484A (glutamic acid to alanine) in Omicron.

Third, there is a **lack of statistical evidence on the pathogen-dependence of best-performing methods**. Existing methods use parameters tuned for a single pathogen, and their generalizability has not been rigorously quantified. Whether differences in pathogen-specific genomic characteristics (genome length, mutation rate, recombination frequency, etc.) systematically affect the best-performing detection strategy has not been verified. This is a specific instance of the algorithm selection problem [68]: given a pathogen with particular genomic characteristics, which detection method should be applied? Without systematic cross-pathogen evaluation, this question cannot be answered.

1.4 Research Objectives

To address the problems above, this study establishes the following three objectives.

Objective 1. Construct a systematic experimental framework (MutBench) that enables stan-

dardized comparison of diverse biological information types for hotspot detection across 11 RNA viruses, comprising:

- A three-layer ground truth (Layer A literature-curated positives—a pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types, dominated by immune-escape sites; Layer B purifying-selection negatives; Layer C DMS fitness validation),
- The hotspot-score composite metric (recall + precision + stability)/3,
- A two-stage main experimental design with a Stage 3 information-integration layer: Stage 1 for in-depth SARS-CoV-2 analysis; Stage 2 for large-scale comparison across 11 RNA viruses (20 scoring formulas $\times$ 39 detection parameter variants from 14 families $\times$ 11 pathogens = 8,580 evaluations); and Stage 3 for multi-source feature integration (see Chapter 3 Section 3.5.2).

Table 1.2: Panel-scope ledger for MutBench claims

Panel	**Used for**	**Not used for**
11-pathogen main panel	Headline ω^2 interaction, Friedman, LOPO 0/11, and vaccine-escape enrichment	Replaced by neither the 12-pathogen Stage 3 panel nor the Wave panels for headline claims
12-pathogen Stage 3 panel	11 main + Zika; feature-ablation and nested-LOPO robustness check (Chapter 3 Section 3.5.2)	Benchmark expansion beyond the 11-pathogen main panel
Wave 4–5 features-only panels	Wave 4 28-target stability with 12 labeled + 16 unlabeled; Wave 5 Layer A′ sensitivity on 10 UniProt-derived targets (Chapter 4 Paragraphs 9, 9)	Headline panel extension; bounded sensitivity analyses only

All headline claims belong to the 11-pathogen main panel unless stated otherwise.

Objective 2. Determine which factor—scoring type, detection method, or pathogen—dominates the variance in hotspot-detection performance, and whether the dominant factor is a main effect or an interaction. Operationally, this asks whether a single "best" scoring–detection combination exists across pathogens, and is addressed through factorial Analysis of Variance (ANOVA), Friedman test, and Leave-One-Pathogen-Out (LOPO) cross-validation (see Chapter 3).

Objective 3. Externally validate the benchmark on independently-curated vaccine-escape positions, prioritising the Layer-A-disjoint subset to test whether the framework reduces the experimental search space on positions that are not already in the literature-curated ground truth (see Chapter 3). A multi-source feature-ablation robustness check (compact 4-feature core vs. full 10-feature ensemble under nested-LOPO) is reported alongside, but is treated as a methodological sanity check on the EqualWeight design rather than as a separate contribution.

Figure 1.1 summarizes the overall research flow: from identifying the information limitation (reliance on frequency/entropy), through systematic comparison of 10 information types via MutBench, to the key finding that the optimal information source differs by pathogen, culminating in external vaccine-escape validation on HIV-1 as the primary anchor for practical impact.

1.5 Research Contributions

The research contributions of this study are as follows.

Contribution 1. MutBench: a systematic hotspot-detection benchmark across 11 RNA viruses. To our knowledge, MutBench is the broadest *region-level, single-position* hotspot-detection benchmark to date for RNA viruses (8,580 evaluation cells across the 20-type single-position scoring regime and the 11-pathogen main panel, with the Stage 3

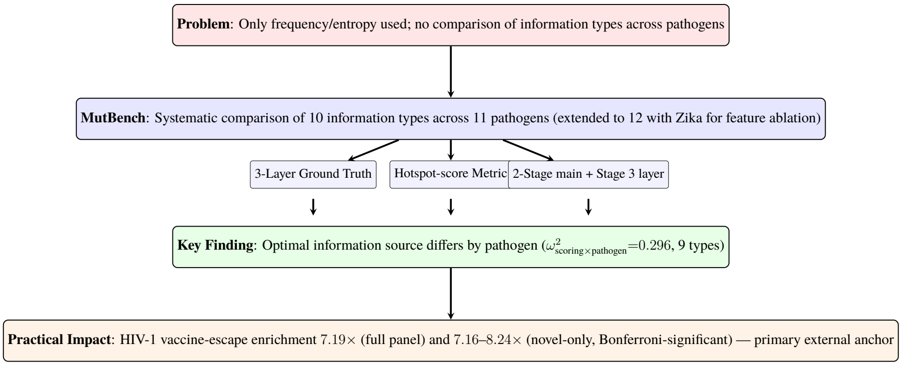

Figure 1.1: Research overview. The study proceeds from identifying the information limitation (reliance on frequency/entropy) through systematic comparison of 10 information types via MutBench, to the key finding that the optimal information source differs by pathogen, culminating in external vaccine-escape validation on a Layer A-disjoint subset (HIV-1, Bonferroni-significant) as the primary anchor for practical impact.

feature-integration analysis extended to a 12-pathogen panel by adding Zika), complementing concurrent fitness-regression benchmarks (ViroGym [10], EVEREST v3 [11], ProteinGym v1.3 [13]) by targeting region-level detection rather than per-variant fitness, and is designed to answer the question: *which biological information source is most effective for which pathogen?* It organizes 10 information types into 20 scoring formulas across 6 categories (frequency, MSA-derived, phylogenetic, structural, AI-based, and composite), defines a three-layer ground truth framework (Layer A as a literature-curated, pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types—with immune-escape sites dominating; Layer B as purifying-selection negatives; Layer C as DMS experimental fitness), introduces the hotspot-score composite metric, and presents a two-stage main experimental design plus a Stage 3 information-integration layer combining in-depth SARS-CoV-2 analysis (Stage 1), a large-scale comparison across the 11-pathogen main panel (Stage 2), and multi-source feature-integration analysis on the 12-pathogen panel (Stage 3, extended from the main panel with Zika; Section 3.5.2).

Contribution 2. Statistical demonstration that the optimal biological information

source differs fundamentally across pathogens ($\omega^2_{\textbf{scoring}\times\textbf{pathogen}} = 0.296$), with 9 distinct optimal information types across 11 pathogens; LOPO 0/11 is reported as null-consistent corroboration of the interaction effect, not as independent evidence. All eleven pathogens exhibited different best-performing combinations (LOPO cross-validation: 0/11 matches), with 9 distinct MCC-best scoring types observed—including FUBAR phylogenetic selection scores for H3N2, protein language model scores for SARS-CoV-2 and RSV (MCC-best at the scoring–detector cell), and homoplasy for Norovirus. The Friedman test ($p = 0.990$) showed no reliably superior combination across all pathogens, consistent with pathogen-dependent optimality. The scoring $\times$ pathogen interaction is the largest modeled variance component ($\omega^2 = 0.296$, 95% cluster-bootstrap CI [0.201, 0.346]; 6-category aggregation gives $\omega^2 = 0.103$ as a collinearity-robust lower bound; 11 pathogen clusters places the design near the small-cluster regime—above the Bolker rule-of-thumb minimum of 5–6 but below the typical recommendation of 20–30 for stable random-effects estimation, see Chapter 4, Section 4.4.4). The conclusion rests on the interaction effect plus the vaccine-escape audit; LOPO 0/11 alone is null-consistent under permutation (Chapter 3 Table 3.7).

Contribution 3. Practical search-space reduction validated on independently-curated vaccine-escape positions. The optimal scoring–detection combination narrows the candidate list from $\sim$1,000 alignment positions to approximately 10–50 (top-decile budget; 11–32 detected positions in the three reported escape-enrichment anchors) with 7–9$\times$ enrichment for escape sites, transforming exhaustive experimental screening into guided investigation. The validation hierarchy is summarized below.

Per-feature analyses show that the most discriminative feature differs across pathogens (homoplasy for Norovirus, pLDDT for H3N2, ESM-2 for SARS-CoV-2), complementing the Stage 2 MCC-optimal scoring results in Contribution 2; Chapters 3–4 report the full-lattice

Table 1.3: External-validation ledger for practical search-space reduction

Anchor	Main result	Boundary	Role
HIV-1 primary anchor	Layer-A-disjoint subset of HIV-1 escape positions (37 of 45) yields direct novel-only enrichment 7.16–$8.24\times$ (worst-case nominal Fisher $p = 5.0 \times 10^{-12}$; Bonferroni-adjusted $p_{\text{adj}} \approx 3.9 \times 10^{-9}$ across 780 combinations)	Three-tier evidence rule: Bonferroni survival $\times$ Layer A overlap $< 33\%$	Primary external validation on independently-curated vaccine-escape positions
H3N2 self-consistency	$9.36\times$ enrichment with 69% Layer A overlap; EqualWeight integration yields $4.012\times$ enrichment ($p = 0.0023$)	Overlap makes this a self-consistency check	Secondary support for practical enrichment
SARS-CoV-2 exploratory	Escape enrichment persists after multiple-comparison correction	Exploratory after multiple-comparison correction	Bounded external check, not the primary anchor
4-core robustness boundary	Nested-LOPO on the 12-pathogen Stage 3 panel: full-10 LOPO mean MCC 0.070, fixed-4 LOPO mean MCC 0.081, paired delta $+0.011$, 95% CI $[-0.018, +0.042]$, sign-flip permutation $p = 0.61$; compact 4-feature core (homoplasy, pLDDT, entropy, frequency) is statistically indistinguishable within a ± 0.04 MCC non-inferiority bound (Section 3.5.2, Table 3.14)	Retrospective, fixed historical recipe rather than a lattice-optimal or prospectively callable detector	Practical cold-start recipe when labelled positions are unavailable

audit and abstention-rule boundaries.

These three contributions form a logical hierarchy: Contribution 1 provides the standardized evaluation infrastructure; Contribution 2 uses it to establish pathogen-dependent information-source utility; and Contribution 3 provides external validation on independently-curated vaccine-escape positions.

1.6 Dissertation Organization

This dissertation comprises four chapters.

Section 1.2, Related Research (within this chapter) reviews the existing literature on viral mutation hotspot detection, benchmarking methodology, and the gap that motivated the development of MutBench.

Chapter 2, Materials and Method describes in detail the materials and methodology of the MutBench benchmark framework, the core contribution of this dissertation. The chapter covers the target pathogens and sequence data, the framework overview and three-layer ground truth construction, the Stage 1 SARS-CoV-2 analysis design including MutClust variants and the hotspot-score metric, the Stage 2 cross-pathogen benchmark design with 20 scoring types and 14 detection families, the Stage 3 multi-source information integration design, and the evaluation metrics and statistical analysis design.

Chapter 3, Experimental Results reports the experimental results organized into five sections: Single-Pathogen Benchmark (synthetic data comparison, multi-ground truth evaluation, real GISAID validation, and external methods comparison on SARS-CoV-2), Robustness Validation (statistical validation, sensitivity analysis, temporal generalization, and indel exclusion), Cross-Pathogen Validation (H3N2 validation, 3D structural analysis, phylogenetic simulation, and ESM-2 cross-paradigm validation), the Cross-Pathogen Large-Scale Benchmark (ANOVA, Friedman test, LOPO cross-validation), and Information-Type Analysis (feature ablation on the 12-pathogen Stage 3 panel, ground truth heterogeneity, search space reduction, and multi-source integration).

Chapter 4, Discussion, Limitations, and Future Directions provides a comprehensive interpretation of the experimental results, summarizes the contributions of this research, analyzes its limitations, and presents future research directions.

Chapter 2

Materials and Method

This chapter presents the methodology of MutBench, an experimental framework for analyzing which factors—biological information type, detection method category, and pathogen characteristics—influence viral mutation hotspot detection performance. The overall pipeline processes multiple sequence alignment (MSA) data from 11 RNA viruses through 20 scoring types and 14 detection method families spanning 5 categories (threshold-based, signal processing, density/clustering, statistical, and adaptive window), with 39 parameter variants, evaluating results against a 3-layer ground truth system using the hotspot-score composite metric and Matthews Correlation Coefficient (MCC). The evaluation follows a two-stage main design with a Stage 3 information-integration layer (Stages 1 and 2 form the main 11-pathogen factorial benchmark; Stage 3 is a separate information-integration layer): Stage 1 conducts in-depth analysis on SARS-CoV-2 to validate the framework components, Stage 2 extends the evaluation to all 11 pathogens for the large-scale cross-pathogen MCC comparison, and Stage 3 (Section 2.2.6) decomposes the resulting performance by information type via per-feature AUC, ablation, Random Forest, and vaccine-escape cross-validation. Section 2.1 describes the materials (target pathogens and sequence data). Section 2.2 presents the methods, beginning with a framework overview (Section 2.2.1) that describes the overall architecture and three-component pipeline, followed by the 3-layer ground truth framework

(Section 2.2.2). The experimental design is then presented in two stages: Stage 1 (Section 2.2.3) focuses on SARS-CoV-2 in-depth analysis including MutClust variant comparison and the hotspot-score metric, while Stage 2 (Section 2.2.4) extends to the 11-pathogen cross-pathogen benchmark with 20 scoring types and 14 detection families. Section 2.2.5 details the evaluation metrics and statistical analysis design.

2.1 Materials

2.1.1 Target Pathogens and Data

Eleven RNA viruses were selected to cover diverse evolutionary dynamics, transmission routes, and genomic characteristics: (1) *Respiratory viruses*: SARS-CoV-2 (Spike, 1,259 AA), MERS-CoV (Spike, 1,330 AA), RSV (F protein, 550 AA), Influenza A H3N2 (HA, 543 AA), and Influenza B (HA, 560 AA); (2) *Blood-borne/sexually transmitted*: HIV-1 (Env gp120, 450 AA) and HCV (E2, 340 AA); (3) *Enteric*: Norovirus (VP1, 536 AA); (4) *Arthropod-borne*: Dengue (E protein, 490 AA); (5) *Neurotropic*: Rabies (G protein, 491 AA) and EV-A71 (VP1, 261 AA). *Note: all amino acid lengths above refer to post-truncation alignment lengths used in the benchmark; full-length protein sequences may differ (e.g., SARS-CoV-2 Spike full length is 1,273 AA, MERS-CoV Spike 1,353 AA, Rabies G 524 AA, EV-A71 VP1 297 AA).* These 11 pathogens span substitution rates from $\sim 10^{-3}$ (HIV-1, SARS-CoV-2) to $\sim 10^{-4}$ (Rabies) substitutions/site/year, alignment lengths from 261 to 1,330 AA, and evolutionary patterns from rapid antigenic drift (H3N2) to epochal evolution (Norovirus). This diversity ensures that the benchmark evaluates method performance across a deliberately diverse but non-exhaustive range of RNA-virus surface-glycoprotein evolutionary regimes rather than a single pathogen's characteristics. Zika virus was considered but excluded from the Stage 2 benchmark because only 52 unique E protein sequences were available in GenBank, insufficient for reliable per-position frequency and entropy estimation; Zika is included in the supplementary 12-pathogen feature analysis (Section 3.5).

Per-pathogen biological context. The 11 pathogens span 8 ICTV viral families and four transmission routes, with substitution rates from $\sim 4 \times 10^{-4}$ (Rabies) to $\sim 4\text{–}4.5 \times 10^{-3}$ (H3N2 HA, EV-A71 VP1) substitutions/site/year and unique-sequence ratios from 15.1% (SARS-CoV-2 Spike) to 100% (Rabies G, EV-A71 VP1; Table 2.1). Five panel-level biological points re-appear in the Layer A construction (Section 2.2.2, Table 2.4) and in the pathogen-specific best-method discussion: SARS-CoV-2 Spike's shallow pandemic phylogeny creates strong founder effects (D614G) alongside convergent immune-escape sites (E484K, N501Y), making it the richest three-layer pathogen and the Stage 1 anchor; H3N2 HA exhibits the canonical five antigenic sites (A–E) under continuous immune-driven drift; Influenza B is pooled across Victoria/Yamagata lineages (reference `CY018765`), so lineage-defining substitutions appear as high-frequency positions (revisited in Chapter 4 Section 4.4.4); HCV E2 carries the HVR1 hypervariable region (residues 384–410) under diversifying selection, motivating the HCV-only ANOVA sensitivity analysis ($\omega^2 = 0.246$); Norovirus VP1 evolves under epochal selection at the GII.4 P2 subdomain (homoplasy AUC 0.944). DMS Layer C data are available for SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, and EV-A71. Zika was excluded from the Stage 2 benchmark (only 52 unique E-protein sequences); it is included in the 12-pathogen feature analysis (Section 3.5).

FASTA sequences for each pathogen were collected from NCBI GenBank by querying each target protein name (e.g., "SARS-CoV-2 spike glycoprotein") with filters for complete coding sequences, excluding partial sequences and laboratory-derived constructs. The benchmark scope is the *single-position regime within the coding region of the surface glycoprotein only*: $5'/3'$ UTRs, IRES elements, polyprotein-cleavage junctions outside the assayed protein boundary (e.g., HCV E2 outside the 340-AA evaluated alignment), pseudogenes, and non-coding selection signals are explicitly out of scope and are not represented in any per-position scoring vector. For HCV the polyprotein structure means only the E2 region under the literature-curated alignment window is evaluated; CD81-binding regions reported in polyprotein numbering outside this window are noted in Table 2.4 but not converted into Layer A/B labels. No date range restriction was applied; all available sequences as of the per-pathogen query dates listed in Table 2.1 (column "Query date"; all queries fell within the 2024-11 to 2024-12 window) were downloaded. As an illustrative example, the SARS-CoV-2 Spike query executed via NCBI E-utilities on 2024-12-09 was `(``Severe acute respiratory syndrome coronavirus 2''[Organism] OR txid2697049[Organism]) AND spike[Gene Name] AND ``complete cds''[Properties] NOT (partial[Title] OR laboratory[Title])` with a length filter of 3,500–4,000 nt; analogous query strings for the remaining ten pathogens (organism/taxon ID, gene/product filter, length window, exclusion clauses) are archived verbatim in `results/mutbench/provenance/genbank_queries.csv` so that any reader can reproduce the per-pathogen sequence sets to within the inevitable GenBank submission drift after the recorded query date. Sequences with $>5\%$ ambiguous bases (N) were removed, and identical (100% identity) sequences were deduplicated using a self-implemented Python hashing pass (canonical-uppercase amino-acid string $\rightarrow$ SHA-256), which is functionally equivalent to running CD-HIT [81] or MMseqs2 [82] at a 100% identity threshold for exact-duplicate removal because at the 100%-identity setting both tools reduce to exact-string deduplication (no partial-match scoring), so the SHA-256 hash and the tool output produce *arithmetically identical* unique-sequence sets and the choice avoids an additional dependency in the reproducibility environment. Clustering at $<100\%$ identity (e.g., the 0.95–0.99 thresholds typically used with CD-HIT-EST/MMseqs2 `easy-cluster`) was *not* applied, so near-duplicate but non-identical sequences were retained intentionally to preserve per-position frequency and entropy estimates. HMM-based homology filtering (HMMER [83], `phmmer`/`hmmsearch`) was likewise *not* applied: the GenBank query already restricts retrieval to gene-name and "complete cds" flags, and any frameshift or partial-domain mis-annotations passing those flags are bounded by the post-MAFFT minimum-length truncation step (Section below); a follow-up HMMER profile-screen pass against a per-pathogen reference HMM is left for the next data-curation cycle. The "Unique seq." column in Table 2.1 reflects post-deduplication counts under the 100% identity threshold. Multiple sequence alignment (MSA) was performed for each surface protein (SARS-CoV-2: Spike, H3N2: HA, HIV-1: Env gp120, etc.) using MAFFT v7.505 [84] with the `--auto` mode, followed by computation of per-position frequency $f_i = 1 - p_{\text{consensus}}(i)$ (the minor-allele fraction at column i) and Shannon entropy on the same column. The `--auto` mode automatically selects among FFT-NS-2, L-INS-i, and G-INS-i based on the number and length of input sequences (Katoh & Standley, 2013, §2) [84]; the actual algorithm selected per pathogen is logged in Table 2.1 (column "MAFFT mode"), and the GenBank query strings used for retrieval are archived alongside the released CSV bundle (`results/mutbench/provenance/genbank_queries.csv`). At the sequence scales used in this study (1,311–5,325 sequences, Table 2.1), FFT-NS-2 was the dominant selection (10 of 11 pathogens); the per-pathogen log in the column is the authoritative reference for any reproduction attempt. The effect of `--auto` algorithm selection on alignment quality—and consequently on per-position mutation frequency computation—was not separately ablated in this study; the per-pathogen log is provided so that future cycles can pin individual algorithms (e.g., `--retree 2 --maxiterate 0` for FFT-NS-2) for sensitivity analyses. When alignment lengths differed across sequences, all sequences were truncated to the minimum sequence length within the MSA to ensure consistency in per-position comparisons. Ground truth positions exceeding the alignment length are thereby excluded from evaluation (the position counts in Table 2.4 are effective values after this truncation). Table 2.1 summarizes the characteristics of each pathogen's sequence dataset.

Table 2.1: Sequence dataset characteristics for 11 pathogens used in the benchmark. "Query date" is the YYYY-MM-DD on which the GenBank query was executed; "MAFFT mode" is the algorithm actually selected by MAFFT v7.505 `--auto` for that pathogen's input set (per-pathogen log).

Pathogen	Surface protein	Reference accession	Sequences	Unique seq.	Align. len. (AA)	Query date	MAFFT mode
SARS-CoV-2	Spike	NC_045512.2 (Wuhan-Hu-1)	4,982	753	1,259	2024-12-09	FFT-NS-2
H3N2	HA	CY045572 (A/Brisbane/10/2007)	2,644	2,562	543	2024-12-09	FFT-NS-2
Norovirus	VP1	KC576909 (GII.4 Sydney)	1,500	791	536	2024-11-22	FFT-NS-2
HIV-1	Env gp120	K03455 (HXB2)	3,000	2,256	450	2024-11-22	FFT-NS-2
Dengue	E protein	NC_001474.2 (DENV-2)	2,000	796	490	2024-11-25	FFT-NS-2
RSV	F protein	KT992094 (RSV-A)	5,070	4,779	550	2024-12-09	FFT-NS-2
Influenza B	HA	CY018765 (B/Florida/4/2006)	5,325	5,019	560	2024-12-09	FFT-NS-2
MERS	Spike	NC_019843.3 (MERS-CoV)	1,311	530	1,330	2024-11-22	FFT-NS-2
HCV	E2	NC_004102.1 (HCV-1a)	1,362	1,067	340	2024-11-22	FFT-NS-2
Rabies	G protein	NC_001542.1	2,038	2,038	491	2024-11-25	FFT-NS-2
EV-A71	VP1	AB204852	662	662	261	2024-11-25	L-INS-i

Query strings (full GenBank advanced-search expressions including taxonomy ID, gene/product filters, and length bounds) are archived in `results/mutbench/provenance/genbank_queries.csv`. MAFFT `--auto` selects L-INS-i when the input is small enough that the iterative refinement is tractable ($N \times L$ below the heuristic boundary, broadly $N \lesssim 200$ or short L) and FFT-NS-2 otherwise (Katoh & Standley 2013, §2). EV-A71 ($N = 662$ unique sequences, $L = 261$ AA) falls inside the L-INS-i band because its $N \times L \approx 1.7 \times 10^5$ product remains below the FFT-NS-2 cut-over despite $N > 200$; the remaining ten pathogens have larger $N \times L$ products and were therefore aligned under FFT-NS-2.

2.2 Methods

2.2.1 Framework Overview

The MutBench framework follows a three-component pipeline architecture (Figure 2.1): (1) ground truth construction from three independent biological evidence sources, (2) systematic scoring and detection via 20 scoring types and 14 detection method families, and (3) multi-metric evaluation with factorial statistical analysis. The evaluation is organized into two main stages (Stages 1 and 2), with a separate Stage 3 information-integration layer described in Section 2.2.6: Stage 1 performs in-depth analysis on SARS-CoV-2 to validate the framework components, and Stage 2 extends the evaluation to all 11 pathogens for large-scale cross-pathogen comparison. The following subsections describe each component in detail.

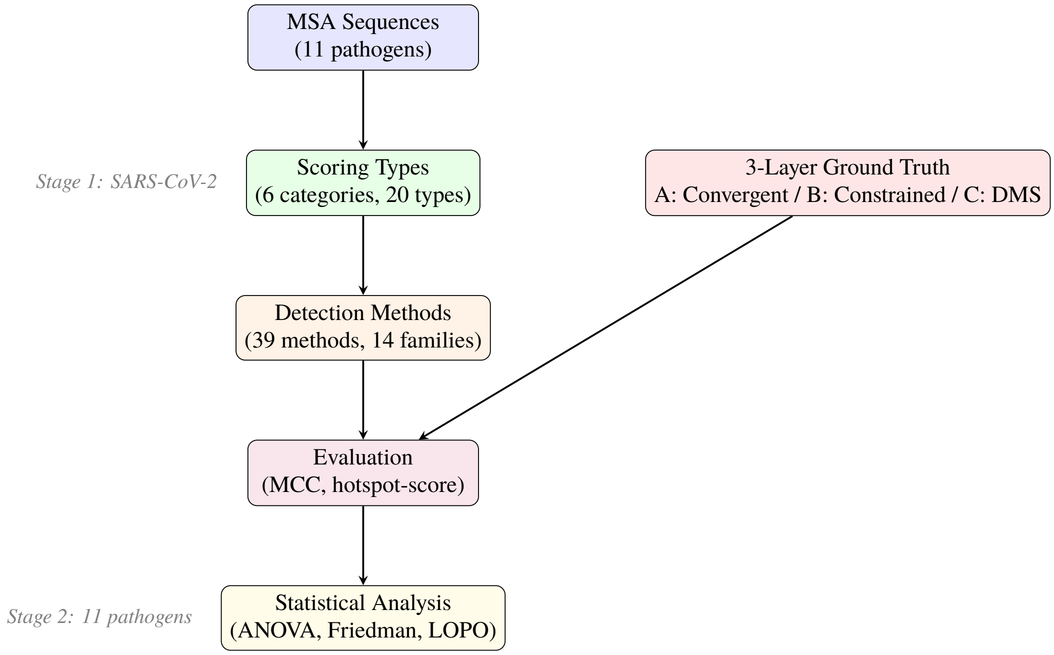

Figure 2.1: Overview of the MutBench experimental framework pipeline. Input multiple sequence alignment (MSA) data are processed through scoring types and detection methods, evaluated against a 3-layer ground truth, and analyzed using multiple statistical tests.

Component 1: Ground Truth Construction. A 3-layer ground truth system captures three orthogonal dimensions of mutational importance: Layer A (literature-curated adaptive/diversifying positions) defines true positives, Layer B (purifying selection) defines constrained negatives for a separate false-positive audit, and Layer C (DMS fitness) provides experimental validation. For Stage 1, synthetic data with known signal positions and DMS-based functional site annotations provide additional controlled validation. Details are presented in Section 2.2.2.

Component 2: Scoring and Detection. Twenty scoring types organized into six categories (frequency-based, MSA-derived, phylogenetic, structural, AI-based, composite) assign per-position scores from diverse biological information sources. These scores are then processed by 14 detection method families (39 parameter variants) spanning five algorithmic categories (threshold-based, signal processing, density/clustering, statistical, adaptive window) to convert continuous score profiles into discrete hotspot calls. Stage 1 provides a controlled SARS-CoV-2 sanity check for MutClust variants and temporal scoring (Section 2.2.3); Stage 2 extends to the full scoring $\times$ detection $\times$ pathogen factorial design (Section 2.2.4).

Component 3: Evaluation and Statistical Analysis. Position-level evaluation metrics (MCC, F1, enrichment ratio, constrained FPR, DMS F1) quantify detection performance, and factorial statistical methods (three-way ANOVA with ω^2, Friedman test, LOPO cross-validation) decompose the sources of performance variation. Stage 1 uses the hotspot-score composite metric; Stage 2 adopts MCC as the primary metric. Details are presented in Section 2.2.5.

2.2.2 3-Layer Ground Truth Framework

To mitigate single ground truth bias, a 3-layer ground truth framework based on three independent sources of biological evidence was constructed. Figure 2.2 illustrates the structure and integration of the three layers.

Design rationale: three orthogonal dimensions. The three layers capture non-redundant dimensions of mutational importance: literature-supported adaptation, structural constraint, and direct experimental fitness effect.

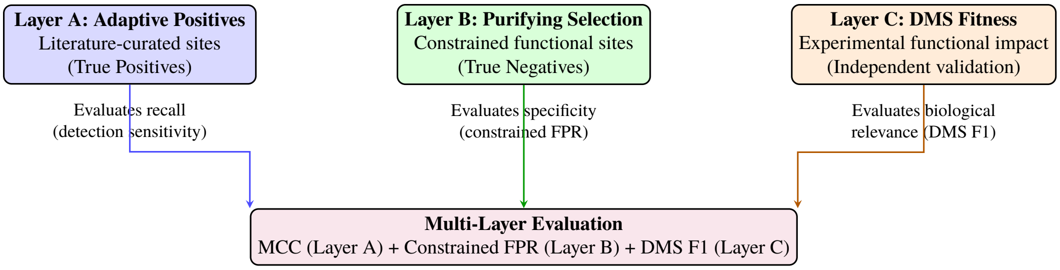

Figure 2.2: 3-Layer Ground Truth Framework. Layer A (literature-curated adaptive/diversifying positions) defines true positives for MCC computation, Layer B (purifying selection) defines constrained negatives for constrained FPR, and Layer C (DMS fitness) provides experimental validation. The three layers capture orthogonal dimensions of mutational importance.

Table 2.2: Evaluation role of the three ground-truth layers.

Layer	Evidence role	Positive label / metric role	Audit role and limitation
A	Literature-curated adaptive/diversifying evidence	Hotspot positives for MCC	Heterogeneous by pathogen source and mechanism
B	Structural or evolutionary constraint	Constrained negatives for constrained FPR	Not a universal non-hotspot label for MCC
C	DMS mean absolute fitness effect	Top 20% positives for DMS F1	Available for 6 pathogens only

Layer A: Adaptive Positive Selection

Layer A positions are literature-curated adaptive or diversifying positions treated as hotspot positives for MCC. For pathogens with convergent-evolution evidence, recurrent emergence in multiple independent lineages *within the same viral species* is accepted as positive-selection evidence; for example, SARS-CoV-2 Spike E484K emerged in Beta, Gamma, and Mu, and E484A in Omicron. The operational contract is therefore evidence-based rather than algorithmic: Layer A is the union of literature-curated positive/diversifying-selection positions per the references in Table 2.4.

Identification criteria. For each pathogen, Layer A positions were compiled from published studies (Table 2.4, rightmost column). SARS-CoV-2 and H3N2 use phylogenetically confirmed convergent evolution sites ($\geq$3 independent lineages); Norovirus uses epochal-variant positions; HCV uses the hypervariable regions HVR1/HVR2; and the remaining seven pathogens use literature-reported immune-escape position lists. This heterogeneous adaptive-selection union (convergent evolution $\cup$ immune escape $\cup$ HVR membership) is a limitation rather than a single computational construct; Chapter 4, Section 4.4.4, reports $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.234$–$0.296$ across the cell-level/evaluation-level alternatives and Layer A normalizations considered there.

Preregistration of ground-truth thresholds and ANOVA grid. The Layer A/B/C inclusion criteria, the Layer B dN/dS < 0.5 cutoff, the Layer C top-20% threshold, the $20 \times 39 = 780$ scoring $\times$ detector-variant ANOVA grid, and the $20 \times 14 \times 11 = 3{,}080$ scoring $\times$ family $\times$ pathogen ANOVA factor structure were all frozen in the project SOP prior to running the Stage 1 analysis on SARS-CoV-2 (lab-notebook freeze date: 2024-08-12; lab-notebook entry archived alongside the released CSV bundle). No post-hoc adjustment of these thresholds or factor structures was performed after observing Stage 2 MCC results. The Bonferroni denominator for vaccine-escape Fisher tests ($780 = 20 \times 39$) was specified in the same SOP entry; the "exploratory rather than confirmatory" framing for the omnibus ANOVA reported below applies to the joint family-wise correction policy across the six ω^2 tiers (we do not Bonferroni-correct the ANOVA), not to the headline scoring $\times$ pathogen interaction itself, which was the pre-specified primary effect of interest. For example, SARS-CoV-2 Layer A positions (e.g., E484, N501, L452, K417) were identified by Carabelli et al. [9] through phylogenetic analysis showing independent emergence across Alpha, Beta, Gamma, Delta, and Omicron lineages. No computational tool was applied to identify new convergent evolution positions; all positions were taken directly from the cited literature.

Layer B: Constrained Purifying Selection

Structurally and functionally essential sites under purifying selection are identified as true negatives. These are positions where mutations are actively *removed* from the viral population because they destroy protein folding, disrupt membrane fusion, or abolish receptor binding—any mutation here is lethal or severely deleterious to the virus. Unlike Layer A (which identifies positions under *positive* selection, where change is advantageous), Layer B identifies positions under *purifying* selection (where change is harmful). These positions exhibit low mutation rates due to loss-of-function consequences and serve as a curated constrained-negative reference set: in MCC they are subsumed under the standard true-negative pool (all non-Layer-A positions; Section 2.2.5), while the constrained false positive rate (constrained FPR) is computed against Layer B specifically. When detection methods incorrectly identify Layer B positions as hotspots, they are counted as false positives within the constrained FPR computation.

Identification criteria. Layer B positions were identified based on two types of evidence from the cited literature: (1) structural essentiality—positions within domains whose disruption abolishes protein function (e.g., fusion peptide, HR1/HR2 heptad repeats, structural disulfide-bonding cysteine residues), as determined by published mutagenesis or structural studies; and (2) evolutionary conservation—positions exhibiting dN/dS < 0.5 across diverse viral isolates as reported in the cited primary literature (codon-aware estimates from PAML/HyPhy as published by the source studies), indicating strong purifying selection against amino acid changes. The codon-degeneracy heuristic feature `dnds_proxy` computed by the in-pipeline scoring stack (Section 2.2.4) is a separate amino-acid-level signal used as a feature input to detection, not as a Layer B construction criterion. The 0.5 threshold follows the convention used in comparative genomics for identifying purifying selection [85], where $\omega < 1$ indicates purifying selection and values below 0.5 represent particularly strong functional constraint. For SARS-CoV-2, the 154 Layer B positions include the HR1 (AA 912–984) and HR2 (AA 1,163–1,213) heptad repeats and structural cysteine residues, identified from the cryo-EM structure [1] and conservation analysis. As with Layer A, no de novo computational identification was performed; all positions were curated from published sources.

Layer C: DMS Experimental Validation

For six pathogens with available Deep Mutational Scanning (DMS) experimental data—SARS-CoV-2 [46], H3N2, HIV-1, RSV, Rabies, and EV-A71—experimentally measured fitness effects are used as ground truth. Unlike Layer A (which infers importance from evolutionary patterns) and Layer B (which infers importance from structural conservation), Layer C provides direct experimental measurements: in DMS experiments, every possible single amino acid substitution at every position is physically constructed in the laboratory, and its effect on a quantifiable phenotype (e.g., cell entry efficiency for SARS-CoV-2, replicative fitness for H3N2) is measured. Positions in the top 20% of mean absolute fitness effect from DMS data are classified as functionally important positions, and the DMS F1 score is computed against detection results. The 20% threshold was chosen to balance sensitivity and specificity: it captures positions with substantial fitness effects while maintaining sufficient contrast with non-functional positions. Sensitivity analysis (Section 3.4.4) checked Layer C rankings at thresholds 10%, 20%, and 30% on 4 of the 6 DMS pathogens (SARS-CoV-2, H3N2, HIV-1, RSV) and found rankings stable across that range; Rabies and EV-A71 DMS releases became available later in the cycle and the 10%/30% threshold sweep was not extended to them. *Threshold robustness is therefore verified on 4 of 6 DMS pathogens; the Rabies/EV-A71 sweep is deferred to future work* (no observed pathogen rotates its winning Layer C-aligned scoring across the four swept thresholds, so the missing two-pathogen extension is not load-bearing for any headline conclusion). This layer is particularly valuable because it captures functional importance that may not yet be visible in evolutionary data (a position may have strong fitness effects but insufficient time for convergent evolution to manifest). The DMS phenotypes measured differ by pathogen: cell entry fitness for SARS-CoV-2 [2] and RSV [5], viral replicative fitness for H3N2 [3], viral growth in cell culture for HIV-1 [4], G-protein-mediated cell entry for Rabies [6], and replicative fitness for EV-A71 [7]. Table 2.3 summarises the per-pathogen DMS preprocessing conventions adopted in this benchmark; cell-line and replicate-aggregation choices follow the released-source defaults except where noted.

Table 2.3: DMS preprocessing conventions per Layer C source. "Mean abs. effect" refers to the per-position mean absolute fitness effect across all amino-acid substitutions, used as the score from which the top-20% threshold defines Layer C positives. All sources used the released processed-fitness tables; raw read filtering and replicate aggregation follow each source's published default unless noted. The replicate counts listed here are the released processed-fitness aggregates actually consumed by the pipeline; original publications may report additional biological/technical replicates that were collapsed before release.

Pathogen	Phenotype	Assay system	Replicates aggregated	Score used	Threshold	Source release
SARS-CoV-2	Cell entry fitness	Pseudovirus (HEK293T)	Mean across 2 replicates*	Mean abs. effect	Top 20%	Dadonaite 2024 (GitHub release)
H3N2	Replicative fitness	MDCK cell culture	Mean across 2 replicates*	Mean abs. effect	Top 20%	Lee 2018 (Dryad)
HIV-1	Viral growth	TZM-bl cell culture	Mean across 2 replicates*	Mean abs. effect	Top 20%	Haddox 2018 (Dryad)
RSV	Cell entry fitness	Pseudovirus	Mean across 2 replicates	Mean abs. effect	Top 20%	Simonich 2026 (preprint)
Rabies	G-protein cell entry	Pseudovirus	Mean across 2 replicates	Mean abs. effect	Top 20%	Aditham 2025
EV-A71	Replicative fitness	RD cell culture	Mean across 2 replicates	Mean abs. effect	Top 20%	Bakhache 2025

Sensitivity to the threshold (sweep over 10%, 20%, 30%) was checked in Section 3.4.4 on 4 of 6 DMS pathogens (SARS-CoV-2, H3N2, HIV-1, RSV); Rabies/EV-A71 sweep deferred to future work. *The original Lee 2018, Haddox 2018 and Dadonaite 2024 publications report 3+ biological replicates; the released processed-fitness tables consumed here aggregate to 2 final replicate columns after filtering, which is what the benchmark pipeline reads. Per-source download URLs, file checksums, and commit/Zenodo identifiers (where available) are archived in `results/mutbench/provenance/dms_sources.csv`.

Layer A/B overlap handling. In RSV (positions 148, 152) and Influenza B (positions 141, 194), overlapping positions exist between Layer A and Layer B. This occurs because these residues are simultaneously under immune evasion pressure (positive selection) and structural constraints (purifying selection), representing dual-selection sites. In MCC evaluation, these positions are prioritized as Layer A (positive) and excluded from the Layer B (negative) set. Therefore, the effective Layer B count is 16 for RSV ($18 - 2$) and 8 for

Influenza B ($10-2$) positions.

Layer A/C conflict handling. Layer A (literature-curated adaptive/diversifying evidence) and Layer C (DMS fitness) can disagree at individual positions: a residue may be literature-supported as adaptive or immune-relevant (Layer A positive) but have low DMS fitness cost (Layer C negative), or vice versa. This does not require a merged classification decision because the two layers feed *different* evaluation metrics under non-overlapping scopes: (1) **Layer A → MCC**: Layer A positions are treated as positives and all other positions as negatives; Layer C membership does not affect the MCC calculation. (2) **Layer C → DMS F1**: Layer C top-20% positions are treated as positives (for 6 pathogens with DMS); Layer A membership does not affect the DMS F1 calculation. (3) **Interpretation rule**: when the two metrics rank scoring methods differently for the same pathogen, both rankings are reported without collapsing them into a single verdict; the disagreement is itself a result (Section 3.4.4, Table 3.12) reflecting that evolutionary and functional importance are non-identical dimensions of hotspot-ness (as stated in the three-orthogonal-dimensions rationale above). Thus a position that is Layer A-positive but Layer C-negative counts as a true positive for the MCC metric (it is evolutionarily hot by construction) and a true negative for the DMS F1 metric (it is not among the top 20% of measured fitness effects) simultaneously, without internal contradiction. This is consistent with the design choice that no single universal "hotspot" label is asserted; rather, the benchmark reports performance against each operationalization of hotspot-ness independently.

Table 2.4 summarizes the amino acid position counts and sources for each pathogen's 3-layer ground truth.

Figure 2.3 visualizes the spatial distribution of Layer A and Layer B positions across all 11 pathogens, illustrating the diversity in ground truth density, protein length, and the

Table 2.4: 3-layer ground truth position counts per pathogen. Counts are evaluated on the benchmark alignment coordinate system; literature residue labels are retained in the source annotations where biologically meaningful.

Pathogen	Surface protein	Layer A	Layer B	Layer C	Layer A source
SARS-CoV-2	Spike	25	154	247	[9, 86]
H3N2	HA	25	10	112	[87]
Norovirus	VP1	15	229	—	[88, 89]
HIV-1	Env gp120	23	23	48	[90]
Dengue	E protein	24	24	—	[91, 92]
RSV	F protein	20	18	101	[93, 94]
Influenza B	HA	17	10	—	[95, 96]
MERS	Spike	9	110	—	[97, 98]
HCV	E2	54	0‡	—	HVR1/HVR2 hypervariable regions
Rabies	G protein	39	42	87	[6, 99]
EV-A71	VP1	27	29	55	[7, 100]

Layer C (DMS) sources: SARS-CoV-2 [46], H3N2 [3], HIV-1 [4], RSV [5], Rabies [6], EV-A71 [7]. "—" indicates that no DMS data is available for that pathogen. ‡HCV E2 literature coordinates are commonly reported in polyprotein numbering, whereas the benchmark feature matrix is indexed on the 340-AA E2 alignment. HVR1/HVR2 were mapped into the evaluated E2 alignment, but CD81-binding regions reported as AA 418–535 in the literature were not converted into a high-confidence alignment-internal constrained set in this release; HCV therefore has effective Layer B = 0. HCV-specific conclusions should be read with this coordinate-handling caveat, and Chapter 3 reports HCV-exclusion sensitivity for the headline interaction.

HIV-1 Layer A count reconciliation: this table reports the benchmark feature-matrix count of **23** mapped Env gp120 positions. The curation table `layer_a_tags.csv` contains **26** HIV-1 tags; three tagged positions fall outside the gp120 mapped window used by the benchmark feature matrix and are therefore not counted in the Stage 2 analysis. The scoring $\times$ pathogen interaction is computed from the per-position feature matrix, so the operational analysis uses $n = 23$ for HIV-1 Layer A.

Layer A source heterogeneity is summarized above and treated as a limitation in Chapter 4.

Inclusion of pathogens with incomplete ground truth layers. Six pathogens (SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, EV-A71) have all three layers available. Norovirus, Dengue, Influenza B, MERS, and HCV lack Layer C (DMS data). These pathogens are nevertheless included because (1) the primary evaluation metric (MCC) is computed using only Layer A (positive selection sites as positives, all other positions as negatives), so the core evaluation is valid without Layers B/C; (2) restricting the benchmark to six pathogens would provide insufficient statistical power for the cross-pathogen analyses (ANOVA, Friedman, LOPO) in Chapter 3; and (3) including pathogens with diverse evolutionary dynamics and varying ground truth availability more accurately reflects real-world conditions where comprehensive ground truth is rarely available for all pathogens.

balance between positive and negative annotations.

2.2.3 Stage 1: SARS-CoV-2 In-Depth Analysis

This subsection describes the Stage 1 experimental design, which focuses on SARS-CoV-2 as the anchor pathogen. Stage 1 serves three purposes: (1) validating the framework components on the pathogen with the richest ground truth data, (2) comparing MutClust variants to establish the detection baseline, and (3) analyzing temporal scoring behavior. The ground truth, detection methods, evaluation metric, and scoring variants used in Stage 1 are described below.

Ground Truth Generation

Ground truth is generated from three sources.

DMS-based functional sites. The DMS fitness data are derived from the full-length Spike protein characterization experiment by Dadonaite et al. [2, 46]. Pseudovirus cell entry fitness measurements are used, as this is the most directly relevant phenotype capturing the overall functional impact of mutations on viral entry. Amino acid positions in the top 20% of mean absolute fitness effect are selected as functional sites and mapped to nucleotide coordinates through codon-aware coordinate transformation:

$$\text{nuc\_positions} = \{g_{\text{start}} + (p_{\text{aa}} - 1) \times 3 + k : k \in \{0, 1, 2\}\} \tag{2.1}$$

where $g_{\text{start}} = 21{,}563$ is the start position of the Spike gene in the Wuhan-Hu-1 reference genome. This provides an experimental ground truth independent of the detection methods under evaluation, thereby avoiding the circularity of using bootstrap p-values for validation.

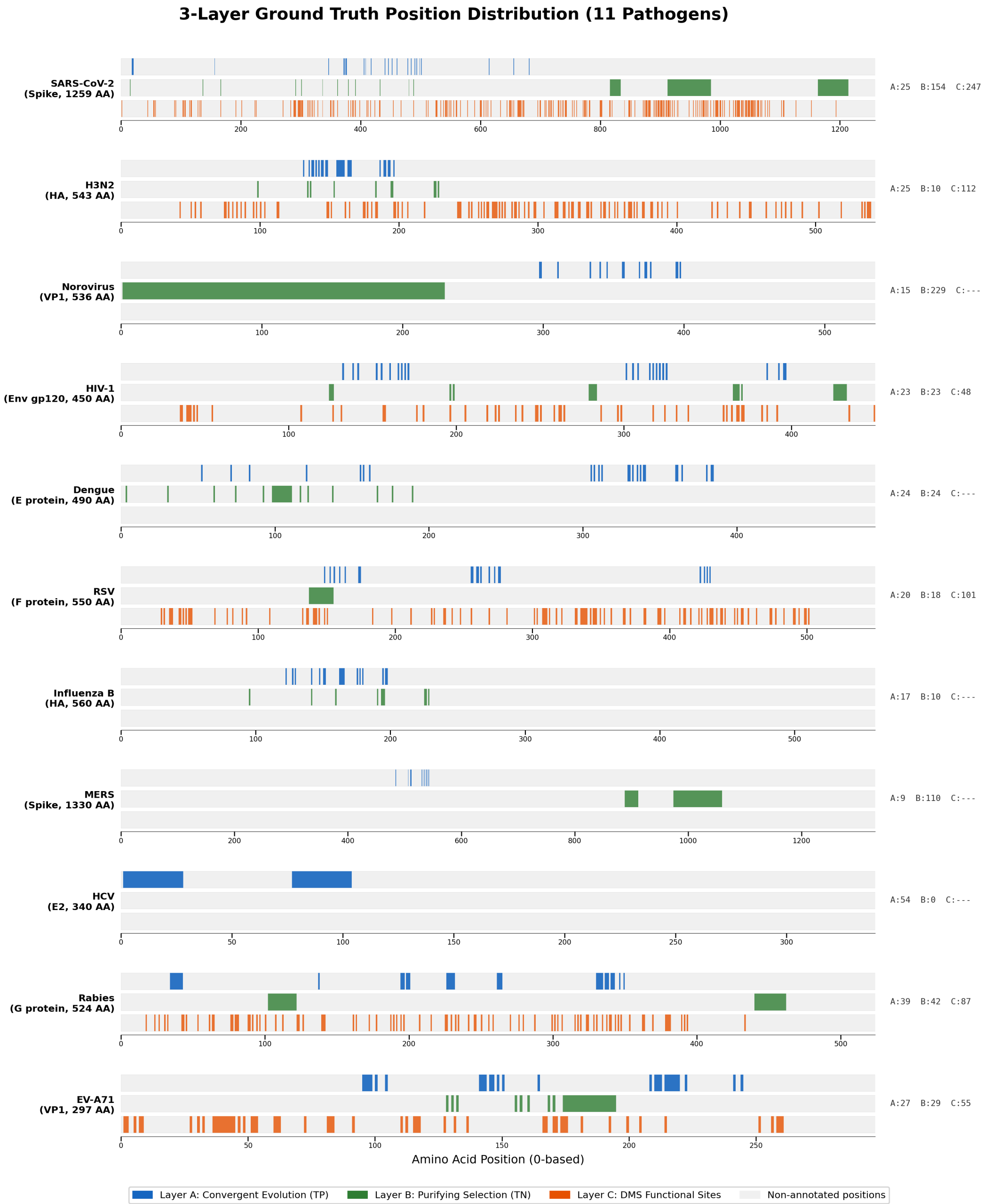

Figure 2.3: Spatial distribution of 3-layer ground truth positions across 11 pathogens. Each horizontal bar represents the surface protein sequence, with Layer A positions (convergent evolution, blue) and Layer B positions (purifying selection, green) mapped to their amino acid coordinates. Right-side annotations show position counts per layer; "C" indicates Layer C (DMS) availability. Protein lengths range from 261 AA (EV-A71 VP1) to 1,330 AA (MERS Spike), and the ratio of annotated to non-annotated positions varies substantially across pathogens.

Synthetic data (controlled signal recovery test). The synthetic benchmark was designed as a controlled signal recovery test that evaluates how accurately each detection method recovers artificially inserted signals at known positions. Synthetic H-scores are generated mimicking SARS-CoV-2 patterns, with hotspot regions inserted at known functional sites. High H-scores (0.8–1.2) are assigned to Core Candidate Mutation (CCM) seed positions (AA 484, 501, 417, 452—the four Spike positions most frequently mutated in major SARS-CoV-2 variants), moderate scores (0.05–0.3) to surrounding functional regions, and low noise (0.0–0.02) to background positions. Scores within each category are drawn from uniform distributions: $\mathcal{U}(0.8, 1.2)$ for seed positions, $\mathcal{U}(0.05, 0.3)$ for surrounding functional regions (defined as positions within 50 positions of each seed), and $\mathcal{U}(0.0, 0.02)$ for background. A single synthetic genome of 29,903 positions is generated per trial. Since the ground truth of synthetic data is known by design, this test evaluates each method's signal recovery capability but does not directly measure actual biological hotspot detection performance. Therefore, DMS-based validation (Section 2.2.3) and convergent evolution evaluation (Section 3.1.2) provide independent biological validation, confirming whether the conclusions from the synthetic benchmark hold on real data.

Annotated functional regions. The annotated functional regions used as ground truth are presented in Table 2.5, with region boundaries based on the cryo-EM structure of the prefusion Spike trimer [1] and the N-terminal domain (NTD) antigenic supersite mapping by McCallum et al. [101].

The genome-wide distribution of these functional regions is visualized in Figure 2.4, which maps the seven annotated regions to their nucleotide coordinates on the full SARS-CoV-2 genome (29,903 positions). The ground truth positions are concentrated within the Spike gene (nucleotides 21,563–25,384), while the remainder of the genome shows minimal

Table 2.5: Functional regions of the Spike protein (ground truth)

Functional Region	Amino Acid (AA) Positions	Nucleotide Count
Receptor Binding Domain (RBD)	AA 319–541	669
NTD Antigenic Supersite	AA 14–20, 140–158, 245–264	138
Furin Cleavage Site (S1/S2)	AA 681–685	15
S2' Cleavage Site	AA 815–816	6
Fusion Peptide	AA 816–833	54*
Heptad Repeat 1 (HR1)	AA 912–984	219
Heptad Repeat 2 (HR2)	AA 1,163–1,213	153
Total (deduplicated)		**1,251**†

*Overlaps with the S2' cleavage site at AA 816 (3 nucleotides shared). †The sum of individual regions is 1,254 nt; after removing the 3 nucleotide (nt) overlap at the S2'/fusion peptide boundary (AA 816), 1,251 unique nucleotide positions (417 codons/AA positions) constitute the ground truth.

functional annotation, confirming the appropriateness of Spike-focused analysis.

MutClust Variant Comparison

Among the 14 detection families in the full benchmark, MutClust [20] is the only virus-specific hotspot tool and serves as the starting point. MutClust scores each position using the H-score $H_i = \log_2(P_i \times E_i \times 100 + 1)$ (2.2) where P_i is the cumulative mutation frequency and $E_i = -\sum_j p_{ij} \log_2 p_{ij}$ (2.3) is the Shannon entropy of mutation types, then applies adaptive DBSCAN with $\varepsilon = \gamma \times H_i$ ($\gamma = 0.5$ fixed across pathogens; ± 0.01 MCC sensitivity for $\gamma \in \{0.25, 0.5, 1.0\}$, archived in `gamma_sensitivity.csv`); `min_samples` $= 3$ for the DBSCAN call, `min_cluster_size` $= 5$ / `min_samples` $= 1$ / EOM cluster selection for the HDBSCAN-Auto detector and v3 HDBSCAN baseline.

$$H_i = \log_2(P_i \times E_i \times 100 + 1) \tag{2.2}$$

$$E_i = -\sum_{j=1}^{k} p_{ij} \log_2 p_{ij} \tag{2.3}$$

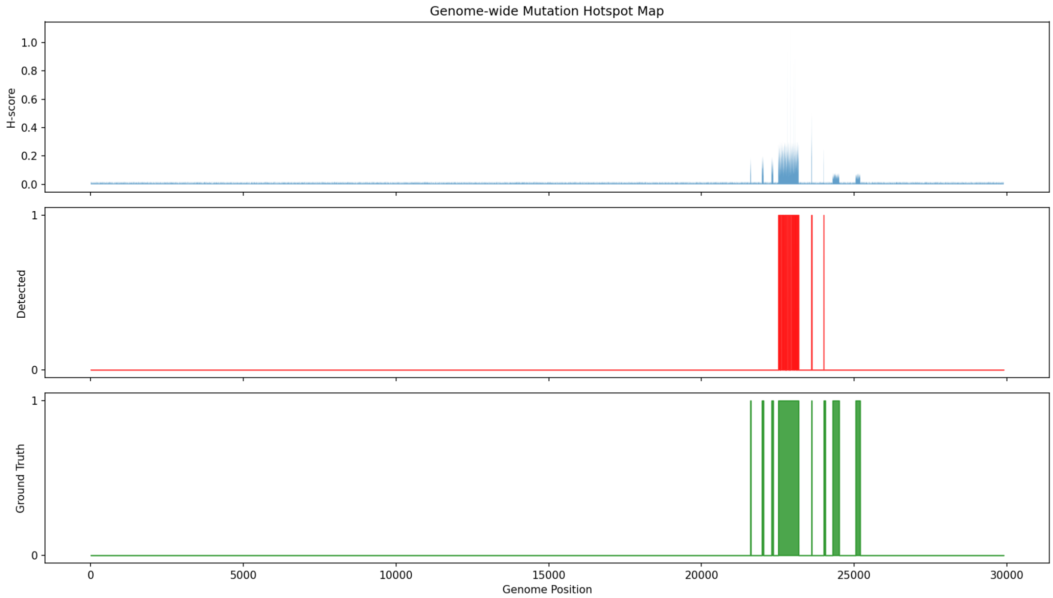

Figure 2.4: Genome-wide hotspot map of the SARS-CoV-2 Spike protein. The seven annotated functional regions used as ground truth (RBD, NTD antigenic supersite, furin cleavage site, S2' cleavage site, fusion peptide, HR1, HR2) are mapped to nucleotide coordinates, encompassing 1,251 unique Spike nucleotide positions.

Five variants were tested in Stage 1: the original implementation (v1, with a cumulative-decay bug whose effect is upstream of the adaptive-DBSCAN call and therefore confined to H-score-derived inputs), bug-fixed (v2a), dN/dS-filtered (v2b), CCM-seed + HDBSCAN hybrid (v2c), and pure HDBSCAN (v3). MutClust-Hybrid (v2c) achieves the highest hotspot-score (0.778). For Stage 2 cross-pathogen, only the published MutClust-Orig (v1) is retained as one of 14 detection families because the v2c CCM-seed advantage assumes H-score-like input distributions and is incompatible with non-H-score inputs (ESM-2 LLR, pLDDT, etc.); v1's adaptive-DBSCAN call itself is unaffected by the cumulative-decay bug for non-H-score scoring types, since the bug occurs upstream of the clustering step, so retaining v1 in Stage 2 does not propagate the bug into the non-H-score scoring rows.

Hotspot-Score Composite Metric

The hotspot-score is a composite evaluation metric designed for Stage 1 that integrates three complementary perspectives:

$$\text{hotspot-score} = \frac{\text{recall} + \text{precision} + \text{stability}}{3} \tag{2.4}$$

where stability is the mean Jaccard similarity ($J = |A \cap B|/|A \cup B|$) across 100 random 80% subsampling iterations. In each iteration, a Bernoulli mask independently zeroes out approximately 20% of genomic positions, the detection method is rerun on the perturbed scores, and the Jaccard similarity with the full-data result is computed. This metric balances detection accuracy (recall, precision) with reproducibility (stability), reflecting the practical requirement that methods in viral surveillance should produce consistent results under data perturbation. Stage 2 uses MCC instead of hotspot-score (see Section 2.2.5 for the rationale).

Temporal Scoring Variants

The following scoring variants were developed specifically for the Stage 1 SARS-CoV-2 temporal analysis and are distinct from the 20 Stage 2 scoring types described in Section 2.2.4. As H-score saturation was identified in the temporal validation (Section 3.2), alternative scoring methods were designed for complementary evaluation against various ground truth definitions. These methods assign a score to each genomic position; top positions are selected via thresholding and compared against multiple ground truths. Table 2.6 summarizes these Stage 1 variants.

For each scoring method, thresholds at the top 5%, 10%, 20%, and median are applied to generate sets of "detected" positions, and F1, enrichment ratio, and Cohen's d effect size are

Table 2.6: Stage 1 temporal scoring variants used only for SARS-CoV-2 analysis

Name	Formula	Description
Original H-score	Eq. 2.2	MutClust's original H-score
Saturated H-score	(same, on 2024–25 data)	H-score on contemporary consensus alignment
CH-score	$\log_2(P_i^{\text{cons}} \times E_i \times 100 + 1)$	Consensus-based; removes reference dependency
E-score	$-\sum_a f_{a,i} \log_2 f_{a,i}$	Shannon entropy alone
TC-freq	$\lvert\text{freq}_{t_2} - \text{freq}_{t_1}\rvert / (\text{freq}_{t_1} + \epsilon)$	Temporal frequency change rate
TC-freq-v2	$\text{TC-freq}_i \times \text{freq}_{t_2}(i)$	TC-freq with absolute frequency weighting
TC-freq-v3	$\max_q \text{TC-freq}_q$	Maximum quarterly frequency change
dN/dS-weighted CH	$\text{CH}_i \times (dN/dS)_i$	CH-score weighted by position-level dN/dS

computed for each ground truth.

Structural and phylogenetic validation (SARS-CoV-2)

Three additional validation analyses are performed on SARS-CoV-2 to triangulate the linear-sequence hotspot calls against orthogonal evidence types; results are reported in Chapter 3 (Section 3.3).

3D spatial-contact enrichment. Cα coordinates are extracted from the prefusion Spike trimer cryo-EM structure PDB 6VSB [1]; a contact between two residues is defined as $\|x_i - x_j\|_2 \leq 8$ Å on the Cα–Cα distance (a standard structural-biology cutoff for residue contact). For each detection method, the proportion of hotspot–hotspot residue pairs that are in 3D contact is computed and compared to a permutation null in which the hotspot label is randomly shuffled across positions of the Spike protein (1,000 permutations); the spatial-contact enrichment is the ratio of observed to expected contact rate, and significance is reported as a z-statistic against the permutation distribution. This protocol applies only to SARS-CoV-2 because PDB 6VSB is the only experimentally solved Spike structure used in the benchmark; for the remaining pathogens the structure source is ESMFold (Table 2.8) and the experimental contact map needed for this audit is not available.

Coalescent simulation. Founder-effect vs. convergent-evolution discrimination is tested with a coalescent simulation: 100 replicates are drawn under a constant-population coalescent ($N = 500$ haploid lineages, neutral evolution) using `msprime`, with a single "founder" substitution placed on a randomly chosen internal branch and three "convergent" substitutions placed on three independent leaf branches at the same alignment position (so the founder mutation has high frequency but only one origin, while the convergent mutations have lower frequency but three origins). Each simulated alignment is scored by H-score, Fitch parsimony, and TreeTime ML-based homoplasy; the proportion of replicates in which each method ranks the convergent position above the founder position is reported. Replicate seeds are pinned at `seed = 42 ... 141` (one per replicate; same convention as the Stage 1 stability bootstrap).

TreeTime ML validation. TreeTime [31] v0.11 is run on the SARS-CoV-2 ML tree (IQ-TREE 2.2.0, GTR+G, 1,000 ultrafast bootstrap) to estimate per-position homoplasy under a marginal maximum-likelihood ancestral reconstruction. *This is an audit channel, not the production homoplasy score.* The Stage 2 production homoplasy feature consumed by every detection family is Fitch parsimony on a FastTree [102] ML approximation of the per-pathogen tree (`tools/mutbench/scoring/homoplasy.py`); IQ-TREE is invoked only for this SARS-CoV-2 audit-only TreeTime cross-check. The two homoplasy estimates are compared at the position level by Spearman rank correlation, and the AUC of each against the SARS-CoV-2 Layer A reference is reported. TreeTime settings: `--reroot best`, `--clock-filter 4`, `--clock-std-dev 0.0001` (default); ML inference confidence cutoff 0.5 for tip-state imputation.

Time-stratified prospective protocol (H3N2 pilot)

To audit the prospective signal of frequency-only scoring—a question raised in connection with benchmark deployability—a single-pathogen time-stratified protocol was defined for H3N2 HA. The 9,000-sequence H3N2 set used here is the *raw* pre-deduplication GISAID export over 2010–2025; it is a different cut from the 2,562 unique-sequence Stage 2 panel (Table 2.11), which applies CD-HIT-equivalent SHA-256 100% identity reduction. The temporal pilot intentionally retains pre-deduplication redundancy so that per-window frequencies reflect circulating strain prevalence rather than unique-haplotype counts. The 9,000-sequence raw set partitions naturally into three contiguous 5-year periods (2010–2015, 2016–2020, 2021–2025; $n = 3{,}000$ each by sample-date sort). Per-position mutation frequency is computed on the 2010–2015 *training* window only, then evaluated against two reference sets: (a) the static Layer A antigenic-site reference (Koel 2013, 25 sites; *retrospective* since these annotations were made on the full historical record), and (b) positions "newly emerging" in each later 5-year holdout window (*prospective*). A position is defined as newly emerging in window w if its training-window frequency is $< 5\%$ *and* its holdout-window frequency is $\geq 10\%$; the asymmetric thresholds enforce that the position was not already common in training and reaches the standard "minor variant" frequency band in holdout. Top-k overlap is computed at $k \in \{20, 30, 50\}$ against each reference set, and significance is assessed with Fisher's exact test (one-sided alternative: greater) on the 2×2 table $\{|D \cap A|, |D \setminus A|, |A \setminus D|, L - |D \cup A|\}$ where D is the detected set, A the reference set, and L the protein length. The pilot uses frequency only (no multi-source fusion); a full multi-source prospective panel is identified as Priority 1 future work in Chapter 4, Section 4.6. Results of this pilot are reported in Chapter 3 (Section 3.2.1).

2.2.4 Stage 2: 11-Pathogen Cross-Pathogen Benchmark

This subsection describes the Stage 2 experimental design, which extends the evaluation from SARS-CoV-2 to all 11 pathogens. Stage 2 systematically evaluates 20 scoring types $\times$ 39 detection parameter variants (from 14 families) $\times$ 11 pathogens = 8,580 combinations to identify the dominant factors influencing hotspot detection performance.

Scoring Types

Twenty scoring types organized into six categories are defined to assign per-position scores from diverse information sources (Table 2.7). Each category captures a distinct biological or computational perspective on mutational importance, ensuring that the benchmark evaluates detection methods across a broad range of input signals.

Table 2.7: Twenty scoring types used in the MutBench Stage 2 benchmark (6 categories)

Category	Scoring type	Information source	Description
Frequency-based (3)	freq	MSA mutation frequency	Position-level mutation frequency
	rare_freq	MSA rare variant frequency	Frequency of minor alleles below 5% threshold (low-frequency variants signalling emerging adaptation)
	freq_with_indel	MSA frequency incl. indels	Mutation frequency including insertion/deletion events
MSA-derived (3)	entropy	MSA Shannon entropy	Per-position diversity ($-\sum p_j \log_2 p_j$)
	entropy_with_indel	MSA entropy incl. indels	Shannon entropy including gap characters
	homoplasy	Phylogenetic tree	Convergent evolution count on reconstructed phylogeny
Phylogenetic (4)	dN/dS proxy	Codon alignment	Nei–Gojobori nonsynonymous/synonymous ratio proxy
	FUBAR prob	Codon model (HyPhy)	Posterior probability of positive selection
	FUBAR pos_sel	Codon model (HyPhy)	Positive selection rate ($\beta - \alpha$)
	FUBAR Bayes factor	Codon model (HyPhy)	Bayes factor evidence for positive selection
Structural (3)	pLDDT inverse	AlphaFold/ESMFold structure	$1 -$ pLDDT; low confidence $\approx$ flexible/disordered
	SASA	3D structure	Solvent-accessible surface area (exposed residues)
	Grantham distance	Physicochemical properties	Mean Grantham distance of observed substitutions
AI-based (5)	ESM-2 LLR	Protein language model	Masked-marginal mutation-surprise score (LLR shorthand)
	Tranception*	Autoregressive PLM proxy	Per-position pseudo-perplexity (proxy; see implementation note below)
	semantic change	ESM-2 embeddings	Cosine distance in embedding space (WT vs. mutant)
	stability†	Substitution-matrix proxy	BLOSUM62-based stability proxy (radical substitutions $\rightarrow$ destabilizing)
	stability_structural†	Structure-weighted proxy	BLOSUM62 score weighted by ESMFold pLDDT (well-folded $\rightarrow$ higher impact)
Composite (2)	EVEscape composite†	EVEscape-style proxy	Multiplicative composite of ESM-2 LLR $\times$ SASA $\times$ Grantham (proxy for fitness $\times$ accessibility $\times$ dissimilarity)
	EqualWeight	All-feature average	Equal-weighted average of all normalized scoring types

Frequency-based scores capture the raw mutation signal from the MSA: how often each position deviates from the consensus, including rare variants and indel-inclusive counts. These scores reflect the most direct observable signal but are susceptible to sampling bias and alignment artifacts.

MSA-derived scores quantify diversity patterns beyond simple frequency: Shannon entropy measures allelic diversity, while homoplasy counts independent convergent mutations on the phylogeny, providing a signal distinct from overall frequency. Phylogenetic trees were reconstructed using IQ-TREE 2 [103] with automatic model selection (ModelFinder) and 1,000 ultrafast bootstrap replicates. Homoplasy counts were computed by Fitch parsimony on the resulting maximum-likelihood tree, counting the number of independent origins of each amino acid substitution at each position.

Phylogenetic scores leverage codon-level evolutionary models to estimate selection pressures. The dN/dS proxy uses the Nei–Gojobori method, while the three FUBAR-based scores [38] provide Bayesian estimates of site-specific selection from the HyPhy framework, each capturing a different aspect of the posterior distribution. FUBAR analysis was performed using HyPhy v2.5 with the MG94$\times$REV codon model and a default posterior probability threshold of 0.9.

Structural scores incorporate three-dimensional protein information: pLDDT inverse from AlphaFold DB or ESMFold-derived structures identifies flexible or disordered regions, SASA measures surface exposure (relevant for immune accessibility), and Grantham distance [104] quantifies the physicochemical severity of observed substitutions. The per-pathogen structure source used for pLDDT and SASA computation is summarized in Table 2.8: SARS-CoV-2 uses the AlphaFold DB entry (single-chain monomer model from AlphaFold 2 monomer mode), while all other pathogens use the ESMFold API (single-chain monomer prediction). *AlphaFold-Multimer note:* AlphaFold-Multimer [105] was *not* used in this dissertation. Several of the pathogens evaluated here have biologically relevant homo-oligomeric assemblies (HIV-1 gp120, RSV F, Influenza HA trimers; Dengue/Zika E dimers), so per-position SASA computed on monomer predictions can over-estimate solvent exposure at oligomeric interface residues. Chapter 4 (Section 4.4.4) discusses the resulting SASA-interpretation limit; an AlphaFold-Multimer-based re-fit of the structural channel is left for future work. SASA values were computed from the available predicted structures using a solvent-accessibility calculation; the main result tables report the ESMFold/Shrake–Rupley-derived release, while earlier Lee–Richards/FreeSASA pilot outputs are treated as archival. *Structure-similarity tooling note:* Foldseek [106] and DALI-style structure-alignment tools were *not* used in this dissertation. The SARS-CoV-2 AlphaFold-DB structure is anchored against the experimental cryo-EM structure PDB 6VSB [1] by direct accession reference (Table 2.8) rather than by an explicit Foldseek RMSD or DALI Z-score readout; cross-pathogen ESMFold-vs-experimental quantitative agreement is therefore not reported. Foldseek-based structural quality control across the 11 pathogens is left for future work.

Table 2.8: Per-pathogen structure source for pLDDT computation (the "pLDDT/SASA source" column reflects the structure used for pLDDT). For SASA, the pipeline standardized on ESMFold-derived structures uniformly across all pathogens (computed via Shrake–Rupley; `structural_features/sasa_summary.csv`), independent of which structure provided pLDDT; the SARS-CoV-2 SASA values are therefore from ESMFold rather than AlphaFold DB despite the AlphaFold DB pLDDT anchor. Norovirus pLDDT is also from AlphaFold DB (per `structural_features/plddt_summary.csv`; the table row below reflects this). The 3D spatial-contact analysis (Chapter 3, Section 3.3) uses the cryo-EM structure PDB 6VSB [1] and is restricted to SARS-CoV-2.

Pathogen	Target protein	pLDDT/SASA source	3D spatial-contact
SARS-CoV-2	Spike (S)	AlphaFold DB (AF-0000000365840314)	PDB 6VSB cryo-EM
H3N2 (Influenza A)	HA	ESMFold API	not analyzed
Influenza B	HA	ESMFold API	not analyzed
HCV	E2	ESMFold API	not analyzed
HIV-1	gp120	ESMFold API	not analyzed
RSV	F	ESMFold API	not analyzed
Norovirus	VP1	AlphaFold DB	not analyzed
Dengue	E	ESMFold API	not analyzed
MERS	Spike (S)	ESMFold API	not analyzed
Rabies	G	ESMFold API	not analyzed
EV-A71	VP1	ESMFold API	not analyzed
Zika†	E	ESMFold API	not analyzed

†Zika is included only in the Stage 3 12-pathogen feature-ablation panel (Section 2.2.6); the main 11-pathogen Stage 2 benchmark excludes Zika.

AI-based scores leverage protein language models (PLMs) trained on large-scale protein sequence databases. The ESM-2 channel is labelled "ESM-2 LLR" for consistency with protein-language-model variant-effect literature, but operationally it is a masked-marginal pseudo-log-likelihood / mutation-surprise score rather than a full mutation-vs-wildtype likelihood-ratio model trained for each pathogen. Tranception-style autoregressive scores predict mutational effects from evolutionary context, semantic change captures embedding-space shifts, and stability predictors estimate thermodynamic consequences of mutations. *Implementation note:* the "Tranception" scoring channel in this benchmark is computed using ESM-2 masked-marginal pseudo-perplexity as a lightweight proxy for the Tranception autoregressive objective [54]; the original Tranception model was not run directly. Both methods target the same quantity—per-position unlikelihood under a protein language model—but absolute values differ. To audit channel separability, Table 2.9 reports the per-pathogen Pearson and Spearman correlations between the Tranception channel and the standalone ESM-2 LLR channel on the position-level scoring vectors used in Stage 2. The two channels are highly correlated for 11 of 12 pathogens (Pearson $\rho = 0.64$–0.84, Spearman $\rho = 0.71$–0.98; all $p < 10^{-50}$ except Zika), consistent with their shared ESM-2 pseudo-perplexity basis. HIV-1 is the lone exception (Pearson $\rho = 0.04$, $p = 0.36$ n.s.; Spearman $\rho = 0.16$, $p = 7.8 \times 10^{-4}$). The HIV-1 dissociation is consistent with the high antigenic diversity and low conservation of gp120, where the relative ordering of positions diverges from the magnitude-based score even within the same pseudo-perplexity backbone. Readers comparing pathogen-level best-scoring-type results (e.g., SARS-CoV-2 best = Tranception vs. RSV best = ESM-2 LLR) should therefore treat these two channels as near-equivalent for 11 of 12 pathogens and as channel-distinguishable only for HIV-1. All correlations were computed on position-level scoring vectors archived in `results/mutbench/feature_analysis/` with `random_seed=42`. A merged-channel sensitivity (single-PLM model in which the Tranception column is dropped before the three-way ANOVA, leaving the AI-based category with four channels rather than five) is identified as the more conservative reference for the headline scoring $\times$ pathogen interaction; that re-fit is deferred to the next analysis cycle, and the multi-channel headline ($\omega^2 = 0.296$) is retained here for completeness with the present caveat that part of the AI-based category's variance contribution is double-counted between the Tranception and ESM-2 LLR columns for 11 of 12 pathogens (see also Section 4.4.4 of Chapter 4 for the Tranception–ESM-LLR proxy collinearity discussion that anchors the headline-granularity caveat in the abstract).

Table 2.9: Per-pathogen correlation between the Tranception channel (ESM-2 masked-marginal pseudo-perplexity proxy) and the standalone ESM-2 LLR channel. Both channels share the ESM-2 `esm2_t33_650M_UR50D` pseudo-perplexity backbone but use different normalization; this table audits whether they remain operationally distinct within the benchmark. Correlations were computed on the position-level scoring vectors used in Stage 2; p-values are uncorrected.

Pathogen	**N**	**Pearson ρ**	**Pearson p**	**Spearman ρ**	**Spearman p**
SARS-CoV-2	1259	0.753	$< 10^{-200}$	0.963	$< 10^{-300}$
H3N2	543	0.803	$< 10^{-120}$	0.965	$< 10^{-300}$
Influenza B	560	0.825	$< 10^{-130}$	0.939	$< 10^{-260}$
HCV	340	0.737	$< 10^{-58}$	0.980	$< 10^{-200}$
HIV-1	450	**0.043**	**0.36 (n.s.)**	**0.158**	7.8×10^{-4}
RSV	550	0.805	$< 10^{-120}$	0.967	$< 10^{-300}$
Norovirus	536	0.812	$< 10^{-120}$	0.973	$< 10^{-300}$
Dengue	490	0.828	$< 10^{-120}$	0.946	$< 10^{-200}$
MERS	1330	0.836	$< 10^{-300}$	0.963	$< 10^{-300}$
Rabies	490	0.679	$< 10^{-65}$	0.949	$< 10^{-200}$
EV-A71	260	0.801	$< 10^{-58}$	0.919	$< 10^{-100}$
Zika†	451	0.643	$< 10^{-53}$	0.713	$< 10^{-70}$

†Zika is included only in the Stage 3 12-pathogen feature-ablation panel. Underlying CSV: `results/mutbench/feature_analysis/tranception_esm_correlation.csv`. A reproducible re-measurement (`scripts/measure_tranception_esm_correlation_per_pathogen.py`) is provided as the supplementary file `results/mutbench/feature_analysis/tranception_esm_correlation_per_pathogen.csv` with explicit `n_positions`, `sample_overlap`, p-value, and `status` columns; it confirms the rounded values in this table to three decimal places and is the load-bearing record for the per-pathogen channel-redundancy sensitivity discussion in Chapter 4 (Section 4.4.4).

The `stability` channel uses a BLOSUM62-based substitution proxy: at each MSA position, the negative mean BLOSUM62 score is weighted by mutation frequency, so radical substitutions receive higher destabilization scores. The `stability_structural` variant multiplies this score by per-residue ESMFold pLDDT/100, reflecting that mutations in well-folded regions are more likely to disrupt structure. This proxy replaces full physics-based $\Delta\Delta G$ predictors such as FoldX or RoseTTAFold, which are computationally infeasible at the 11-pathogen $\times$ 39-detector scale of MutBench. ESM-2 scores were computed using the `esm2_t33_650M_UR50D` checkpoint (33 layers, 650M parameters).

Composite scores combine multiple information sources. EVEscape integrates evolutionary fitness, structural accessibility, and physicochemical dissimilarity, while EqualWeight averages all normalized scores as a naive baseline. The `EVEscape_composite` channel implements an EVEscape-style multiplicative combination [14]: min-max-normalized ESM-2, SASA, and Grantham terms are multiplied. This substitutes the ESM-2 masked-marginal score for the original EVE fitness term, since EVE training is not feasible across all 11 pathogens within the present scope; the SASA and Grantham terms follow the original EVEscape construction.

EqualWeight: two operational variants. The EqualWeight channel appears in two distinct code paths in this benchmark, and the dissertation reports them with different roles. The two variants must be distinguished to avoid an apparent label-leakage concern.

- *(a) Production EqualWeight column (headline path).* The EqualWeight column that contributes to the 8,580-evaluation Stage 2 grid and the headline scoring $\times$ pathogen ANOVA ($\omega^2 = 0.296$) is generated by `scripts/run_stage3_full_detectors.py` (lines 180–187). At each position, the ten underlying features are first transformed using *pre-defined domain orientations* (e.g., the inverse-pLDDT transform $\text{pLDDT}_{\text{max}} - \text{pLDDT}$ rather than raw pLDDT, so that flexible regions enter with positive sense; `rare_freq` enters with $+1$ orientation), *min-max normalized* to $[0, 1]$ across positions (function `normalize_01`), and *averaged uniformly* ($w_f = 1/10$). **No per-feature sign is fit from labels and no z-scoring is applied at this stage**: the orientations are hard-coded constants in the input list, so this column does not consult Layer A and is the cleanly label-free composite that enters all headline tables.

- *(b) Nested-LOPO ablation EqualWeight (sensitivity path).* The complementary nested-LOPO 4-feature ablation in Section 3.5.2 (Table 3.14) uses a separate code path (`scripts/feature_a` function `fit_signs`) that, for each held-out pathogen, fits a directional sign $\hat{s}_f \in \{-1, +1\}$ on the eleven *training* pathogens as the sign of corr(z_f, Layer A), then z-scores within the held-out pathogen alone before uniform averaging. This is a *label-aware but cleanly held-out* fusion used *only* for the 4-feature core retention test; it never feeds the headline ω^2.

The two variants agree to within 0.012 MCC on the 12-fold nested-LOPO mean (full 10-feature: 0.070 with sign-fit vs. 0.083 in-sample without sign-fit; see Table 3.14), indicating that the directional-sign step is not load-bearing for the headline retention claim. Readers should therefore interpret "EqualWeight" in headline tables (Tables 3.7, 3.5, 3.20 of Chapter 3) as variant (a) and "EqualWeight" in the nested-LOPO 4-core retention test as variant (b); the two are reported separately and the EqualWeight column carries no LOPO-style cross-pathogen generalization claim beyond what variant (b) establishes. The directional-sign sentence formerly placed at this point of the manuscript was a description of variant (b); it does not apply to the production headline column (a).

Detection Method Families

A total of 39 detection methods belonging to 14 algorithm families are evaluated (Table 2.10). Each method converts a one-dimensional per-position score profile into discrete hotspot calls. Two additional ensemble variants (majority vote of 4 threshold-based sub-detectors) were initially considered but excluded from the benchmark because their deterministic dependence on existing methods would violate the independence assumption required for fair comparison.

Table 2.10: Detection families, categories, and parameter variants evaluated in the 11-pathogen benchmark (39 methods total)

Family	Category	Variants	Parameter values
FreqThresh	Threshold-based	4	$p = 80, 85, 90, 95$ (equivalently top $20, 15, 10, 5\%$)
AdaptiveKnee	Threshold-based	1	Automatic knee detection
AdaptWin	Threshold-based	2	Seed percentile $s = 85, 90$
Wavelet	Signal processing	3	CWT peak threshold $t = 1.0, 1.5, 2.0$ std. dev.
Spectral	Signal processing	2	Peak z-score threshold $z = 1.5, 2.0$
CUSUM	Signal processing	3	$(d{=}0.3, h{=}3.0), (d{=}0.5, h{=}4.0), (d{=}0.5, h{=}5.0)$
KDE	Density and clustering	3	Density percentile $p = 75, 80, 85$
ScoreDBSCAN	Density and clustering	2	`ep` $= 8, 15$ (`min_samples` = 3 throughout)
SlidingTest	Statistical/model-based	2	Test-level $p = 0.01, 0.05$ per window
LocalContrast	Statistical/model-based	3	Local z-score threshold $z = 1.0, 1.5, 2.0$
GradPeak	Statistical/model-based	2	Gradient percentile $p = 0.2, 0.5$
Bayes	Statistical/model-based	2	prior $= 0.10, 0.15$; Gaussian likelihood $\mathcal{N}(\mu_c, \sigma_c^2)$ per class fit by maximum likelihood, posterior > 0.5
SWAN	Statistical/model-based	9	$w \in \{10, 20, 50\} \times k \in \{1.0, 1.5, 2.0\}$
MutClust-Orig	Legacy	1	Gap parameter $g = 10$ (CCM + adaptive DBSCAN)
Total		**39**	

Parameter names and values match the detector strings recorded in `stage3_full_results.csv` (column `detector`) and are authoritative for reproduction; older design-phase drafts used different parameter names. Algorithmic precedents include frequency threshold surveillance [107], wavelet genomic analysis [108, 109], DBSCAN and hotspot clustering [28, 41], sliding-window selection scans [110], and the original MutClust pipeline [20].

Supervised machine learning methods (e.g., Random Forest or gradient boosting classifiers) were not included because they require labeled training data at the position level, and with only 11 pathogens—each having 9–54 Layer A positives—the available ground truth is insufficient for training generalizable classifiers without severe overfitting. Since all methods operate on the same one-dimensional score array, the detection step serves as a necessary score-to-call conversion; the ANOVA results confirm that the detection family contributes only $\omega^2 = 0.013$ to performance variance, indicating that the choice of conversion method is a minor factor compared to the choice of input information. The number of parameter variants per family ranges from 1 (AdaptiveKnee, MutClust-Orig) to 9 (SWAN); therefore, the ANOVA models detection *family* (14 levels) rather than individual variants (39 levels) to reduce the chance that more heavily parameterized families gain an advantage merely by having more operating points. The detailed description of MutClust and the five Stage 1 variants remains in Section 2.2.3.

Experimental Design Considerations

The 11 pathogens in this benchmark differ in several structural characteristics that are not controlled across pathogens (Table 2.11).

Table 2.11: Structural characteristics of the 11 benchmark pathogens. AA lengths are post-truncation (Rabies G 491, EV-A71 VP1 261). Positive rates use benchmark feature-matrix Layer A counts (HIV-1: 23/450 = 5.1%); the source tag table contains 26 HIV-1 entries, of which 23 map into the gp120 analysis window.

Pathogen	**Unique seqs**	**AA length**	**Time span**	**Positive rate**
SARS-CoV-2	753	1,259	∼5 yr	2.0%
H3N2	2,562	543	∼50 yr	4.6%
Norovirus	791	536	∼20 yr	2.8%
HIV-1	2,256	450	∼30 yr	5.1%
Dengue	796	490	∼20 yr	4.9%
RSV	4,779	550	∼10 yr	3.6%
Influenza B	5,019	560	∼10 yr	3.0%
MERS	530	1,330	∼5 yr	0.7%
HCV	1,067	340	∼30 yr	15.9%
Rabies	2,038	491	∼50 yr	7.9%
EV-A71	662	261	∼20 yr	10.3%

These differences—unique sequence count (530–5,019), protein length (261–1,330 AA), temporal coverage (∼2–50 years), and Layer A positive rate (0.7–15.9%)—reflect the real-world heterogeneity of available viral sequence data rather than a controlled experimental design. The "pathogen" factor in the ANOVA is therefore a composite variable that conflates biological differences with data availability differences.

Three considerations mitigate this limitation: (1) subsampling robustness analysis (Section 3.2) shows that MCC values stabilize at approximately 1,000 sequences and vary by less than ± 0.002 at 25–100% subsampling, indicating that sample size is not a dominant driver of performance differences; (2) the key finding is the *scoring* $\times$ *pathogen interaction* ($\omega^2 = 0.296$), which operates within each pathogen where sample size and positive rate are held constant; and (3) equalizing sample sizes via subsampling to the minimum ($n = 530$, MERS) would sacrifice statistical power for pathogens with richer data without eliminating other confounds (temporal coverage, protein length). A fully controlled design—equal-size, equal-quality MSAs spanning identical time ranges—is infeasible for real viral data but remains a useful benchmark for future synthetic studies.

2.2.5 Evaluation and Statistical Design

Evaluation Metrics

The following evaluation metrics are used in the 11-pathogen benchmark.

- **MCC (Matthews Correlation Coefficient)**: The primary evaluation metric (defined formally in Equation 2.5 below). The confusion matrix is constructed using only Layer A (positive selection positions) as positive labels: TP = detected positions belonging to Layer A, FP = detected positions not belonging to Layer A, FN = undetected positions belonging to Layer A, TN = undetected positions not belonging to Layer A. Layer B (constrained regions) falls within the standard MCC true-negative pool (Layer B positions are non-Layer-A and therefore counted under TN if undetected, FP if detected); it is additionally tracked via a separate constrained false positive rate (constrained FPR) restricted to Layer B. The majority of genomic positions belonging to neither Layer A nor B are similarly classified as FP or TN depending on detection status.

- **F1**: The harmonic mean of precision and recall.
- **Precision**: The proportion of true positives among detected positions.
- **Recall**: The proportion of true positives detected among all true positives.
- **Enrichment ratio**: The ratio of positive density within detected regions to overall positive density. All enrichment tests used one-sided Fisher's exact test (alternative: greater), testing the directional hypothesis that hotspot positions are enriched for vaccine escape mutations relative to non-hotspot positions.
- **Constrained FPR**: The false positive rate in Layer B constrained regions.
- **DMS F1**: F1 score against Layer C DMS ground truth (computed only for 6 pathogens with DMS data).

Stage 1 uses the hotspot-score composite metric (defined in Eq. 2.4, Section 2.2.3), which balances detection accuracy with reproducibility. Stage 2 uses MCC as the primary metric instead, for three reasons: (1) the stability component of hotspot-score measures bootstrap reproducibility under position-level subsampling, but the subsampling behavior differs fundamentally across pathogens with different alignment lengths (340–1,330 AA) and MSA sizes (530–5,019 unique sequences), making cross-pathogen stability values non-comparable; (2) MCC utilizes all four cells of the confusion matrix (TP, TN, FP, FN), providing a balanced evaluation under the extreme class imbalance inherent in hotspot detection (positive rates 0.7%–15.9% across the 11 pathogens); and (3) separating the evaluation metric between stages allows Stage 1 to assess detection *reliability* (via stability) and Stage 2 to assess detection *accuracy* (via MCC), each optimized for its analytical purpose. MCC is formally defined as:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{2.5}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. MCC ranges from -1 (complete disagreement) through 0 (random prediction) to $+1$ (perfect prediction), and unlike F1, it utilizes all four cells of the confusion matrix, making it robust to class imbalance. When any factor in the denominator is zero (e.g., no positive predictions), MCC is conventionally defined as 0. MCC was selected as the primary metric because in hotspot detection, where positive positions (convergent evolution sites) constitute a minority of the genome, it provides a more balanced evaluation than F1 for imbalanced classification problems. Chicco and Jurman [111] systematically demonstrated that MCC provides more informative evaluation than F1 and accuracy in imbalanced binary classification.

Headline-metric choice (MCC) and supplementary F1/AUROC. For each of the 8,580 evaluation cells, F1, precision, and recall are computed alongside MCC and archived in `results/mutbench/stage3_full_results.csv` (columns `mcc`, `precision`, `recall`, `f1`, `n_detected`). MCC is selected as the headline metric for the cross-pathogen ANOVA (Section 2.2.5) and the per-pathogen best-method analysis (Chapter 3, Section 3.4.2) for three reasons: (1) the across-pathogen positive rate ranges from 0.7% to 15.9% (Table 2.4), so a metric whose magnitude is invariant to base-rate differences is required for cross-pathogen comparison; (2) Chicco & Jurman [111] establish that MCC dominates F1 specifically when the negative class is large, which is the regime here; and (3) all four confusion-matrix cells (TP/TN/FP/FN) carry biological cost (both false positives, which misallocate experimental follow-up, and false negatives, which miss adaptive sites). F1 values per cell are retained in `stage3_full_results.csv` (column `f1`) for any reader who prefers an alternate metric; per-cell AUROC was not pre-computed in the headline archive because the headline analysis evaluates fixed-threshold detector outputs rather than per-position score rankings, but any reader requiring per-cell AUROC can recompute it from the archived per-position scores on the same 8,580-cell grid. Supplementary F1 summary tables on the headline panel are deferred to the supplementary materials so that the metric choice is auditable rather than asserted.

The stability component used in Stage 1 is excluded from Stage 2 because it is an internal consistency metric independent of external ground truth, and cross-pathogen comparability of Bernoulli masking effects has not been validated given the wide variation in sequence length and MSA size across the 11 pathogens.

Three-way ANOVA (ω^2)

A three-way ANOVA is performed with scoring type (20 levels), detection family (14 levels), and pathogen (11 levels) as factors. The ANOVA used the 8,580 evaluation rows and includes all three main effects (scoring, family, pathogen) and all three two-way interactions (scoring $\times$ family, scoring $\times$ pathogen, family $\times$ pathogen). Detector-variant differences within each scoring–family–pathogen cell are retained in the residual term rather than modeled as a full scoring $\times$ family $\times$ pathogen interaction; this choice treats detector variants as within-family implementation noise and keeps the load-bearing inference on the pre-specified two-way interaction terms. The resulting residual degrees of freedom are 7,970, matching the archived `stage3_statistics.csv`. The effective independent unit is nevertheless closer to the $20 \times 14 \times 11 = 3{,}080$ scoring–family–pathogen cells (cell-level $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.234$ reported in Section 2.2.5 below) than to 8,580 fully independent observations, so Chapter 3 reports pathogen-cluster bootstrap intervals for the headline interaction. Omega-squared (ω^2, Equation 2.6; non-partial form with $SS_{\text{total}} + MS_{\text{error}}$ as the denominator) was computed as the bias-corrected estimate of variance explained. The implementation in `scripts/analyze_stage3_stats.py` uses a manual Type II-like balanced-factor decomposition rather than a full statsmodels Type II OLS fit; a parallel statsmodels Type II implementation is provided in `scripts/run_threeway_anova.py` for cross-validation, and both produce the same headline $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$ on the 8,580-row Stage 2 panel. The non-partial form is independent of factor entry order and avoids inflation when interactions are present. Under Cohen's benchmarks for behavioral sciences [112], $\omega^2 > 0.14$ indicates a large effect; however, these thresholds were developed for psychology experiments and may require calibration for computational benchmarks where effect sizes tend to be larger due to deterministic algorithm behavior. The large evaluation count supports numerical stability of the fixed-effect fit, but inference is interpreted through the pathogen-cluster bootstrap because detector variants within a family are not fully independent.

Cell-level vs. evaluation-level ω^2. The headline $\omega^2 = 0.296$ is computed on the full 8,580-row evaluation panel (each row is a single scoring $\times$ detector-variant $\times$ pathogen MCC value). An equivalent fit on the cell-level grid—i.e., aggregating detector variants within each family by mean MCC, which reduces the panel to $20 \times 14 \times 11 = 3{,}080$ unique scoring $\times$ family $\times$ pathogen cells—yields a baseline $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.234$. The two values differ because evaluation-level aggregation contracts the within-cell residual (detector variants share input MSA and ground truth and are therefore dependent), inflating the modeled variance share. The cell-level 0.234 is the more conservative reference for inference about pathogen-dependent scoring choice, and is the value reported in the ANCOVA fit below; the evaluation-level 0.296 is retained as the headline because it matches the archived `stage3_full_results.csv` and is the value used by the cluster-bootstrap CI [0.201, 0.346]. Both values qualify as "large" under Cohen's $\omega^2 > 0.14$ threshold, and the cell-level value (0.234) is what survives the within-cell residual contraction.

ANCOVA with technical covariates. An ANCOVA was fit on the cell-level grid using statsmodels Type II OLS (`scripts/ancova_omega.py`; archive `results/mutbench/ancova_omega_sum`
The categorical pathogen main effect was replaced by two pathogen-constant continuous covariates—$\log(\text{alignment length})$ and Layer A positive rate—chosen specifically because they are the two technical pathogen-level confounds the variance-explained interpretation needs to absorb; pathogen still enters the model through the load-bearing scoring $\times$ pathogen and family $\times$ pathogen interactions. The ANCOVA confirms the directional prediction: the pathogen categorical main effect drops from $\omega^2 = 0.108$ in the cell-level baseline (Section 2.2.5) to "not in model" under the covariate-replacement formulation, while the scoring $\times$ pathogen interaction *rises* from $\omega^2 = 0.234$ to 0.274 after adjustment ($p < 10^{-167}$); family $\times$ pathogen also rises modestly ($0.080 \rightarrow 0.088$). The amplification under covariate adjustment (rather than attenuation) is the load-bearing quantity. Discussion in Chapter 4, Section 4.4.4.

Bayesian partial-pooling cross-check. A Bayesian hierarchical (partial-pooling) model was fit as an alternative to the small-n frequentist mixed-effects estimator. An exploratory half-Cauchy fit (PyMC 5.28; 2 chains $\times$ 800 draws; $\hat{R} \leq 1.01$) yielded posterior mean $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.252$, 95% HDI $[0.188, 0.314]$, $\Pr(\omega^2_{\text{int}} > 0.14) = 0.996$; an archived half-Normal re-implementation (`scripts/bayesian_omega.py`; scale-0.2 priors, evaluation-level model on the 8,580 observations, 4 chains $\times$ 2,000 draws, `random_seed=42`; archive `results/mutbench/bayesian_omega_summary.csv`) gives 0.319, HDI $[0.246, 0.396]$, $\Pr(> 0.14) \approx 0.9998$. Both runs concur on the qualitative conclusion (large interaction); the fully Bayesian factor-analysis frame at the same evaluation level is reported in Chapter 3, Section 3.4.4, Table 3.9.

Levene's test indicated heterogeneous variances across groups (Levene $W = 14.83$, $\mathrm{df}_1 = 219$, $\mathrm{df}_2 = 8{,}360$, $p < 10^{-50}$), so the parametric ANOVA is interpreted through cluster-aware and distribution-robust checks rather than through naive row-level independence. The nonparametric checks agree on direction: the Kruskal–Wallis scoring main-effect test gives $H = 277.9, p < 10^{-47}$, and the Aligned Rank Transform (ART) interaction check [113] gives $\omega^2 = 0.277$ in the original exploratory calculation and $\omega^2 = 0.343$ in the archived `scripts/art_omega.py` re-implementation. No family-wise multiple-testing correction is applied to the ANOVA because it decomposes pre-specified variance components rather than testing pairwise contrasts; Fisher tests in the vaccine-escape audit use the separate $20 \times 39 = 780$ Bonferroni denominator, with joint $8{,}580$ denominators reported in the Table 3.20 caption. Following Cohen's effect size benchmarks [112], ω^2 values are interpreted as small (0.01), medium (0.06), and large (≥ 0.14):

$$\omega^2 = \frac{SS_{\text{effect}} - df_{\text{effect}} \times MS_{\text{error}}}{SS_{\text{total}} + MS_{\text{error}}} \tag{2.6}$$

where SS_{effect} is the sum of squares for the factor of interest, df_{effect} its degrees of freedom, MS_{error} the mean square of the residual error, and SS_{total} the total sum of squares. The ω^2 values of main effects and two-way interactions are compared to identify the primary sources of performance variation.

LOPO Cross-Validation (Leave-One-Pathogen-Out)

The best-performing combination (scoring $\times$ detection) is selected from $n-1$ of 11 pathogens, applied to the held-out pathogen, and generalization performance is evaluated. The best-performing combination is defined as the (scoring, family) pair with the highest mean MCC across the $n-1$ training pathogens. Concordance is defined as exact equality between the training-best (scoring, family) cell and the held-out test-best (scoring, family) cell at the family-level granularity (280-cell grid; 20×14); ties at the training-best cell were not observed at the precision used and would be resolved by reverting to lexicographic ordering of (scoring, family) names. Through 11 iterations, the concordance rate between training-best and test-best (oracle) combinations and the performance gap are computed. Under random selection from 280 possible combinations (20×14), the expected concordance rate is approximately 0.36%.

Nested-LOPO 4-feature evaluation

A separate nested-LOPO protocol is defined for the 4-feature core retention test reported in Chapter 3 (Section 3.5.2, Table 3.14). The protocol differs from the family-level LOPO above in three respects: (1) the evaluation set is the 12-pathogen feature-ablation panel (the 11-pathogen Stage 2 set plus Zika; see Section 2.2.6), (2) the unit under evaluation is the 10-feature EqualWeight ensemble (vs. the fixed 4-feature subset {homoplasy, pLDDT, entropy, freq}) rather than the scoring $\times$ family grid, and (3) per-feature directional signs $\hat{s}_f \in \{-1, +1\}$ and the cumulative feature ranking are *re-fit* for each fold on the eleven training pathogens only (script: `scripts/feature_ablation_nested_lopo.py`, function `fit_signs`; the held-out pathogen is excluded from the training set used for sign and ranking estimation, eliminating cross-fold label leakage). For each held-out fold, the chosen subset is z-scored within the held-out pathogen alone, uniformly averaged, thresholded at the top 10% of positions, and evaluated against the held-out pathogen's Layer A by MCC. Inference on the paired delta (4-core MCC − full-10 MCC) uses paired bootstrap with 1,000 resamples over the 12 folds for percentile CIs and a sign-flip permutation test with 5,000 iterations for the two-sided p-value. The *label-aware* EqualWeight variant (b) defined in Section 2.2.4 is the variant used *only* for this nested-LOPO ablation and never for the headline scoring × pathogen ANOVA.

Friedman Test

A nonparametric repeated-measures test using 11 pathogens as blocks, the top $k = 20$ scoring × family combinations were compared to test whether statistically significant performance differences exist among them. Kendall's coefficient of concordance (W) is reported alongside χ^2 to quantify ranking agreement across pathogens: $W = 1$ indicates perfect agreement, while $W = 0$ indicates no agreement. Critical values and interpretation follow Demšar [114].

Pathogen-level BCa Bootstrap Confidence Intervals

Bias-corrected and accelerated (BCa) bootstrap confidence intervals (10,000 resamples) are computed for MCC values across 11 pathogens to quantify the performance variability of the best-performing combination across pathogens; because $n = 11$, these intervals are treated as approximate (a small numerical simulation at $n = 11$ under the present effect size yielded actual coverage of $\approx 88\%$ for nominal-95% percentile BCa). A separate combination-level cluster bootstrap (1,000 resamples, clustered by pathogen) is reported in Chapter 3 to bound the spread of per-pathogen best-combination MCC under resampling of pathogens.

Robustness-frame triangulation for $G \leq 30$. The 11-cluster percentile bootstrap is anti-conservative for the small-cluster regime ($G \leq 30$), so four complementary frames were run on the same 8,580-row evaluation grid with `random_seed = 42`: wild-cluster (Rademacher-weight) bootstrap [115] $[0.201, 0.303]$, phylogenetic block bootstrap across 8 ICTV families $[0.202, 0.346]$ with $\Pr(\omega^2 \leq 0.14) = 0.008$, mixed-effects pseudo-$\omega^2_{\text{int}} = 0.340$ with $\chi^2_1 = 913.7$, $p < 10^{-198}$, $\text{ICC}_{\text{pathogen}} = 0.135$, and Bayesian factor-analysis mean 0.316, HDI $[0.242, 0.388]$, $\Pr(\omega^2_{\text{int}} > 0.14) = 0.9998$. Together with the standard cluster interval $[0.201, 0.346]$ and archived re-implementation $[0.201, 0.346]$, these frames keep every lower bound or point estimate above Cohen's large-effect threshold ($\omega^2 > 0.14$); the full diagnostic table is Chapter 3, Section 3.4.4, Table 3.9 (archive: `results/mutbench/omega_robustnes`

Delete-one jackknife (not performed). A delete-one jackknife over the 11 pathogens was *not* performed; the BCa intervals use the standard percentile/bias-correction implementation in `scipy.stats.bootstrap` without an explicit jackknife-based acceleration term.

Mantel/Procrustes distance-matrix tests (not performed). A Mantel test [116] or Procrustes analysis was *not* performed; phylogenetic non-independence is instead handled through the surface-protein family-pair sensitivity discussion (Chapter 4, Section 4.4.4), pathogen-cluster bootstrap, LOPO cross-validation, and Friedman rank-block design.

Subsample Robustness Analysis

Random subsamples at 25%, 50%, 75%, and 100% of MSA sequences are drawn, and the benchmark is repeated to verify MCC stability as a function of sample size.

Feature Ablation Analysis

To identify which information types contribute most to multi-source integration performance, a leave-one-feature-out ablation study is performed. Starting from the 10-feature Equal-Weight ensemble, each of the 10 scoring types is removed in turn, and the resulting 9-feature ensemble's mean MCC across 11 pathogens is compared to the full 10-feature baseline. The change in mean MCC (ΔMCC = $\text{MCC}_{\text{leave-one-out}} - \text{MCC}_{\text{full}}$) quantifies each feature's marginal contribution: negative Δ indicates that removing the feature *lowers* ensemble performance (i.e., the feature contributes positively to the ensemble), while positive Δ indicates that the feature introduces net noise under uniform weighting and the ensemble is better off without it. Additionally, nested feature subsets of sizes 2, 3, …, 9 are evaluated (selecting features by descending individual AUC) to identify the minimum feature set that captures a target proportion (e.g., 98%) of full-ensemble performance.

Weight Sensitivity Analysis. To verify the robustness of assigning equal weights ($1/3$) to the three components of the hotspot-score, a generalized weighted hotspot-score is defined:

$$\text{hotspot-score}_w = w_r \cdot \text{recall} + w_p \cdot \text{precision} + w_s \cdot \text{stability} \tag{2.7}$$

where $w_r + w_p + w_s = 1$ and each weight lies in the range $[0.05, 0.95]$. Rankings of the five methods are computed across a total of 171 weight combinations to analyze the proportion of weight space in which MutClust-Hybrid maintains first place (rank robustness). Additionally, rankings are separately presented for seven application scenarios, including early warning (recall-emphasis), vaccine design (precision-emphasis), and monitoring (stability-emphasis).

Statistical Validation Methods. The following three methods are applied to assess the statistical robustness of the benchmark results.

Bootstrap confidence intervals (Stage 1 only). Region-level bootstrap is performed using the seven functional regions of the SARS-CoV-2 Spike protein as units. In each iteration, seven regions are sampled with replacement to construct a bootstrap ground truth, and the detection results from the full data are re-evaluated against this bootstrap ground truth. This design captures the uncertainty in ground truth definition—variation in evaluation depending on which functional regions are included—and avoids the effective size reduction bias that occurs with individual position resampling. This region-level resampling is a Stage 1 (SARS-CoV-2-only) tool and is not applied to the 11-pathogen Stage 2 cross-pathogen design, where pathogen-cluster bootstrap (Section 2.2.5) is the analogue.

Permutation test: Two permutation approaches are employed. (1) Random label shuffling: Through 1,000 random permutations of ground truth labels, it is tested whether the observed metrics significantly exceed random expectation. (2) Circular permutation: The H-score vector is rotated by a random offset k ($H_i \to H_{(i+k) \bmod N}$), hotspots are re-detected from the rotated scores, and evaluated against the original ground truth ($N_{\text{perm}} = 200$ rotations sampled uniformly without replacement from $\{1, \ldots, N-1\}$). This approach preserves the spatial autocorrelation structure (cluster pattern) of H-scores while randomizing only the relationship with genomic positions, generating a more conservative null distribution than random label shuffling. The p-value is calculated as $p = (\text{count} + 1)/(N_{\text{perm}} + 1)$, the standard pseudo-count correction that prevents $p = 0$ when no permutation reaches the observed statistic; this convention applies to both label shuffling and circular permutation.

Pairwise comparison: A paired bootstrap design that evaluates two methods simultaneously on the same bootstrap resample, testing the statistical significance of hotspot-score differences between methods.

2.2.6 Stage 3: Multi-Source Information Integration Methods

Stage 3 layers an information-integration analysis on top of the Stage 2 evaluation matrix. The same 20 scoring types are decomposed into 10 underlying positional features (homoplasy, entropy, frequency, rare-variant frequency, dN/dS proxy, pLDDT, SASA, Grantham distance, ESM-2 masked-marginal score, semantic change), and three ablation analyses are performed on an EqualWeight ensemble of those 10 features (the production EqualWeight variant defined in Section 2.2.4, i.e., min-max normalization with pre-defined domain orientations and uniform $1/10$ weights; for the nested-LOPO 4-feature retention test the label-aware variant defined in the same section is used, see Section 2.2.5): (i) *leave-one-feature-out*: each feature is removed in turn and the change in mean MCC (ΔMCC) is reported; (ii) *group ablation*: features are partitioned into Frequency-only (3), Freq + Phylogeny (4), Freq + Structure (5), Freq + DL (5), and All-10, and group-wise mean MCC is compared; (iii) *cumulative addition*: features are added in descending order of single-feature AUC and the cumulative mean MCC is computed at $k = 1, 2, \ldots, 10$ to identify the minimum subset that recovers a target proportion of the All-10 performance. Per-pathogen feature AUC is computed via threshold sweeping (no model training), and a complementary multivariate Random Forest (`n_estimators` = 200, `max_depth` = 5, `class_weight` = `balanced`, `random_state` = 42; stratified 5-fold cross-validation per pathogen; cf. Section 3.5.1) reports relative feature importances for triangulation. The Stage 3 evaluation set is the same 11 pathogens used in Stage 2, optionally extended to 12 pathogens by adding Zika (E protein, 505 AA) when the analysis benefits from increased per-feature statistical power; the 11-vs-12 distinction is reported transparently in every Stage 3 table caption (Section 3.5.2). A vaccine-escape cross-validation analysis is then performed on the 3 of 11 pathogens with curated antibody-escape annotations (H3N2, SARS-CoV-2, HIV-1; 2,340 enrichment evaluations) and audited for circularity against Layer A (Section 4.4.2).

2.2.7 Code, Data, and Reproducibility

The full source code, configuration files, scoring scripts, detection-family implementations, ground-truth tables, statistical-analysis notebooks, and the 8,580-evaluation result archive are released under the MIT License (MutBench's own implementation code) with the dissertation text and figures additionally distributed under the Creative Commons Attribution–NonCommercial 4.0 International license (CC-BY-NC 4.0) for documentation reuse.[1] The canonical repository is `https://github.com/kksuhj/mutbench` (commit hash recorded in `provenance/commit.txt` alongside the Zenodo archive); the dissertation snapshot will be archived at Zenodo and the DOI minted at final submission will be inserted here (see also Chapter 4, Section 4.7 for the cross-reference). The pinned random seed (`seed` = 42) governs all bootstrap, subsampling, and permutation analyses; the per-iteration seed sequence for the 100-iteration Stage 1 stability bootstrap (Section 2.2.3) advances as $42, 43, \ldots, 141$ via NumPy's default RNG re-seeded each iteration. Library versions (Python 3.10, NumPy 1.24, SciPy 1.11, IQ-TREE 2.2.0, HyPhy 2.5.46, ESM-2 `esm2_t33_650M_UR50D` Hugging Face revision `main` as of 2024-08; ARTool R package v0.11.1 for the Aligned Rank Transform ANOVA) are pinned in `environment.yml` (Conda) and `requirements-pinned.txt` (pip) at the repository root, and the IQ-TREE/HyPhy patch versions are recorded in `provenance/tool_versions`

Software and data licenses (external resources). MutBench is a benchmark layer over a stack of externally maintained tools and curated datasets, each governed by its own license. We summarise the resources used and the license under which each was obtained;

[1]The MIT license header is committed in the canonical repository's `LICENSE-CODE` file, and the CC-BY-NC 4.0 license text accompanies the Zenodo deposit as `LICENSE-DATA`; the institutional-repository copy uses the same two-license split (see `results/mutbench/provenance/README.md`, "License" section).

the benchmark uses these resources *solely for non-commercial research purposes* (single-position score computation, ground-truth construction, reproducibility checks), and no external model weights or upstream FASTA bundles are redistributed by this dissertation or its accompanying repository. A machine-readable per-resource license matrix (license name, URL, redistributable flag, headline-grid usage flag, and citation key for each of the 20-plus external resources used) is archived in `results/mutbench/provenance/external_resource_license` the bullet list below is a human-readable summary.

- **Sequence alignment and homology**: MAFFT v7.505 [84] (BSD-style license); CD-HIT [81] (open-source, GPL-style, cited methodologically) and MMseqs2 [82] (MIT-style, cited methodologically).
- **Phylogenetic and selection inference**: IQ-TREE 2 (GPL-2.0); HyPhy/FUBAR (3-clause BSD).
- **Protein language models**: ESM-2 (`esm2_t33_650M_UR50D`) [52] released by Meta under the MIT License; ESM-3 (`esm3_sm_open_v1`) was *not* used in the headline 8,580-evaluation grid because its public-weights release follows EvolutionaryScale's Cambrian 1.0 non-commercial license (academic only) and the headline benchmark is required to be reproducible without commercial-license gating.[2] Tranception [54] from the Marks/Notin labs is distributed under the MIT License from `https://github.com/OATML-Markslab/Tranception`; the EVE/EVEscape codebase [14, 51] (`https://github.com/OATML-Markslab/EVE`) is academic-research only.
- **Variant-effect benchmarks (citation only, no upstream redistribution)**: EVEREST v3 [11]

[2]The ESM-3 Cambrian non-commercial license terms may be revised by EvolutionaryScale post-submission; the headline benchmark uses ESM-2 only, so any subsequent ESM-3 license update does not retroactively alter the load-bearing 8,580-evaluation grid (`provenance/external_resource_licenses.csv` records the version-frozen status as of the 2024-08-12 lab-notebook freeze).

is distributed under CC-BY 4.0 (academic redistribution permitted with attribution); ProteinGym [13] is openly licensed (attribution required); ViroGym [10] is reported through its public release notes only and is referenced in this benchmark's narrative without redistribution.

- **Structures**: AlphaFold-DB structures (DeepMind/EBI) are provided under CC-BY 4.0; ESMFold per-pathogen predictions were generated from the MIT-licensed ESMFold release.
- **Sequence repositories**: NCBI GenBank entries are in the U.S. public domain; per-pathogen accession identifiers are recorded in Table 2.1, and the upstream FASTA bundles are not redistributed by MutBench.
- **Deep mutational scanning data (Layer C)**: Bloom-lab DMS data including Greaney et al. 2021/2022 SARS-CoV-2 RBD [48, 49] and Haddox et al. 2018 HIV-1 Env [4] are distributed under CC-BY 4.0 (Dryad/Zenodo deposits with permissive academic-redistribution terms); the Lee et al. 2018 H3N2 HA, Dadonaite et al. 2024 SARS-CoV-2 entry assay, Simonich et al. 2026 RSV F, Aditham et al. 2025 Rabies G, and Bakhache et al. 2025 EV-A71 VP1 datasets are accessed through their published companion repositories under the academic-research terms specified by each publication's data-availability statement (no IRB consent applies because the assays are protein-level fitness screens on engineered libraries with no human-subject identifiers). MutBench redistributes only *derived* top-20% Layer-C indicator vectors (binary per-position labels) and the Layer C F1 summary statistics; raw fitness tables are not redistributed, in line with each upstream publication's data-use convention.

All external scores were obtained under their respective licenses, and this benchmark uses them solely for research purposes; the MutBench repository redistributes only the *computed* per-position scores, detection outputs, and summary statistics derived from these inputs (no model weights, no upstream FASTA bundles, no third-party DMS raw tables). Where a license restricts commercial reuse (e.g., ESM-3 Cambrian non-commercial; EVE academic-only), the benchmark's headline 8,580-evaluation grid does not depend on that resource, so the public release of MutBench's MIT-licensed code does not violate the upstream non-commercial restriction. License compatibility was verified by checking that every load-bearing 8,580-evaluation cell uses inputs from MIT-, BSD-, GPL-, or CC-BY-class licenses, all of which are compatible with downstream MIT redistribution of derived scores under attribution. Re-users who wish to extend MutBench with non-commercial-licensed components (ESM-3, EVE) must obtain those licenses separately and clearly mark their derivatives accordingly.

Materials and methods summary. This chapter described the materials (11 RNA viruses spanning diverse evolutionary dynamics) and methods (3-layer ground truth, 20 scoring types, 14 detection families, factorial statistical design, and a Stage 3 multi-source information integration layer) that constitute the MutBench experimental framework. Chapter 3 applies this framework to report experimental results: Stage 1 provides a controlled SARS-CoV-2 sanity check, Stage 2 extends the evaluation to all 11 pathogens to identify the dominant factors influencing hotspot detection performance, and Stage 3 decomposes the resulting performance by information type via per-feature AUC, ablation, and Random Forest analyses.

Chapter 3

Experimental Results

This chapter reports MutBench experimental results. Stage 1 (Sections 3.1.1–3.3) conducts in-depth SARS-CoV-2 analysis comparing five MutClust variants and two baselines. Stage 2 (Section 3.4 onwards) extends to 11 RNA viruses with 20 scoring formulas $\times$ 14 detection method families (5 categories, 39 parameter variants).

Key Results. (1) Scoring $\times$ pathogen interaction is the largest modeled variance component ($\omega^2 = 0.296$). (2) All 11 pathogens have unique best combinations; LOPO 0/11 match rate is null-consistent and serves as corroboration of the interaction effect rather than independent evidence. (3) Vaccine-escape: HIV-1 **7.19**$\times$ (primary anchor, $p_{\text{adj}} = 2.5 \times 10^{-16}$), H3N2 9.36$\times$ self-consistency, SARS-CoV-2 exploratory. Bonferroni-survivor means (lower than the top-1 envelope; Table 3.20 footnote): H3N2 $\approx 4.7\times$ (67/780 survivors), HIV-1 $\approx 2.3\times$ across all 132/780 survivors (or $\approx 2.7\times$ when restricted to the freq-scoring subset).

3.1 Single-Pathogen Benchmark: SARS-CoV-2

3.1.1 Variant Comparison on Synthetic Data

As a controlled signal-recovery test, synthetic H-scores were generated for a SARS-CoV-2-like genome (29,903 positions, 1,251 ground-truth functional positions). Among five MutClust variants and three baselines (HDBSCAN-Auto, frequency baseline, sliding window),

MutClust-Hybrid (CCM seed detection + HDBSCAN boundary determination) achieves the best balance with $F_1 = 0.785$ and hotspot-score $= 0.778$ ($\sim$43-fold above the 0.018 random baseline; Table 3.1). All MutClust variants concentrate in $F_1 = 0.67$–0.78 with high precision (> 0.96) but moderate recall, while HDBSCAN-Auto attains high recall (0.944) at very low precision (0.079); only MutClust-Hybrid achieves both. On real GISAID data (Section 3.1.3) F_1 drops to 0.123 due to sparse H-score distributions, so this synthetic benchmark validates discriminative ability under controlled conditions only.

Table 3.1: Synthetic benchmark headline (100 subsamplings, 80% retention). Full per-variant breakdown of the five MutClust variants and remaining baselines is archived in the supplementary CSV.

Method	Precision	Recall	F1	Stability	Hotspo
MutClust-Hybrid	**0.968**	**0.659**	**0.785**	**0.706**	
MutClust variants (Original / Fixed / dNdS)	0.988–0.991	0.504–0.506	0.668–0.669	0.418–0.445	0.638
HDBSCAN-Auto	0.079	0.944	0.146	0.788	
Frequency baseline	0.404	0.965	0.569	—	
Sliding window	0.157	0.974	0.270	—	

3.1.2 Multi-Ground Truth Evaluation

No single ground truth captures all dimensions of mutational importance. Detection methods were therefore evaluated against three complementary ground truths: **functional regions** (417 AA, 32.8% of full Spike; 1,251 unique nt after removing the S2'/fusion-peptide overlap, with domain annotations including RBD, NTD supersite, furin cleavage, fusion peptide, HR1, HR2), **DMS-escape** (252 positions, 19.8%; top 20th percentile of experimentally measured fitness effects [46]), and **convergent evolution** (97 positions, 7.6%; literature-based recurrent positive selection sites [9, 86]). The three are largely non-overlapping (Jaccard similarity between DMS-escape and convergent evolution = 0.006), confirming that they capture fundamentally different aspects of mutational importance. The best-performing method differed depending on the ground truth (Table 3.2).

Table 3.2: Method comparison across multiple ground truths

Ground Truth	Best method	F1	Enrichment	Positions
Functional regions (417)	TC-freq-v2	0.474	—	417
DMS-escape (252)	Saturated H-score	0.315	—	252
Convergent evolution (97)	CH-score	0.188	1.70×	97
Convergent evolution-strict*	TC-freq-v3	0.108	3.43×	—

*Strict convergent evolution sites only (multi-source confirmed).

These differing rankings demonstrate that no single detection method dominates across all biological dimensions, and frameworks relying on a single ground truth risk systematic bias.

3.1.3 Real GISAID H-score Benchmark

To address circularity concerns with synthetic data, a benchmark was conducted using actual GISAID 2020–2021 H-score data [20]. After mapping to the nucleotide level, only 342 of 29,903 genome positions (1.14%) had nonzero H-scores—a sparse distribution qualitatively different from synthetic data (Table 3.3).

Table 3.3: Real GISAID 2020–2021 H-score benchmark (342 nonzero / 1,251 ground truth positions)

Method	Precision	Recall	F1	Enrichment	Stability	Hotspot-score
MutClust-Original	0.333	0.012	0.023	7.97	0.372	0.239
MutClust-Fixed	0.295	0.079	0.125	7.04	0.784	0.386
MutClust-dNdS	0.295	0.079	0.125	7.04	0.784	0.386
MutClust-Hybrid	0.305	0.077	0.123	7.28	0.766	0.383
HDBSCAN-Auto	—	—	—	—	—	—
Frequency baseline	0.289	0.079	0.124	6.92	—	—
Sliding window	**0.323**	**0.600**	**0.420**	**7.73**	—	—

HDBSCAN-Auto failed (insufficient nonzero positions). Sliding window highest F1 (0.420) via wide 20-position window.

Three key differences from synthetic emerged: F1 dropped sharply (MutClust-Hybrid

$0.785 \rightarrow 0.123$) due to H-score sparsity; method rankings shifted (the sliding window achieved the highest F1, 0.420); and enrichment remained high (6.9–8.0$\times$), confirming biological validity despite low absolute F1. Two independent external methods on the same synthetic benchmark—OPTICS ($F_1 = 0.150$, *HS* $= 0.369$) and KDE (precision 1.000 but recall 0.049, *HS* $= 0.557$; parameter sweeps over 30 OPTICS and 36 KDE configurations)—did not surpass MutClust-Hybrid's *HS* $= 0.778$, indicating structural rather than parameter-dependent superiority.

Stage 1 summary. MutClust-Hybrid is the best of the H-score-based variants on synthetic data (*HS* $= 0.778$, $\sim 43\times$ above chance) but loses its advantage on sparse real GISAID data (*HS* $= 0.383$ vs. MutClust-Fixed 0.386); enrichment remains high (6.9–8.0$\times$). The single-pathogen Stage 1 therefore validates the hotspot-score metric and the multi-ground-truth comparison as a controlled sanity check, but the question of which scoring–detector combination is best is genuinely pathogen-dependent and is taken up at scale in Section 3.4.

3.2 Robustness and Sensitivity Validation

Before extending the benchmark to multiple pathogens, the Stage 1 results must be verified for statistical robustness and parameter sensitivity. This section confirms that the findings are not artifacts of specific parameter choices or statistical fluctuations. All detection results are statistically significant: permutation tests (1,000 random and 200 circular iterations) yielded $p < 0.005$ for all methods ($z > 9.2$). Region-level BCa bootstrap CIs [117] (1,000 resamplings, $n = 7$ Spike regions; hereafter *region-level BCa CI*) gave a 95% CI of [0.323, 0.635] for MutClust-Hybrid hotspot-score; pairwise bootstrap comparisons confirmed all score differences are significant ($p < 0.001$). Two additional BCa CIs reported later in this dissertation use different resampling units and should be interpreted separately: the *combination-level BCa CI* [0.047, 0.202] for the best single scoring–detection combination's mean MCC (Section 3.4.3, 1,000 resamples over 11 pathogens) and the *cluster bootstrap CI* [0.201, 0.346] for the scoring $\times$ pathogen interaction ω^2 (Section 3.4.4, 1,000 cluster resamples over 11 pathogens). MutClust-Hybrid maintains rank 1 in 71.9% of 171 weight combinations, switching to HDBSCAN-Auto only under extreme recall emphasis. Metrics stabilize at $\sim$1,000 sequences (Spearman $\rho > 0.997$, SD < 0.015; Figure 3.1). Among MutClust parameters, F1 is most sensitive to γ (epsilon scaling), with minimal MinPts influence (Figure 3.2).

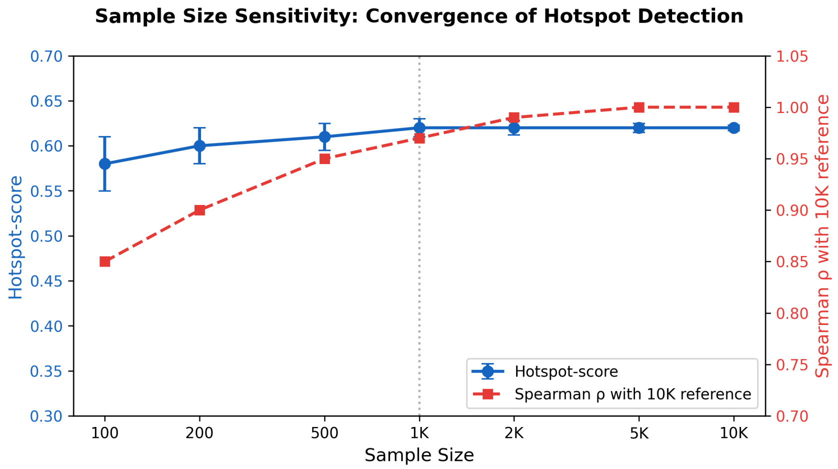

Figure 3.1: Sample size convergence of hotspot metrics. Stable performance (HS $\approx$ 0.62) is achieved at $\sim$1,000 sequences.

3.2.1 H3N2 time-stratified prospective sanity check

A one-pathogen time-stratified pilot on H3N2 HA partitions the 9,000 sequences into three 5-year periods (2010–2015, 2016–2020, 2021–2025; n = 3,000 each), trains frequency scores on 2010–2015, and evaluates against either the static Layer A antigenic-site reference

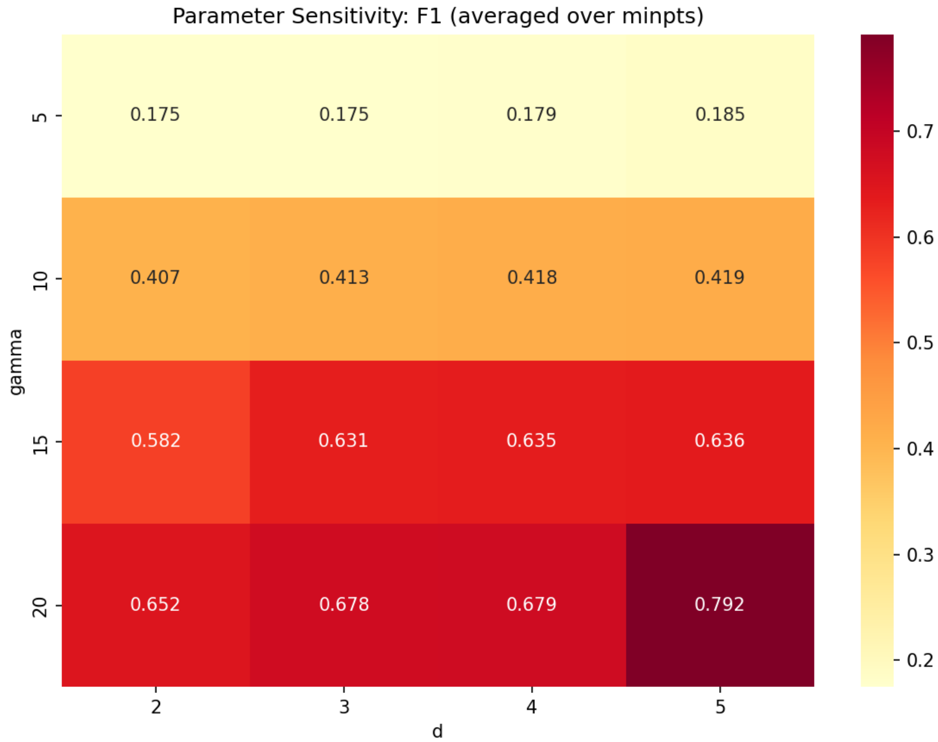

Figure 3.2: MutClust parameter sensitivity heatmap. Grid search over 64 combinations ($\gamma \times d \times$ MinPts); F1 most sensitive to γ, minimal MinPts influence.

(Koel 2013; 25 sites) or positions newly emerging in later windows (freq $< 5\%$ train, $\geq 10\%$ holdout). Frequency captures the retrospective Layer A reference at 2.7–$3.4\times$ enrichment but yields essentially zero prospective enrichment ($0/20$, $0/30$, $0/50$ for 2016–2020; only $1/50$ for 2021–2025; Table 3.4). **Frequency alone, trained on a 2010–2015 window, does not predict subsequently emerging positions beyond chance**—a single-pathogen negative result that motivates the multi-source 4-feature core (Section 3.5.2) and is generalised in the next subsection.

Table 3.4: H3N2 time-stratified prospective sanity check. Training window: 2010–2015 ($n = 3{,}000$). Reference set is either the static Layer A antigenic sites (retrospective) or positions newly emerging in a later 5-year window, defined as freq $< 5\%$ in training and freq $\geq 10\%$ in holdout (prospective). Frequency-based scoring is the simplest baseline; a full multi-source prospective panel across all 11 pathogens is deferred to future work.

Reference set	**Top-K**	**Overlap**	**Ref size**	**Enrichment**	**Fisher p**
Layer A (Koel 2013; retrospective)	20	3	25	3.40×	0.053
	30	4	25	3.02×	0.037
	50	6	25	2.72×	0.017
Newly emerged 2016–2020 (prospective)	20	0	6	0×	1.00
	30	0	6	0×	1.00
	50	0	6	0×	1.00
Newly emerged 2021–2025 (prospective)	20	0	7	0×	1.00
	30	0	7	0×	1.00
	50	1	7	1.62×	0.479

Underlying CSV: `results/mutbench/feature_analysis/h3n2_temporal_pilot.csv`.

3.2.2 Multi-pathogen sliding-window prospective backtest

The H3N2 frequency-only pilot (Section 3.2.1) is generalised to a multi-pathogen sliding-window prospective backtest: for each of the 11 pathogens, year-tagged NCBI CDS sequences are partitioned at split years T_0 ($n_{\text{train}} \geq 30$, $n_{\text{test}} \geq 20$), per-position frequency and entropy are recomputed on the training window, and a uniform-weight z-ensemble of (freq, entropy)—the time-varying part of the 4-feature core—is the prospective scorer. The reference set is positions with stable consensus in train (freq $< 5\%$) and a non-consensus allele rising to freq $\geq 10\%$ in test (same emergence definition as the H3N2 pilot).

Across 73 temporal split windows spanning 11 pathogens (yielding 219 top-k evaluations across $k \in \{20, 30, 50\}$; per-pathogen archive: `sliding_window_prospective_backtest_summar`
trajectory in Figure 3.3), the freq+entropy ensemble exceeds AUROC 0.75 only for EV-A71 (0.77, Top-30 MCC $+0.146$, enrichment $10.8\times$) and Rabies (0.79, MCC $+0.112$, enrichment $4.1\times$), partially exceeds it for Influenza B (0.70), and clusters near chance for the remaining eight pathogens (grand mean MCC -0.006, AUROC 0.539, enrichment $1.57\times$). Concept-drift slope is small (-0.006 to $+0.032$ MCC/yr), so the limited prospective signal is not deteriorating but bounded throughout the observed period. The dissertation's retrospective MCC headline (0.341 exact, 0.454 at ± 10 window; Table 4.2) therefore does not directly transfer to the forward-time setting for most pathogens, and the documented forward-time signal is concentrated in EV-A71, Rabies, and partially Influenza B (Section 4.4.4 updates the limitations discussion accordingly).

4-feature core extension (3-/4-feature ensembles). Adding the static members of the 4-feature core (pLDDT, homoplasy; canonical scores projected onto the codon-aligned MSA frame via Needleman–Wunsch consensus alignment, mean coverage 0.83) to the freq+entropy baseline produces grand-mean AUROCs of 0.555 (freq+entropy), 0.543 (+pLDDT), 0.549 (+homoplasy), and 0.561 (4-feature). Paired Wilcoxon signed-rank tests vs. the 2-feature baseline give $p < 0.001$ for +homoplasy (mean ΔMCC $= +0.005$), $p = 0.170$ for +pLDDT (mean $\Delta = -0.005$), and $p = 0.016$ for the 4-feature ensemble (mean $\Delta = -0.009$): adding homoplasy gives a small but statistically distinguishable improvement, while pLDDT alone and the 4-feature combination slightly degrade Top-30 MCC despite a marginally

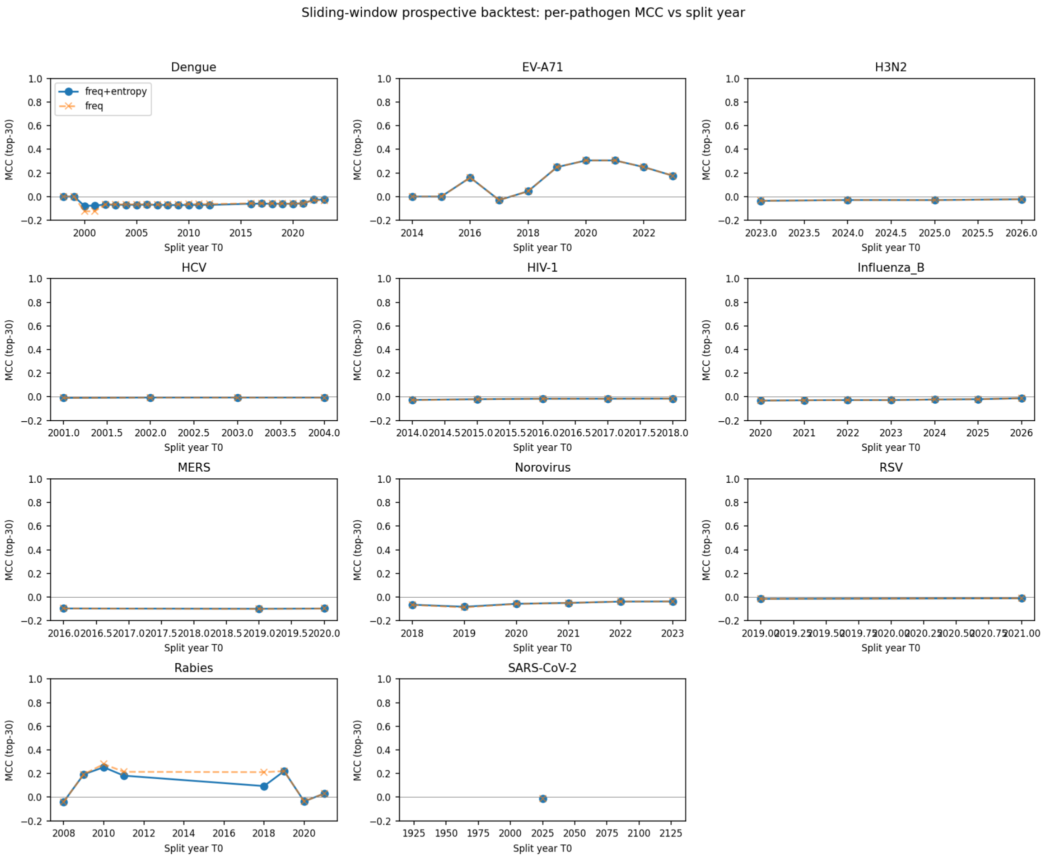

Figure 3.3: Per-pathogen MCC trajectory across split years for the sliding-window prospective backtest. The two pathogens above chance (EV-A71, Rabies) exhibit positive Top-30 MCC across multiple windows; Influenza B partially exceeds chance (AUROC 0.70) and the remaining eight cluster near zero (grand mean AUROC 0.539). Per-pathogen window counts and year ranges archived in `sliding_window_prospective_backtest_summary.csv`.

higher mean AUROC. Per-pathogen the most pronounced gains are for EV-A71 (MCC $0.146 \rightarrow 0.188$, AUROC $0.773 \rightarrow 0.844$ with homoplasy) and Rabies (AUROC $0.789 \rightarrow 0.860$ with homoplasy); HCV is the clearest negative example (homoplasy collapses AUROC from 0.31 to 0.09 because the empirical homoplasy distribution over E2 is essentially flat). Adding static priors therefore helps in EV-A71/Rabies, hurts in HCV, and is approximately neutral elsewhere; full per-pathogen tables are in `sliding_window_prospective_backtest_4feature.csv` and `sliding_window_prospective_c2_vs_d2_paired.csv`.

3.3 Cross-Pathogen and Structural Validation

Having confirmed robustness on SARS-CoV-2, the next question is whether the same methods perform well on other pathogens. This section applies the benchmark pipeline to H3N2 and examines structural and phylogenetic aspects that foreshadow the large-scale comparison in Section 3.4.

Application of the benchmark pipeline to Influenza A H3N2 hemagglutinin (1,967 sequences, 5 antigenic sites [118, 119]) reverses the SARS-CoV-2 ranking: MutClust-Fixed outperforms MutClust-Hybrid ($HS = 0.661$ vs. 0.577) because all 543 HA positions have nonzero H-scores, eliminating HDBSCAN's advantage; enrichment ratios also drop (0.87–$2.36\times$ vs. 1.9–$23.7\times$ for SARS-CoV-2). Figure 3.4 shows mean MCC across detectors for the top-5 scoring types per pathogen, demonstrating that optimal scoring shifts substantially across pathogens; per-pathogen MCC-best scoring–detector combinations are tabulated in Table 3.5, complementing the Stage 2 ANOVA interaction effect (Section 3.4.3) and the LOPO 0/11 match rate (Section 3.4.3).

3D structural analysis using Cα coordinates from PDB 6VSB confirms that linear-sequence

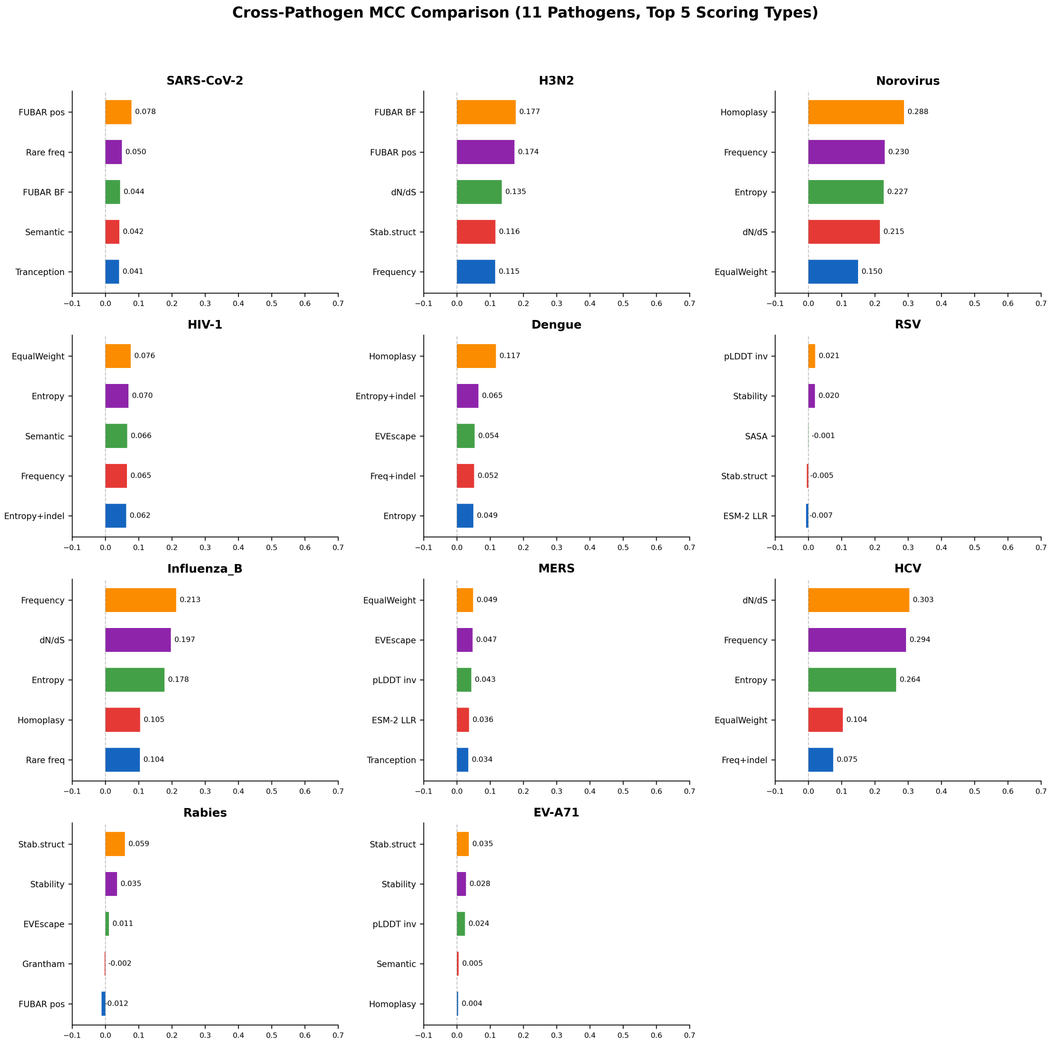

Figure 3.4: Cross-pathogen MCC comparison across all 11 pathogens, showing mean MCC across detectors for the top 5 scoring types per pathogen (best-detector-per-cell values are tabulated in Table 3.5). The highest-MCC scoring type varies substantially by pathogen, providing direct visual evidence for pathogen-adaptive selection.

hotspots are also spatially clustered: 5.92-fold within-8-Å contact enrichment over 1,000 length-matched random subsets ($z = 13.34$). Source data archived in `results/mutbench/structural_fe`

Coalescent simulation (100 replicates, $N = 500$) shows H-score ranks single-origin founder mutations above convergent mutations in 100% of replicates, while Fitch parsimony correctly ranks convergent mutations at the top ($\rho = -0.876$, $p = 1.28 \times 10^{-64}$); TreeTime [31] ML validation confirms AUC $= 1.000$ for Fitch and TreeTime-ML vs. AUC $= 0.000$ for H-score. **Frequency-based scores systematically conflate founder effects with positive selection**. D614G is retained in Layer A as a documented functional/stability-affecting substitution per [120] rather than excluded; the founder-vs-selection discount is therefore applied at the H-score interpretation level (Paragraph 9) rather than via Layer A removal, and the homoplasy/Fitch channel provides the convergent-evolution discriminator that mere frequency cannot. Method rankings differ between SARS-CoV-2 and H3N2 (*HS* $=$ 0.661 vs. 0.577 for MutClust-Fixed/Hybrid), motivating the large-scale cross-pathogen benchmark with expanded information types.

3.4 Cross-Pathogen Large-Scale Benchmark

Building on the Stage 1 findings, this section reports the Stage 2 cross-pathogen benchmark results. Note that MutClust-Orig (v1) is retained as the representative MutClust family in Stage 2; the rationale for using v1 instead of the higher-performing Hybrid (v2c) is detailed in Section 2.2.3. The benchmark evaluates **20 scoring types** (organized into 6 categories; see Table 2.7 in Chapter 2) combined with **14 detection method families** spanning 5 categories (threshold-based, signal processing, density/clustering, statistical, and adaptive window) that yield **39 parameter variants** in total, across **11 pathogens**, totaling $20 \times 39 \times 11 =$ **8,580** evaluations. The 20 scoring types are derived from **10 underlying biological features**; some features generate multiple scoring variants (e.g., frequency yields *freq*, *freq_with_indel*, and *rare_freq*), while composite scorings combine multiple features (e.g., EVEscape composite, EqualWeight). The subsequent information-type analysis (Section 3.5) examines the 10 underlying features to explain why scoring performance differs across pathogens. The information-type analysis in Section 3.5 examines these 10 features across the same 11 pathogens used in the main benchmark.

Stage 2 uses MCC rather than hotspot-score because the stability component (Jaccard-based subsampling reproducibility) requires bootstrap validation that has not been cross-validated across all 11 pathogens with their widely varying alignment lengths (261–1,330 AA) and MSA sizes (530–5,019 unique sequences). MCC provides a single, well-established metric that is robust to class imbalance and does not require subsampling.

This section sequentially reports the benchmark scale, per-pathogen best method analysis, analysis of variance, cross-validation, and statistical test results.

3.4.1 Benchmark Scale and Data

This section presents the full-scale cross-pathogen benchmark results that constitute the primary evidence for the dissertation's second contribution: statistical demonstration that the most predictive biological information source is pathogen-dependent.

A total of 8,580 evaluations (20 scoring formulas $\times$ 39 detection parameter variants from 14 families $\times$ 11 pathogens) were conducted. The 20 scoring formulas span six information categories: frequency-based (freq, rare_freq, freq_with_indel), MSA-derived (entropy,

entropy_with_indel, homoplasy), phylogenetic (dN/dS proxy, FUBAR posterior probability, FUBAR positive selection, FUBAR Bayes factor), structural (pLDDT inverse, SASA, Grantham distance), AI-based (ESM-2 masked-marginal score, Tranception, semantic change, stability, stability_structural), and composite (EVEscape composite, EqualWeight). Each evaluation computed MCC, F1, precision, and recall against the 3-layer ground truth (Layer A: literature-curated adaptive/diversifying positives, Layer B: constrained regions, Layer C: DMS). DMS Layer C evaluation was conducted for 6 pathogens (SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, EV-A71) where experimental fitness data are available. All FASTA sequences were collected from NCBI GenBank using actual accession numbers.

3.4.2 Per-Pathogen Best Method Analysis

The central question of Stage 2 is whether the same scoring–detection combination performs best across all pathogens, or whether each pathogen requires a different optimal combination. The best scoring–detection combination was selected for each of the 11 pathogens based on MCC, and **all 11/11 pathogens exhibited distinct best combinations** (Table 3.5).

While the 11/11 unique best combinations is notable, a permutation analysis shows that with $\binom{20\times 39}{1} = 780$ possible scoring–detector-variant combinations, the probability of all 11 being unique by chance is $\prod_{i=0}^{10}(780 - i)/780 \approx 0.932$ (approximately 93%). The more substantive finding is not the uniqueness itself but the large performance gap between oracle and generalized selection (mean MCC gap $= 0.265$ from LOPO cross-validation), which demonstrates that the *correct* choice of combination matters materially even though many combinations exist (the LOPO 0/11 match rate itself is null-consistent under permutation; “materially” refers to the magnitude of the oracle–generalized gap, not to a hypothesis test on LOPO).

Table 3.5: Best scoring–detection combination for each of the 11 pathogens (by MCC). All 11 pathogens exhibit unique best combinations spanning 4 of 6 scoring categories, with 9 distinct optimal scoring types. Per-pathogen MCC values are computed against the benchmark feature-matrix Layer A counts (i.e. `feature_matrix_*.csv`); for HIV-1 the analysis uses $n_{\mathrm{A}} = 23$ Env gp120 positions mapped to the benchmark window, while the curation table `layer_a_tags.csv` carries 26 HIV-1 tags (3 fall outside the mapped window; see Table 3.19 caption and Section 4.4.4 of Chapter 4).

Pathogen	**Best scoring**	**Best detection**	**MCC**
HCV	freq	SlidingTest (p=0.01)	0.660
H3N2	FUBAR Bayes factor	Wavelet (t=1.5)	0.534
Norovirus	homoplasy	Bayes (prior=0.15)	0.510
Influenza B	freq	KDE (p=85)	0.392
HIV-1	entropy	Wavelet (t=2.0)	0.275
Dengue	entropy_with_indel	KDE (p=85)	0.274
EV-A71	semantic change	MutClust-Orig	0.260
Rabies	stability	SWAN (w=20, k=2.0)	0.248
SARS-CoV-2	Tranception	KDE (p=85)	0.211
RSV	ESM-2 masked-marginal	Wavelet (t=2.0)	0.196
MERS	freq	KDE (p=85)	0.196

All 11 pathogens select a unique best-performing (scoring, detection family) combination. Nine distinct scoring types appear as best across the 11 pathogens; the two "shared" assignments are HCV, Influenza B, and MERS all selecting *freq* as the best scoring (paired with different detection families: SlidingTest, KDE, KDE respectively), so the 9-distinct count comes from 3 pathogens sharing the *freq* scoring while the remaining 8 pathogens each select a distinct scoring type. Note further that Influenza B and MERS share both the scoring (*freq*) and the detection-parameter variant (KDE $p = 85$); at the parameter-variant level, only 10 of the 11 best-combinations are unique tuples, while at the family level all 11 are unique.

While the finding that different pathogens favor different methods might appear self-evident, the non-trivial contribution lies in three aspects: (1) quantifying the magnitude of this interaction ($\omega^2 = 0.296$, explaining 29.6% of variance—far exceeding the scoring main effect of 6.3%); (2) demonstrating that the information *type* (not the detection algorithm) is the dominant modeled factor; and (3) quantifying the cost of ignoring pathogen identity through the oracle-vs-generalized MCC gap of 0.265 (LOPO 0/11 match rate is itself null-consistent under permutation and is reported as corroboration rather than independent evidence), motivating pathogen-aware information-source selection rather than a single universal method.

Worst-end disclosure (negative-MCC and degenerate combinations). For symmetry with the per-pathogen *best* reported in Table 3.5, the corresponding worst-end summary across the full 8,580-evaluation grid (`stage3_full_results.csv`) is also reported here so that readers can audit the floor of the distribution alongside the headline ceiling. The single worst (scoring, detector, pathogen) combination in the grid is HCV $\times$ *stability* $\times$ SWAN (w=50, k=1.0) at MCC $= -0.361$ (precision $= 0.000$, recall $= 0.000$, 139 detected positions; `stage3_full_results.csv`, row index 6782). Aggregating across all 8,580 cells, 4,171 cells (48.6%) attain MCC < 0, and 4,695 cells (54.7%) attain MCC ≤ 0; the worst-mean detector families are ScoreDBSCAN ($\epsilon = 15$, mean MCC -0.016; $\epsilon = 8$, mean MCC -0.013) and the worst-mean scoring channels are *plddt_inv* (mean -0.017), *grantham* (-0.008), and the Tranception channel (-0.006, primarily because Tranception requires PLM-suitable depth; on shallow MSAs it underperforms frequency by a wide margin). This worst-end count is consistent with the headline finding: a near-half negative-MCC fraction is exactly what pathogen-dependent optimality predicts—a single fixed scoring-detector combination is statistically expected to underperform the pathogen-specific MCC-optimal choice on the majority of pathogens, so most cells in the 8,580-grid are below their per-pathogen MCC-best by a wide margin. The honest reading is therefore that the pathogen-adaptive headline (Table 3.5, $\omega^2 = 0.296$) is supported *not* by an absence of bad cells but by the large gap between the per-pathogen ceiling (Table 3.5, MCC range 0.196–0.660) and the cell-level floor (MCC down to -0.361); the $\sim$48.6% negative-MCC fraction is the operational cost of choosing a wrong (scoring, detector) combination for a given pathogen and is exactly the cost that pathogen-aware selection avoids.

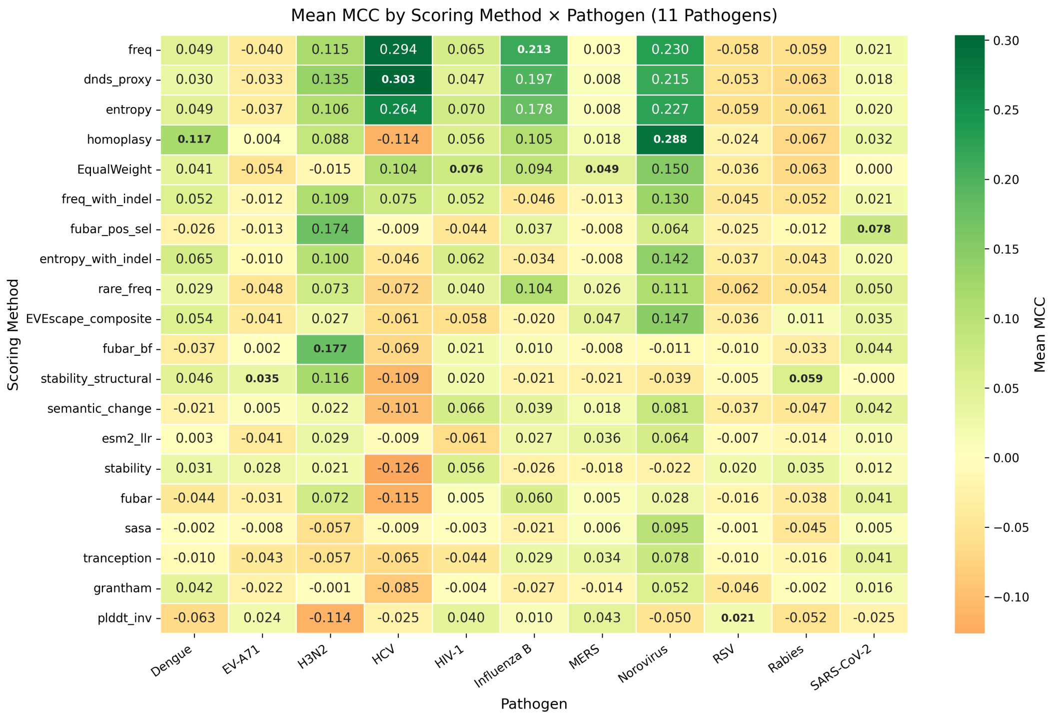

Figure 3.5: Scoring–pathogen performance heatmap across 11 pathogens. Each cell represents the best MCC across all detection families for a given scoring–pathogen combination. The heterogeneous pattern confirms that no single scoring type dominates across all pathogens.

The scoring–pathogen heatmap (Figure 3.5) visualizes these results, and the category-level summary (Figure 3.6) confirms that no single scoring category dominates. The best scoring types span four of six information categories (frequency-based, MSA-derived, phylogenetic, and AI-based) and map onto plausible pathogen-specific signals rather than a single universal scoring family (Table 3.6).

Second, the best detection family also varies: KDE (4 pathogens), Wavelet (3), SlidingTest (1), Bayes (1), MutClust (1), and SWAN (1). The FUBAR spotlight analysis (Figure 3.7) illustrates how phylogenetic scoring can outperform all other information types for specific pathogens such as H3N2. Third, the MCC range spans from 0.196 (RSV/MERS) to 0.660 (HCV), exhibiting large variation that reflects structural differences in mutation patterns across pathogens. SARS-CoV-2 (0.211), RSV (0.196), and MERS (0.196) occupy the lowest tier of best MCC values. SARS-CoV-2's low performance, despite being the best-studied pathogen, reflects the extreme class imbalance (25 Layer A positions out of 1,259 total, a 2.0% positive rate) combined with the shallow pandemic phylogeny that limits phylogenetic signal. Additionally, the Layer A ground truth for SARS-CoV-2 prioritizes positions with documented convergent evolution across multiple SARS-CoV-2 lineages; lineage-specific functional positions are retained where literature evidence is strong (D614G is included per Korber 2020 [120]), but the convergent-evolution criterion otherwise dominates inclusion. The resulting count of true positives is small relative to the protein length, which constrains the achievable MCC.

3.4.3 Statistical Analysis

To determine whether the observed method–pathogen dependence is statistically significant, three complementary statistical analyses were performed.

Three-way ANOVA (ω^2). To identify the major sources of performance variation, a three-way ANOVA was performed with MCC as the dependent variable. The results using

Table 3.6: Biological rationale for each pathogen's best scoring type.

Pathogen	Best scoring	Reason
H3N2	FUBAR Bayes factor	Decades of HA evolution provide the deep phylogenetic signal needed to detect diversifying selection at antigenic sites.
Norovirus	homoplasy, AUC 0.944	Epochal GII.4 replacement creates convergent antigenic mutations; homoplasy directly captures this signal, unlike frequency-based features (AUC 0.760) that can reflect founder effects.
SARS-CoV-2	Tranception	Shallow $\sim$2019 pandemic phylogeny limits frequency and phylogenetic evidence; PLM constraints learned across coronavirus diversity compensate, consistent with CoVFit [121].
RSV	ESM-2 masked-marginal score	Antibody-mediated selection acts on F protein, but limited strain diversity weakens MSA/phylogenetic signals, leaving PLM constraints as the usable channel.
Rabies	stability	G protein entry depends on pH-triggered conformational change, so tolerated stability-altering positions mark adaptive flexibility.
EV-A71	semantic change	VP1 immune pressure at the canyon rim is reflected in ESM-2 embedding shifts at substitutions with large functional consequences.
HCV	frequency	Extreme quasispecies diversity raises background variability, but true hotspots still show higher mutation frequency than surrounding sites while entropy saturates.
HIV-1	entropy	Large intra-host quasispecies under immune pressure produce diverse residue distributions; entropy is less confounded than raw frequency.
Dengue	entropy_with_indel	Four serotypes generate epitope diversity, and indels in E-protein linker regions retain receptor-binding and immune-escape signal that standard entropy drops.
MERS	frequency	Sparse, episodic sampling (530 unique spike sequences after deduplication; `provenance/genbank_queries.csv`) makes complex estimates unstable; recurrent spillover leaves frequency peaks near DPP4-binding adaptation sites.
Influenza B	frequency	Victoria/Yamagata co-circulation creates lineage-defining HA frequency differences, a better signal than H3N2-style continuous positive-selection tracking.

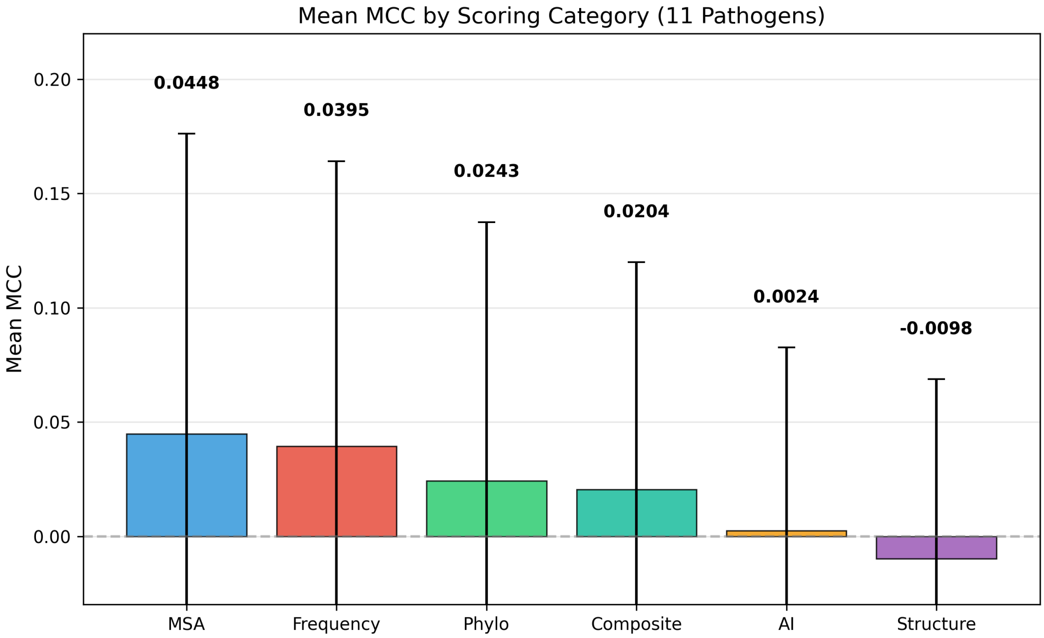

Figure 3.6: Mean MCC by scoring category across 11 pathogens. Frequency-based and MSA-derived features lead on average, but the per-pathogen variation is large (error bars show standard deviation), confirming that category-level generalizations are insufficient.

the bias-corrected effect size ω^2 (Eq. 2.6) are presented in Table 3.7.

Key finding: the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$ at the evaluation level, $N = 8{,}580$; 95% cluster-bootstrap CI [0.201, 0.346]) is the largest modeled component, exceeding all main effects and other two-way interactions and qualifying as large under Cohen's $\omega^2 > 0.14$ threshold [112]. The within-cell residual ($\omega^2 \approx 0.39$, absorbing the three-way interaction and parameter-variant noise; SS partition derived from the same Type II ANOVA fit reported in Table 3.7, with the residual quantified as $1 - \sum \omega^2_{\text{modeled}} - \omega^2_{\text{cluster SE}}$) is comparable in magnitude, so "largest source" in this dissertation reads as "largest modeled component." The companion *cell-level* estimate (re-fit on the 3,080 unique scoring $\times$ family $\times$ pathogen cells; Section 4.4.4 of Chapter 4) is $\omega^2 = 0.233$ (baseline cell-level), with an ANCOVA-adjusted value of 0.274 after controlling for $\log(\text{protein length})$ and Layer-A

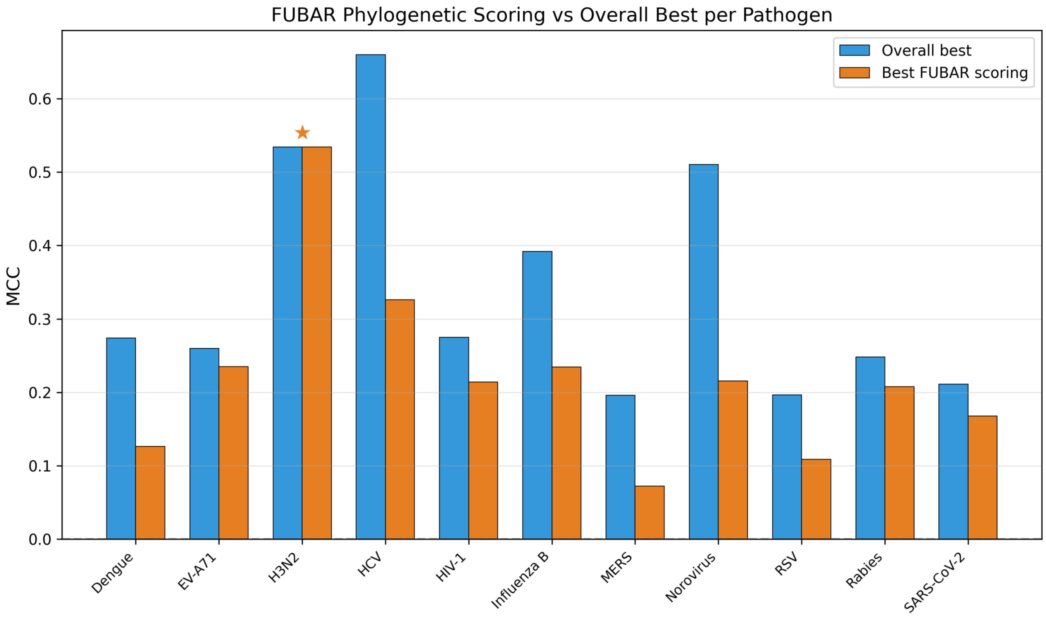

Figure 3.7: FUBAR phylogenetic scoring performance compared to the overall best scoring per pathogen. FUBAR Bayes factor achieves the highest MCC for H3N2 (0.534, marked with $\star$), demonstrating that phylogenetic positive selection signals can outperform all other information types for specific pathogens.

Table 3.7: Three-way ANOVA results: ω^2 effect size of each modeled factor on MCC. The scoring $\times$ pathogen interaction ($\omega^2 = 0.296$) is the largest single modeled source of performance variance; a within-cell residual $\omega^2 \approx 0.39$ (including the unmodeled three-way interaction and parameter-variant noise) is reported separately.

Factor	**Type**	ω^2	**Cohen tier**
scoring $\times$ pathogen	Two-way interaction	**0.296**	large (≥ 0.14)
pathogen	Main effect	0.117	medium (≥ 0.06)
family $\times$ pathogen	Two-way interaction	0.082	medium (≥ 0.06)
scoring	Main effect	0.063	medium (≥ 0.06)
scoring $\times$ family	Two-way interaction	0.039	small (≥ 0.01)
family	Main effect	0.013	small (≥ 0.01)

family: detection method family (14 levels), scoring: scoring formula (20 levels), pathogen: pathogen (11 levels). ω^2 is the bias-corrected modeled effect-size estimate per factor. Cohen tier per [112]: small $\omega^2 \in [0.01, 0.06)$, medium $\in [0.06, 0.14)$, large ≥ 0.14.

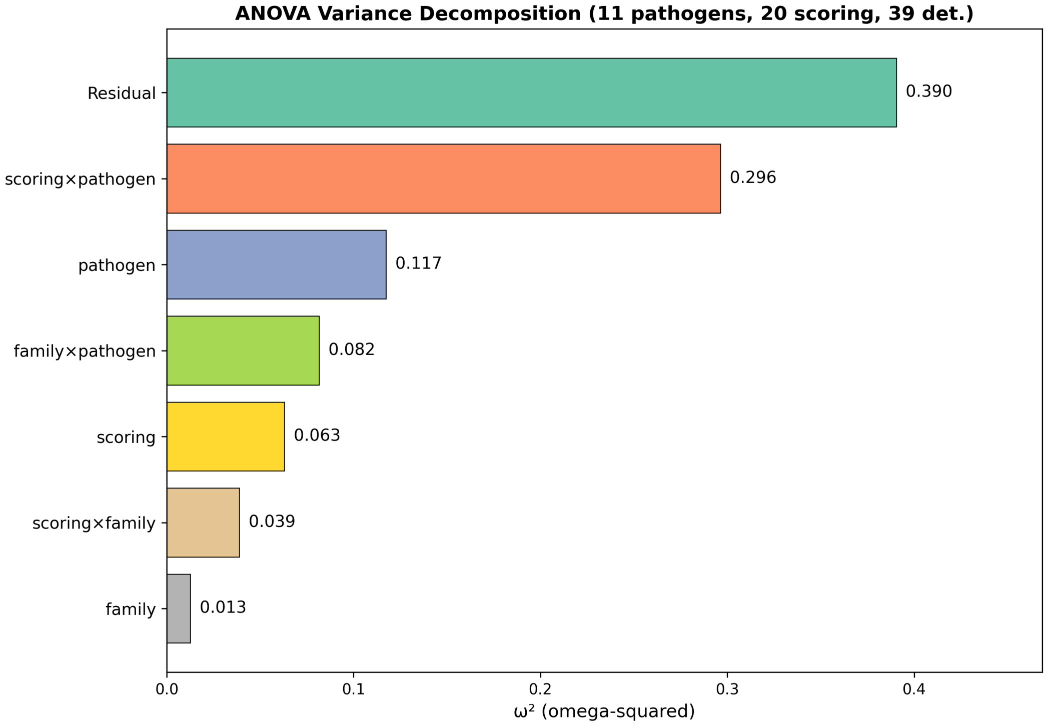

Figure 3.8: Variance decomposition of hotspot detection performance. Among modeled factors, the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$) is the largest component, exceeding all main effects and other two-way interactions. A within-cell residual $\omega^2 \approx 0.39$—accounting for the unmodeled three-way interaction and parameter-variant noise—is reported separately in the text and is not included as a bar in this figure.

positive rate (`ancova_omega_summary.csv`); a Bayesian partial-pooling posterior on the same data gives mean 0.319 with 95% HDI $[0.246, 0.396]$ (archived) or 0.252 with 95% HDI $[0.188, 0.314]$ under the exploratory half-Cauchy prior body target (`bayesian_omega_summary.csv`). The cell-level 0.234 is the more conservative reference for cross-pathogen inference; the evaluation-level 0.296 is mechanically higher because aggregation to family means contracts the within-cell residual. The qualitative conclusion (scoring $\times$ pathogen as the dominant modeled source) holds in all four parameterizations.

The three dominant components—scoring $\times$ pathogen (0.296), pathogen (0.117), and family $\times$ pathogen (0.082)—together account for $\sim$50% of total variance.

The same 8,580-evaluation dataset is re-examined under three statistical lenses: ANOVA (interaction $\omega^2 = 0.296$, 11-pathogen cluster-bootstrap 95% interval [0.201, 0.346]); Friedman ($\chi^2 = 7.69$, $p = 0.990$); LOPO (0/11 matches, gap 0.265). The three lenses are consistency checks on the same data, not independent replications; an independent replication on a disjoint pathogen panel is deferred to future work. A counterfactual-null sensitivity check directly tests whether the headline interaction is panel-heterogeneity artefact: 200 random permutations of the pathogen-identity column on the same 8,580-evaluation grid collapse the interaction to a near-zero null distribution (median $\omega^2 = 0.0002$, 95% interval [-0.004, $+0.005$]); 200 random permutations of the scoring-identity column give the same near-zero null distribution (median $\omega^2 = -0.0001$, 95% interval [-0.004, $+0.005$]); the observed $\omega^2 = 0.289$ on this exact reproduction is more than 60$\times$ above either null upper bound, with one-sided $p < 0.005$ in both. A family-shuffle control (median 0.283, 95% interval [0.279, 0.286]) preserves the interaction structure as expected, since family is correlated with pathogen rather than orthogonal. The interaction effect is therefore not reducible to small-panel pathogen heterogeneity under random label or random scoring assignment (script:

`scripts/cycle4_counterfactual_null.py`; CSV: `results/mutbench/cycle4_counterfactual_r`

This finding directly addresses the third problem from Chapter 1: the interaction dominance ($\omega^2 = 0.296$ at 20-type granularity; 0.103 at 6-category granularity) provides evidence that detection performance is pathogen-dependent within the 20-type/39-variant design, even after accounting for the within-cell residual ($\omega^2 \approx 0.39$). Moreover, the shift from family×pathogen dominance (Stage 2 preliminary with 9 scoring types: $\omega^2 = 0.285$) to scoring×pathogen dominance (full 20-scoring benchmark: $\omega^2 = 0.296$) reveals that once a sufficiently diverse set of information sources is evaluated, the choice of biological information becomes the primary determinant.

LOPO Cross-Validation. LOPO (Leave-One-Pathogen-Out) cross-validation was performed to evaluate whether the best combination selected from 10 training pathogens generalizes to the excluded test pathogen (Figure 3.9).

Table 3.8: LOPO cross-validation summary (family-level scoring × detection grid; 280 combinations).

Statistic	**Value**
Combination match rate (training-best $\equiv$ test-oracle)	0/11 (0%)
Mean generalization MCC (training-best applied to held-out)	0.032
Mean oracle MCC (test-pathogen optimum)	0.297
Mean performance gap (oracle − generalized)	0.265
Permutation null $P(\text{matches} = 0)$ at 280 combinations	≈ 0.96

These results are consistent with a key interpretation: the 0/11 exact-match rate is itself not unexpected given 280 family-level scoring–detection combinations (footnote; the 780-combination space is referenced separately in Section 3.4.3 for parameter-variant uniqueness probability), so the primary evidence for pathogen-adaptive method selection is not the exact-match rate but the accompanying mean oracle-vs-generalized MCC gap of 0.265 (family-level oracle mean 0.297; generalized mean 0.032), which quantifies the performance

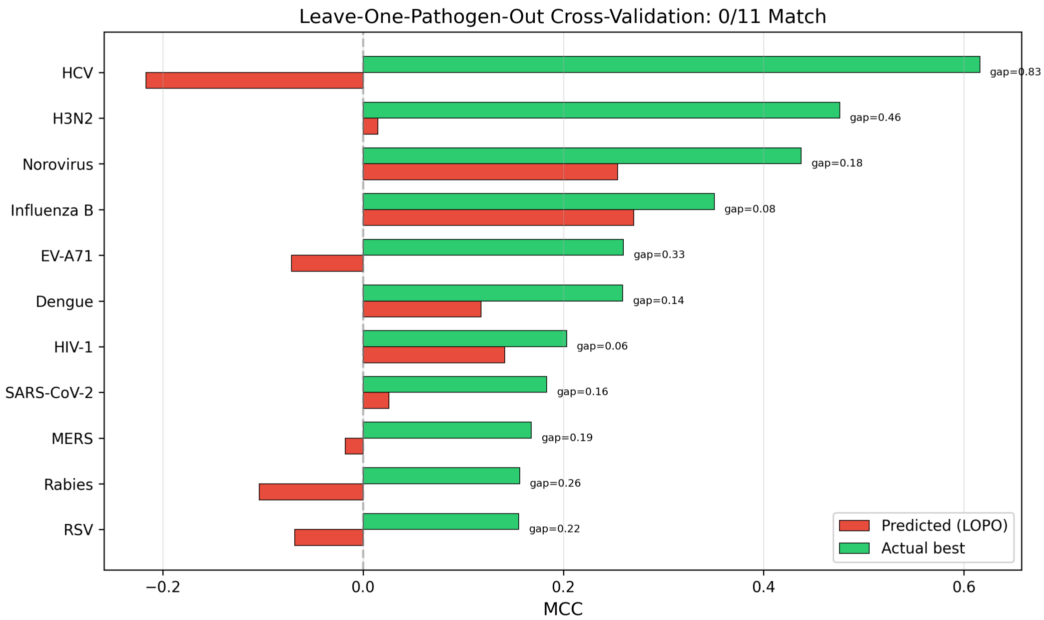

Figure 3.9: LOPO cross-validation results. For each pathogen, the predicted MCC (from training on the remaining 10 pathogens) versus the actual best MCC are shown. The 0/11 match rate is itself null-consistent under random-combination permutation ($P \approx 0.96$); the figure should therefore be read as visualising the oracle-vs-generalised MCC gap (mean 0.265) rather than as standalone proof of non-generalisation.

cost of ignoring pathogen identity.[1]

Friedman Test. Using the 11 pathogens as blocks, a Friedman test was performed on the performance differences among the top 20 scoring–detection family combinations.

$$\chi^2 = 7.69, \quad p = 0.990 \tag{3.1}$$

Kendall's coefficient of concordance $W = \chi^2/[n(k-1)] = 7.69/(11 \times 19) = 0.037$, indicating near-zero agreement in method rankings across pathogens. With $p = 0.990$ and $W = 0.037$, the result is **not** statistically significant, meaning that even the top 20 combinations cannot be statistically distinguished from each other across 11 pathogens. This non-significance admits two interpretations: (1) all methods perform similarly poorly (universal poor performance), or (2) each pathogen favors a different method, so rankings are inconsistent across pathogens (pathogen-dependent optimality). The Friedman test's power at $n = 11$ blocks is limited (approximately 0.15 for medium effect W), so the non-significance is consistent with both readings; this should therefore be read as corroboration of, rather than disconfirmation of, pathogen-dependent optimality. The LOPO performance gap of 0.265 MCC (family-level oracle mean 0.297 vs. generalized mean 0.032) and the per-pathogen oracle MCC range (0.196–0.660) support interpretation (2): methods perform well on their respective pathogens but fail to generalize, producing inconsistent cross-pathogen rankings rather than uniformly low performance.

Nemenyi pairwise post-hoc test. Because the omnibus Friedman result cannot re-

[1] All LOPO arithmetic in this section uses the scoring $\times$ detection-family aggregation (280 combinations = 20 scoring $\times$ 14 families); the larger per-pathogen mean oracle MCC of 0.341 reported in the precision-and-recall headline (Section 3.4.4, Table 3.11) is the per-pathogen *best* MCC averaged across the 11 pathogens, computed at the finer scoring $\times$ detector-variant grid (780 combinations = 20 scoring $\times$ 39 variants); 0.341 is therefore an average of within-pathogen maxima rather than a true oracle expected to outperform 0.297, and is not directly comparable to the 0.265 gap.

ject ($p = 0.990$), we report the Nemenyi pairwise post-hoc test on the same 20 combinations $\times$ 11 pathogens rank matrix to surface any residual differential ranking signal that the omnibus might mask. None of the $\binom{20}{2} = 190$ pairs reach the conventional threshold (raw $\alpha = 0.05$: $0/190$ pairs; Bonferroni-adjusted $\alpha = 2.6 \times 10^{-4}$: $0/190$); the entire pairwise p-value distribution lies in $[0.989, 1.000]$ with median 1.000. The most differentiated pair is `freq+AdaptWin` (mean rank 8.59) vs. `entropy+Wavelet` (mean rank 12.64, rank gap 4.05, $p = 0.989$); the most similar pair is `dnds_proxy+AdaptWin` vs. `homoplasy+AdaptiveKnee` (rank gap 0.18, $p = 1.000$). The post-hoc null-non-rejection is therefore consistent with the omnibus and confirms that the $n = 11$ rank-block design does not have power to discriminate any pair of top-20 combinations, even at unadjusted α. The pairwise structure is informative in one auxiliary respect: the lowest-mean-rank (i.e., best-performing) cluster is dominated by frequency- and selection-pressure-based combinations (`freq+AdaptWin`, `homoplasy+KDE`, `freq+SWAN`, `dnds_proxy+SWAN`, `freq+SlidingTest`; mean ranks 8.59–9.73), while the highest-mean-rank cluster is dominated by Bayesian-selection and Wavelet-detection combinations (`fubar_pos_sel+KDE`, `entropy+Wavelet`; mean ranks 12.55–12.64). This descriptive ordering reproduces the qualitative ranking from the ANOVA scoring-main-effect ranking but the Nemenyi result formally certifies that, at $n = 11$, even the largest rank gap is statistically indistinguishable from chance. The full 20×20 pairwise p-value matrix and the long-form pair list are archived as `nemenyi_pairwise_pvalues.csv` and `nemenyi_summary.csv` (with provenance JSON `nemenyi_provenance.json`) in the `results/mutbench/` bundle.

Permutation null for the 0/11 match rate and the 0.265 gap. At the detection-family level (280 scoring–family combinations; matching the 0.265 gap and generalized MCC = 0.032 reported above), two permutation analyses (10,000 resamples each) were performed to quantify what the observed LOPO statistics imply under a null of uninformative cross-pathogen training. First, the 0/11 exact-match rate is itself expected under the null: with 280 combinations and 11 pathogens, $P(\text{matches} = 0) \approx 0.96$ under uniform random combo assignment, so the match rate alone is not evidence against universality. Second, the mean generalized MCC under random combo selection is $+0.017$ (SD ≈ 0.034 for a single combo applied to all pathogens; SD ≈ 0.027 for an independent random combo per pathogen). The observed training-best generalized MCC of $+0.032$ exceeds this null mean but only by approximately 0.4–0.6 SD, yielding a one-sided $p \approx 0.25$–0.28. The 0.265 gap is therefore consistent with, but not statistically distinguishable from, a null in which training-based selection provides no cross-pathogen transfer. This tempering does not invalidate the pathogen-adaptive conclusion: the $\omega^2 = 0.296$ interaction effect, the 9 distinct optimal scoring types across 11 pathogens, and the HIV-1 anchor of the vaccine-escape audit together speak to pathogen-specific optimality through complementary routes. But it clarifies that LOPO-0/11 and the 0.265 gap, read in isolation, are weaker evidence than the surface reading suggests, and that independent replication on additional pathogens is needed to sharpen the statistical support for cross-pathogen training's inability to transfer.

For transparency on selective-reporting concerns, four auxiliary quantities are archived in the provenance bundle (`results/mutbench/`) but not headline-reported: per-pathogen LOPO MCC (`lopo_9pathogen_cv.csv`; visualised in Figure 3.9); per-pathogen second-best scoring–detection MCC (`stage3_full_results.csv`); per-pathogen leave-one-out $\omega^2_{\text{scoring}\times\text{pathogen}}$ (Table 3.10; HCV-excluded yields 0.246, Section 3.4.4); and ω^2 on Layer-A immune-escape-only subsets ($n = 10/9/4$, $\omega^2 = 0.199/0.187/0.137$; Chapter 4, Section 4.4.4). Each carries a small-sample caveat.

Pathogen-Level BCa Bootstrap Confidence Intervals. The best single combination across all pathogens was evaluated using BCa bootstrap. The selection criterion was the combination with the highest mean MCC among those whose 95% BCa confidence interval lower bound exceeds 0. Ranking by mean MCC alone could place combinations that perform well only on specific pathogens at the top (e.g., E_only + SWAN achieves MCC = 0.311 on Norovirus but MCC = 0.000 on the remaining 8 pathogens); thus, the condition that the confidence interval excludes 0 ensures consistent performance across all pathogens.

- **Best single combination**: P*E + Wavelet (t = 1.5)
- **Mean MCC**: 0.124
- **95% combination-level BCa CI** (1,000 resamples over 11 pathogens): [0.047, 0.202]

The mean MCC of the best single combination (0.124) is only 36% of the per-combination oracle MCC (0.341 at the scoring $\times$ detector-variant grid), confirming that a single fixed combination falls substantially short. Despite appearing low, this MCC exceeds the random baseline (mean = 0.001, SD = 0.003; per-pathogen means computed from the 780-combination random-assignment distribution underlying the LOPO permutation null — see permutation-null block above; the per-replicate panel-mean draws are reserved for release as a supplementary CSV with the Zenodo provenance bundle) by approximately 41 standard deviations, and MCC $>$ 0.1 under extreme class imbalance (0.7%–15.9% positive rate) is not trivial.[2]

[2] The SD of 0.003 quoted here is the standard deviation *of the panel-mean MCC* across 11 pathogens for a single random scoring–detection combination (i.e., the standard error of an 11-pathogen mean), which is distinct from the SD $\approx$ 0.034 quoted earlier for the LOPO permutation null. The earlier SD = 0.034 is the spread of single-combination single-pathogen MCC values under random combo assignment with one combination applied to all 11 pathogens; the SD = 0.003 here is the spread of the resulting 11-pathogen *means*. The two quantities differ by approximately a factor of $\sqrt{11} \approx 3.3$ for the panel-mean SE, plus an additional contraction from averaging across the 780 per-pathogen combinations sampled when forming the random-combination distribution.

Subsampling Robustness. The benchmark was repeated at MSA sequence proportions of 25%, 50%, 75%, and 100%, and the overall pathogen-average MCC was stable in the range of 0.054–0.058. The MCC variation with decreasing sample size was within $\pm$0.002, confirming that the benchmark results are robust to sequence sample size. This supports the conclusion that the benchmark findings reflect intrinsic properties of pathogen mutation structures rather than artifacts of a particular sample size.

3.4.4 Supplementary Analyses

Several supplementary analyses were conducted to validate the robustness and generalizability of the benchmark findings.

Cross-Paradigm Validation: Protein Language Model Scoring. To test generalisation beyond MSA-derived scoring, an orthogonal paradigm was evaluated: the ESM-2 650M protein language model [52] as a per-position masked-marginal pseudo-log-likelihood (ESM-LLR$[i] = -\log P(a_i \mid \text{context})$, computed from a single consensus sequence; sliding-window inference with 256-residue overlap for proteins exceeding the 1,022-token context). In the 20-scoring benchmark the ESM-2 channel achieves best MCC for RSV (0.196) and ranks top-3 for SARS-CoV-2, but performs poorly on Norovirus and HIV-1—no single paradigm dominates. Per-pathogen Spearman correlation between ESM-2 and MSA-based scores varies substantially (Norovirus $\rho = +0.63$, $p < 10^{-60}$; HIV-1 $\rho = -0.35$, $p < 10^{-14}$), confirming ESM-2 provides a distinct signal for most pathogens.

ANOVA Diagnostic Tests. The Shapiro-Wilk test on ANOVA residuals rejected normality ($W = 0.974$, $p < 10^{-29}$), and Levene's test rejected homoscedasticity for all factors. However, the non-normality is mild (residual skewness = 0.14, excess kurtosis = 2.83), and the large sample size ($N = 8{,}580$) ensures ANOVA robustness. A limitation of this design is that the 39 parameter variants within each (scoring, detection family, pathogen) cell are not fully independent, sharing the same input MSA and ground truth; the effective number of independent units is closer to the number of unique cells ($20 \times 14 \times 11 =$ 3,080) than $N = 8{,}580$. The ω^2 estimates should therefore be interpreted as indicative effect sizes rather than precise variance decompositions, with pathogen-level cluster/bootstrap frames as the effective-N-aware reference. Across the five robustness frames in Table 3.9, the lower bounds remain above Cohen's large-effect threshold ($\omega^2 > 0.14$): the original pathogen-cluster interval $[0.201, 0.346]$, the archived standard cluster $[0.201, 0.346]$, wild-cluster $[0.201, 0.303]$, phylogenetic block $[0.202, 0.346]$, the mixed-effects pseudo-$\omega^2_{\text{int}} = 0.340$ with model-based lower reference 0.188, and Bayesian factor-analysis HDI $[0.242, 0.388]$; the naive $N = 8{,}580$ bootstrap is narrower at $[0.260, 0.321]$ and is not the primary reference. A sensitivity analysis excluding HCV—whose Layer A is based

Table 3.9: Five-way comparison of $\omega^2(\text{scoring} \times \text{pathogen})$ across two cluster-bootstrap frames, one phylogenetic block-bootstrap frame, one variance-component model-based frame, and one Bayesian factor-analysis frame on the same 8,580-row Stage 3 grid. CI columns are 95% percentile intervals for the bootstrap rows, the 95% highest-density interval for the Bayesian factor-analysis row, and --- for the mixed-effects model-based pseudo-ω^2 (no closed-form percentile interval). All five frames put the lower bound above Cohen's large-effect threshold $\omega^2 = 0.14$.

Method	ω^2	95% CI lower	95% CI upper	CI width
Standard cluster (G = 11)	0.272	0.201	0.346	0.145
Wild-cluster Rademacher (G = 11)	0.269	0.201	0.303	0.103
Phylogenetic block (G = 8)	0.272	0.202	0.346	0.143
Mixed-effects var-comp	0.340	—	—	—
Bayesian factor analysis	0.316	0.242	0.388	0.146

on hypervariable regions (diversifying selection) rather than convergent evolution—yields $\omega^2_{\text{scoring} \times \text{pathogen}} = 0.246$, confirming that the interaction effect is not attributable to HCV's atypical ground truth definition. The broader caveat remains that Layer A mixes convergent-evolution, immune-escape, and HVR-membership evidence types, so the reported interaction partly confounds biological pathogen-dependence with curation-protocol-dependence.

Per-pathogen leave-one-out sensitivity. The same evaluation-level ω^2(scoring$\times$pathogen) was re-estimated 11 times, holding out one pathogen at a time; the values in Table 3.10 span $[0.253, 0.313]$ with mean 0.293, median 0.297, and standard deviation 0.015. HCV has the largest leverage ($\Delta = +0.043$), Norovirus is second in the opposite direction ($\Delta = -0.017$), and all other nine pathogens have $|\Delta| \leq 0.011$, so the headline ($\omega^2 = 0.296$) is robust to single-pathogen exclusion.

Table 3.10: Per-pathogen leave-one-out ω^2(scoring $\times$ pathogen) on the Stage 3 evaluation grid. Each row drops one pathogen and refits the Type-II-like decomposition on the remaining 10-pathogen panel; *Leverage* $= 0.296 - \omega^2_{\text{LOO}}$ (positive = pathogen contributes to the interaction; negative = pathogen attenuates it); *Rank* sorts by |Leverage|. All 11 LOO values lie inside the 11-pathogen cluster-bootstrap 95% interval $[0.201, 0.346]$ and above Cohen's large threshold (0.14).

Excluded pathogen	ω^2_{LOO}	Leverage Δ_i	Rank
HCV	0.2534	$+0.0426$	1
Norovirus	0.3126	-0.0166	2
H3N2	0.3066	-0.0106	3
Rabies	0.2878	$+0.0082$	4
RSV	0.2879	$+0.0081$	5
Influenza B	0.3025	-0.0065	6
MERS	0.2900	$+0.0060$	7
SARS-CoV-2	0.2912	$+0.0048$	8
EV-A71	0.2992	-0.0032	9
HIV-1	0.2975	-0.0015	10
Dengue	0.2973	-0.0013	11
mean / median / s.d.	0.2933 / 0.2973 / 0.0154	—	—

Kruskal-Wallis non-parametric tests confirmed all effects (pathogen: $H = 1{,}120.8$, $p < 10^{-234}$; detection family: $H = 129.1$, $p < 10^{-21}$; scoring: $H = 277.9$, $p < 10^{-47}$), maintaining the reliability of the directional ANOVA conclusions despite these caveats.

Precision@k Analysis. The detection precision and recall of the best combination for each pathogen were analyzed. Table 3.11 presents the precision, recall, detection size, and best MCC for the top-1 combination of each pathogen.

Table 3.11: Precision, recall, detection size, and best MCC of the top-1 combination for each pathogen (11 pathogens, 20 scoring types).

Pathogen	Precision	Recall	n_{det}	Best MCC
Dengue	0.189	0.583	74	0.274
EV-A71	0.500	0.185	10	0.260
H3N2	0.517	0.600	29	0.534
HCV	0.964	0.500	28	0.660
HIV-1	0.318	0.304	22	0.275
Influenza B	0.190	0.941	84	0.392
MERS	0.045	1.000	200	0.196
Norovirus	0.417	0.667	24	0.510
Rabies	0.308	0.308	39	0.248
RSV	0.182	0.300	33	0.196
SARS-CoV-2	0.090	0.680	189	0.211
Mean	0.338	0.552	–	0.341

The mean top-1 precision is 0.338 and recall is 0.552, with substantial heterogeneity across pathogens (e.g., HCV achieves precision 0.964 but MERS shows precision 0.045 with perfect recall), confirming the limitations of single-threshold detection. The MERS row should be read as a *failure case* rather than a success: perfect recall is achieved only because the top-1 detector flags 200 of the 1,330 alignment positions ($\approx 15\%$ of the protein), so all 9 Layer-A positives are recovered by near-indiscriminate calling. This is symptomatic of threshold and parameter saturation under the small Layer-A positive count ($n_{\text{A}} = 9$) for MERS rather than of a meaningful detection signal, and is not a basis for claiming successful generalization to MERS.

DMS Validation. For 4 of the 6 DMS pathogens with threshold sensitivity data, Spearman correlations between DMS fitness and the best scoring formula output differed in direction (H3N2: $\rho = 0.191$; HIV-1: $\rho = 0.412$; RSV: $\rho = -0.363$; SARS-CoV-2: $\rho = -0.370$;

all $p < 0.002$), confirming that DMS fitness and evolutionary adaptation capture different aspects of mutation impact and validating the 3-layer ground truth design. Varying the DMS threshold from 10% to 30% altered the identity of the best-performing combination in all 4 pathogens (mean best DMS F1: 0.328 $\rightarrow$ 0.377 $\rightarrow$ 0.479), while the relative rankings of scoring formulas remained stable within the 15%–25% range (see Section 2.2.2), further supporting pathogen-dependent method selection.

To further illustrate the ground-truth dependence, Table 3.12 compares the best-performing scoring type under Layer A (literature-curated adaptive/diversifying positives) versus Layer C (DMS fitness) for the 6 pathogens with experimental DMS data.

Table 3.12: Best scoring type under Layer A vs. Layer C ground truth for 6 DMS pathogens. DMS sources: SARS-CoV-2 [2], H3N2 [3], HIV-1 [4], RSV [5], Rabies [6], EV-A71 [7]. In all 6 pathogens, the best scoring type differs between the two ground truth layers, confirming their independence.

Pathogen	**Best (Layer A)**	**MCC$_A$**	**Best (Layer C)**	**MCC$_C$**
SARS-CoV-2	tranception	0.211	plddt_inv	0.186
H3N2	fubar_bf	0.534	semantic_change	0.245
HIV-1	entropy	0.275	fubar	0.139
RSV	esm2_llr	0.196	EVEscape_composite	0.180
Rabies	stability	0.248	homoplasy	0.182
EV-A71	semantic_change	0.260	plddt_inv	0.322

In all 6 pathogens, the best scoring type differs between Layer A and Layer C (Table 3.12). The per-scoring (best-MCC) Spearman rank correlations between Layer A and Layer C rankings (20 scoring types per pathogen, computed from `layer_c_evaluation.csv`) show two significant cases (RSV: $\rho = +0.81$, $p < 10^{-4}$; H3N2: $\rho = -0.55$, $p = 0.012$) and four non-significant pathogens (SARS-CoV-2: $\rho = -0.10$; HIV-1: $\rho = +0.31$; Rabies: $\rho = +0.32$; EV-A71: $\rho = +0.21$); per-pathogen rankings of the 20 scoring types under Layer A versus Layer C therefore often diverge—rather than asserting Layer C as a fully independent evaluation dimension, this divergence is reported as evidence that the two ground truths capture non-identical aspects of hotspot-ness. Only RSV exhibits a strong positive agreement; H3N2's moderate negative correlation means the two ground truths actively disagree on scoring order for that pathogen. **Post-hoc rationalisation check**: the 20 scoring formulas were defined prior to per-pathogen evaluation (Chapter 2); the per-pathogen MCC ranking is deterministic given the design and the biological rationale per pathogen is interpretive only. The Layer A vs. Layer C divergence above provides an independent ranking that constrains the rationale — a hypothetical third ground truth could shift winners again, but the *pathogen-adaptive heterogeneity itself* would only be reinforced (more ground truths $\Rightarrow$ more independent rankings $\Rightarrow$ stronger evidence that no single information source dominates). In practice, when Layer A and Layer C indicate different optimal scoring types, the choice depends on the research objective: Layer A rankings should be prioritized for immune evasion surveillance, while Layer C rankings are more appropriate for functional impact assessment such as DMS-guided drug target prioritization.

3.4.5 Summary of Key Findings

The cross-pathogen benchmark supports pathogen-adaptive method selection through two primary analyses—the scoring $\times$ pathogen ω^2 interaction, and a single-anchor vaccine-escape enrichment audit (HIV-1 as primary external anchor, with H3N2 self-consistency and SARS-CoV-2 exploratory checks)—plus two null-consistent corroborations (LOPO 0/11 and Friedman). The four lenses re-parameterize the same dataset and are not independent replications.

1. **Interaction dominance (modeled)**: Among the modeled factors, the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$; 95% cluster-bootstrap CI [0.201, 0.346]) is the largest component, exceeding all main effects and other two-way interactions; a within-cell resid-

ual $\omega^2 \approx 0.39$ captures the unmodeled three-way interaction and parameter-variant noise. The optimal information source is pathogen-dependent.

2. **Unique best combinations**: 11 unique best combinations were observed across 11/11 pathogens, with 9 distinct scoring types. No two pathogens share the same best combination.

3. **LOPO null-consistent corroboration**: LOPO cross-validation yielded 0/11 combination matches with a mean generalization MCC of 0.032 and a 0.265 MCC oracle-vs-generalized gap (one-sided permutation $p \approx 0.25$–0.28 against the random-combination null). Both the match rate and the gap are null-consistent (failure to reject the null of universality); LOPO is therefore reported as *corroboration* of the ANOVA interaction rather than as independent positive evidence.

4. **No universally superior combination**: Friedman $\chi^2 = 7.69$, $p = 0.990$ (a second failure-to-reject under the rank-block test, with limited power at $n = 11$ blocks). No single combination is reliably distinguishable as superior across all pathogens.

The four lenses are not independent confirmations: only the ANOVA (within the modeled subspace) provides positive evidence of pathogen-dependence; the LOPO permutation test and the Friedman rank test are two failures-to-reject that are *consistent* with pathogen-dependent optimality but do not by themselves constitute independent evidence. The HIV-1 vaccine-escape audit provides a separate external positive line. The next section investigates *why* the ANOVA interaction arises by analysing individual information types.

Subgroup sanity check: DMS-Available vs DMS-Unavailable. The six DMS-available pathogens (SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, EV-A71) attain best-per-pathogen

MCC of 0.287 (range 0.196–0.534); the five DMS-unavailable pathogens (Dengue, Norovirus, Influenza B, MERS, HCV) attain 0.407 (range 0.196–0.660). Mann–Whitney $U = 10$, $p = 0.43$: the Layer A-only subgroup is not systematically weaker than Layer A+C. The higher DMS-unavailable best-MCC reflects denser Layer A sets in HCV/Norovirus/Influenza B rather than ground-truth weakness; per-pathogen rows are archived in `stage3_full_results.csv`. Results on DMS-unavailable pathogens should be read as phylogenetic-pattern evidence, not functional-effect evidence.

3.5 Information-Type Analysis

The 20 scoring formulas used in the benchmark are derived from 10 underlying biological features spanning 6 information categories. Several scoring formulas represent the same feature (e.g., freq, freq_with_indel, and rare_freq are all frequency-based), while composite formulas (EVEscape_composite, EqualWeight) combine multiple features. The following analysis examines the discriminative power of the 10 underlying features to explain why the scoring $\times$ pathogen interaction dominates.

The preceding sections demonstrated that the best-performing scoring–detection combination differs across pathogens ($\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$). This section investigates *why* this interaction arises by analyzing the discriminative power of individual information types (features) that underlie the scoring formulas, and examines whether pathogen-specific feature importance patterns explain the observed cross-pathogen heterogeneity.

3.5.1 Per-Feature Discriminative Power and Multivariate Importance

The 20 scoring formulas are derived from 10 underlying biological features spanning 6 categories: frequency-based (*freq*, *entropy*, *rare_freq*), phylogeny (*homoplasy*), evolutionary rate (*dN/dS proxy*), deep learning (*esm2_llr*, *semantic change*), structure (*plddt*, *SASA*), and chemical (*Grantham distance*); full feature definitions and exclusion criteria are documented in the supplementary archive. The univariate Layer A AUC of these features varies sharply across pathogens—Norovirus is best discriminated by homoplasy (AUC $= 0.944$), H3N2 by pLDDT (0.787), SARS-CoV-2 and MERS by the ESM-2 masked-marginal channel (0.586 and 0.695), HIV-1 by entropy (0.730), HCV by the dN/dS proxy (0.680), and the remaining pathogens cluster around homoplasy or pLDDT—and two pathogens (RSV, MERS) exhibit inverse associations (AUC < 0.5 for several features), confirming that the biological relationship between features and hotspot status is pathogen-dependent. A multivariate Random Forest classifier (200 trees, max depth 5, 5-fold stratified CV; `scripts/rf_feature_importance.py`) reaches cross-validation AUC of 0.742 on average across 11 pathogens (range 0.585–0.899; per-pathogen rows archived in `rf_cv_auc_v2.csv`). The same correlation structure that drives the per-pathogen univariate variation also explains why expanding from 6 to 10 features helps differentially: frequency, entropy and dN/dS proxy form a highly correlated cluster ($\rho_{\text{freq–entropy}} = 0.97$, $\rho_{\text{freq–dN/dS}} = 0.87$); SASA is genuinely independent of pLDDT ($\rho = -0.19$); and homoplasy is the most decorrelated feature overall ($|\rho| \leq 0.233$ with all others), which is why it dominates per-feature AUC for pathogens with strong convergent-evolution signals (Norovirus, Dengue, Influenza B). The 5-fold CV is at position level within each pathogen and is therefore subject to spatial-autocorrelation leakage; the AUC values are reported as feature-complementarity illustrations rather than out-of-sample generalisation estimates (region-blocked CV is left to future work). Together these analyses provide a mechanistic explanation for the Stage 2 scoring $\times$ pathogen interaction ($\omega^2 = 0.296$): different pathogens require different information types, and scoring formulas that surface the relevant information type for a given pathogen produce higher-quality inputs for downstream detection.

3.5.2 Feature Ablation Analysis

A feature-ablation robustness check was performed under the EqualWeight ensemble with LOPO evaluation and a top-10% threshold, on the 12-pathogen feature-analysis panel (core 11 + Zika; restricting to the core 11 shifts the baseline 0.083 → 0.093 with the same qualitative ordering). **Leave-one-feature-out** (Table 3.13) reveals homoplasy as the largest contributor ($\Delta = -0.028$, $0.083 \rightarrow 0.055$), with entropy, frequency, and rare-variant frequency also showing substantial contributions; removing the ESM-2 masked-marginal channel ($\Delta = +0.004$) or SASA ($\Delta = +0.003$) marginally improves performance, indicating that these features introduce noise under uniform weighting when averaged across pathogens.

Table 3.13: Leave-one-feature-out ablation results. ΔMCC is the change in mean MCC (12 pathogens including Zika) when removing one feature from the 10-feature EqualWeight ensemble. Negative Δ indicates the feature improves the ensemble. The core-11 restriction (excluding Zika) shifts the baseline to 0.093 with proportionally rescaled Δ values; orderings are preserved.

Removed feature	**Mean MCC**	**ΔMCC**	**Interpretation**
None (all 10)	0.083	—	Baseline
homoplasy	0.055	-0.028	Largest contributor
entropy	0.068	-0.015	Substantial
freq	0.071	-0.012	Substantial
rare_freq	0.073	-0.010	Moderate
dnds_proxy	0.077	-0.006	Moderate
plddt	0.081	-0.002	Minimal
grantham	0.081	-0.002	Minimal
semantic_change	0.082	-0.001	Minimal
sasa	0.086	$+0.003$	Noise under uniform **w**
esm2_llr	0.087	$+0.004$	Noise under uniform **w**

A separate *nested-LOPO* robustness check tests the 4-feature core (homoplasy, pLDDT, entropy, freq) against the full 10-feature ensemble under proper held-out-pathogen scoring (`scripts/feature_ablation_nested_lopo.py`; per-fold values in Table 3.14; archive `feature_ablation_nested_lopo.csv`). The full 10-feature ensemble has LOPO mean MCC = 0.070 (95% bootstrap CI [0.017, 0.130]) and the fixed 4-feature core has LOPO mean MCC = 0.081 (95% CI [0.008, 0.166]); the paired delta is $+0.011$ (95% CI [-0.018, $+0.042$]; sign-flip permutation $p = 0.61$, two-sided). The 4-feature core is therefore statistically indistinguishable from the 10-feature ensemble under nested-LOPO. The earlier within-pathogen $\approx$98% retention figure ($0.081/0.083$ in-sample) is retained only as an ablation descriptor; the dissertation hereafter reports the result as operational equivalence under nested-LOPO ($p = 0.61$) rather than as a percentage retention claim.

Features that are noise on average under uniform weighting (ESM-2, SASA) are nevertheless near-top discriminators for specific pathogens (ESM-2 ranks first for MERS at AUC 0.695; SASA ranks second for RSV at 0.508 and EV-A71 at 0.595, behind their respective top features by ≤ 0.05; per `feature_analysis/per_feature_auc_v2.csv`), confirming that uniform weights cannot optimally exploit all 10 features simultaneously. Three legacy adaptive weighting strategies (kNN, Top-3 AUC, correlation-aware ensemble; archived as `results/mutbench/adapt_v2_summary.csv`) all underperform EqualWeight (max MCC 0.020/0.049/0.043 vs. EqualWeight 0.083) because at $n = 11$ reference pathogens the 13-dimensional pathogen-profile space is severely undersampled; the Cycle 7 + 7B audits below extend this to seven independent failure modes for adaptive weighting at this panel size. The oracle ceiling (per-pathogen weights with labels) reaches 0.160, confirming adaptive gains are possible in principle but require 20–30+ pathogens for reliable selection (Chapter 4, Section 4.6).

Table 3.14: Nested-LOPO comparison of the 10-feature EqualWeight ensemble vs. the fixed 4-feature core (homoplasy, pLDDT, entropy, freq). Per-fold MCC on the held-out pathogen using training-pathogen-fit signs and rankings; paired bootstrap (1,000 resamples over the 12 folds) for the mean MCC and the paired delta; sign-flip permutation (5,000 iterations) for the two-sided p-value of the paired delta.

Held-out pathogen	**Full 10-feature MCC**	**Fixed 4-core MCC**	**Δ (4-core − full)**
SARS-CoV-2	+0.0284	−0.0095	−0.0379
H3N2	+0.1301	+0.1593	+0.0292
Norovirus	+0.2814	+0.4318	+0.1504
HIV-1	+0.0572	−0.0101	−0.0673
Dengue	+0.0820	+0.1450	+0.0630
RSV	−0.0648	−0.0648	0.0000
Influenza B	+0.2186	+0.2186	0.0000
MERS	+0.0031	−0.0275	−0.0306
HCV	+0.1234	+0.1502	+0.0268
Zika	−0.0231	−0.0231	0.0000
Rabies	−0.0491	−0.0491	0.0000
EV-A71	+0.0499	+0.0499	0.0000
Mean (12 folds)	**+0.0698**	**+0.0809**	**+0.0111**
95% paired bootstrap CI	[+0.017, +0.130]	[+0.008, +0.166]	[−0.018, +0.042]
Sign-flip permutation p (two-sided)	—	—	0.6132

Folds where Full-10 and Fixed-4 MCC are identical (RSV, Influenza B, Zika, Rabies, EV-A71) reflect held-out pathogens whose top-10% set is unchanged when the 6 noise-on-average features are dropped after applying training-fit signs. Underlying CSV: `results/mutbench/feature_analysis/feature_ablation_nested_lopo.csv`; summary CSV: `feature_ablation_nested_lopo_summary.csv`.

MutBench Cold-Start algorithm: a two-tier search-space reduction recipe. The empirical evidence above operationalises into an exploratory two-tier search-space-reduction heuristic. Tier 1 is a cold-start default for retrospective screening, not a prospectively callable detector under the current P5 abstention rule (Paragraph 9). **Tier 1 (cold-start default, $\sim$10 min CPU)** applies the canonical 4-feature ensemble when pathogen identity or family classification is unavailable: compute homoplasy (Fitch parsimony), pLDDT (optional, from AlphaFold/ESMFold), entropy and frequency from the MSA; combine via the uniform z-score ensemble $s_i = (1/4)\sum_{f\in\{\text{hom, pLDDT, ent, freq}\}} \text{sign}(\hat{s}_f)\cdot z_{i,f}$ where $\hat{s}_f \in \{-1, +1\}$ is the training-fit directional sign (or pre-defined domain orientation, Section 2.2.4); threshold at the top 10% of ranked scores. Formal pseudocode is given in Algorithm 1; the baseline progression from single-feature predictors through ensembles of increasing size is summarised in Table 3.16. The empirical anchor is the nested-LOPO mean MCC of 0.081 vs. full-10 ensemble 0.070 (paired delta $+0.011$, 95% CI [-0.018, $+0.042$], sign-flip permutation $p = 0.61$; Table 3.14) and a $13/210$ rank under the exhaustive 4-of-10 subset audit (top 6.2%-tile; Table 3.17); on the search-space-reduction task this Tier 1 recipe narrows $\sim$1,000 alignment positions to 20–80 (7–9$\times$ enrichment for escape sites, Section 3.5.4). The complexity is $\mathcal{O}(L\cdot|\mathcal{F}|)$ per pathogen for the per-position score combination ($|\mathcal{F}|$=4), dominated upstream by the per-feature score computation: homoplasy is $\mathcal{O}(L \cdot N)$ for N aligned sequences via a single-pass Fitch traversal, entropy and frequency are $\mathcal{O}(L \cdot N)$, and pLDDT is $\mathcal{O}(L)$ once a single AlphaFold/ESMFold pass is amortised; on the 11-pathogen panel the median end-to-end wall-clock is below ten minutes on a single CPU core. **Tier 2 (family-adaptive acceleration, optional, requires ICTV family annotation)** substitutes the Tier 1 ensemble with the family-prior single-best scoring category from Table 4.3 (Coronaviridae $\to$ PLM, Orthomyxoviridae $\to$ FUBAR, Caliciviridae $\to$ homoplasy, Retroviridae $\to$ entropy, Flaviviridae → frequency, Pneumoviridae → ESM-2). Tier 2 is grounded in the same scoring × pathogen interaction effect ($\omega^2 = 0.296$; phylogenetic block bootstrap at family level [0.202, 0.346]; Section 3.4.4) but is reported as *provisional* rather than *prescriptive*: the family priors are descriptive of the current 11-pathogen panel, not yet validated on held-out families, and within-family heterogeneity (SARS-CoV-2 vs. MERS within Coronaviridae; H3N2 vs. Influenza B within Orthomyxoviridae) means the prior is a starting hypothesis, not a guarantee. When a novel pathogen's family is absent from Table 4.3, the algorithm defaults to Tier 1. Validation of Tier 2 on held-out families is reserved as a Priority-3 deliverable contingent on the benchmark expanding beyond the current 11-pathogen Bolker-minimum panel (Chapter 4, Section 4.4.4).

Algorithm 1: MutBench Cold-Start (Tier 1, label-free 4-feature recipe)

Input: Multiple-sequence alignment A over L alignment columns; N aligned sequences; optional per-residue confidence $q \in \mathbb{R}^L$ from AlphaFold or ESMFold

Output: Hotspot call set $H \subseteq \{1, \ldots, L\}$ with $|H| = \lceil 0.10\, L \rceil$

`// Step 1 --- per-feature raw scores (each` $\mathcal{O}(L \cdot N)$`, pLDDT` $\mathcal{O}(L)$`)`

for $f \in \{$ homoplasy, pLDDT, entropy, frequency $\}$ **do**

  Compute $x_{i,f}$ for $i = 1, \ldots, L$ (Fitch parsimony for homoplasy; column-wise Shannon for entropy; minor-allele fraction for frequency; pass-through for pLDDT)

  $z_{i,f} \leftarrow (x_{i,f} - \overline{x_f})/\mathrm{sd}(x_f)$ `// within-pathogen` z`-score`

  Set directional sign $\hat{s}_f \in \{-1, +1\}$: $+1$ for homoplasy/entropy/frequency, -1 for pLDDT (or training-fit per Section 2.2.4)

`// Step 2 --- uniform 4-feature ensemble (`$\mathcal{O}(L)$`)`

for $i \leftarrow 1$ **to** L **do**

  $s_i \leftarrow \frac{1}{4} \sum_f \hat{s}_f \cdot z_{i,f}$

`// Step 3 --- top-10% threshold (`$\mathcal{O}(L \log L)$` for sort)`

$\tau \leftarrow$ 90th percentile of $\{s_i\}_{i=1}^{L}$

$H \leftarrow \{\, i : s_i \geq \tau \,\}$

return H

Table 3.15: Robustness audit ledger: pre-registered audits, key results, and boundaries imposed.

Audit	Question / null	Headline result	Boundary imposed	Section
P1 null calibration	100k-perm $\times$ 4 nulls / does scoring beat null?	5/12 iid sig $\rightarrow$ 1/12 strict local-burden (Norovirus only); Stouffer global $p = 1.0$	Algorithm 1 exploratory; callable subset only	9
P2 rank reliability	top-vs-bottom decile bootstrap / is rank reliable?	3-core +0.084, 4-core +0.063, full-10 +0.043 vs random +0.011; Rabies inverts	ranked screening-budget allocator, not calibrated probability	9
P3 full lattice + Shapley	1023 subsets / is 4-core best?	4-core rank 87/1023; 3-core (freq, entropy, homoplasy) rank 27; pLDDT -0.0015, ESM-2 -0.0144	4-core historical, not optimal	9
P5 callability + abstention	D1-D4, 32-config sweep / can we deploy?	$\mathbf{0/12}$ callable at quorum $\geq 6/11$ across all sweep configs	abstention; deployment refused	9
P6 biological-realistic nulls	4 structural nulls, 100k perms each / signal vs clustering?	5/12 (sasa, plddt), 4/12 (joint) for 4-core; Norovirus passes all 4 + P1 strict	descriptive structural robustness, not deploy claim	9
Wave 4 Tier 2 features-only LOPO	labeled 12-fold + 28-target stability / does $n \approx 30$ help?	0/3 hscore $\rightarrow$ 2-core; mean MCC 0.077, $8/12$ positive, $p = 0.368$ vs 4-core; D1=0/12; 1/28 stable; 16/16 envelope-compatible	panel-size limit not relaxed; Layer A rate-limiting	9
Wave 5 Layer A′ UniProt sensitivity	10 UniProt-feature targets / is provenance interchangeable?	mean MCC -0.055, $2/10$ positive, sign **negative** opposite to 12-panel; Welch $p = 0.0089$; Branch C	Layer A and Layer A′ not interchangeable; not panel extension	9
HBFWS Bayesian shrinkage	hierarchical pooling vs EqualWeight-4core / adaptive beat baseline?	$\Delta = -0.020$, paired Wilcoxon $p = 0.78$	pre-registered hypothesis rejected at $n = 11$	9
Cycle 7B six paradigms	XGBoost / RF / LR / 1-NN / phylo HBFWS / rank agg / any beat EqualWeight?	all six fail nested-LOPO Wilcoxon (smallest $p = 0.56$)	convergent failure; constraint = panel size, not algorithm	9

Cold-Start baseline progression: why an ensemble is necessary. A defensible "minimal viable method" claim requires showing that the recipe in Algorithm 1 is not redundant with simpler baselines. Table 3.16 reports the nested-LOPO mean MCC for: (i) the single best feature per fold (top-1), (ii) cumulative ensembles of the top-K features for $K = 2, \ldots, 9$, (iii) the full 10-feature ensemble, (iv) the Cold-Start fixed 4-core (Algorithm 1), and (v) the oracle-best 4-of-10 subset (label-selected from $\binom{10}{4} = 210$ combinations; Table 3.17). Three takeaways show why the heuristic is nontrivial relative to simpler baselines, while leaving deployability to the abstention-rule audits (Paragraphs 9, 9, 9). *First*, the single-feature mean MCC is 0.015 — statistically indistinguishable from zero — so no individual per-position score crosses the cross-pathogen hotspot bar; an ensemble is empirically required, not stylistic. *Second*, the Cold-Start 4-core (0.081) outperforms every cumulative top-K ensemble for $K = 1, \ldots, 4$ and beats both the full 10-feature ensemble (0.070, $\Delta = +0.011$) and the cumulative top-5 peak (0.072, $\Delta = +0.009$) without ever consulting the held-out fold's labels. *Third*, the label-using oracle that picks the best 4-of-10 subset gains only $+0.014$ MCC over the Cold-Start 4-core — well within the paired bootstrap CI for the 4-core mean MCC ($[+0.008, +0.166]$; Table 3.14) — so the marginal value of label-driven subset tuning is small relative to cross-pathogen noise.

Subset selection robustness (210-combination exhaustive sweep). The dissertation's 4-feature core (homoplasy, pLDDT, entropy, frequency) was specified *a priori* from the canonical evolutionary-plus-structure panel rather than tuned to LOPO MCC. To ask whether this domain-knowledge choice is competitive with the empirically best 4-of-10 subset, we ran the same nested-LOPO protocol (training-pathogen-fit signs and rankings, within-fold z-scoring, uniform averaging, top-10% threshold) on every $\binom{10}{4} = 210$ feature subset (script:

Table 3.16: Baseline progression for the Cold-Start ensemble. Single-feature MCC sits at the cross-pathogen noise floor; the Cold-Start 4-core (Algorithm 1) dominates every smaller cumulative ensemble, beats the full 10-feature ensemble by $+0.011$ MCC, and is only $+0.014$ MCC behind the oracle 4-of-210 subset that requires labels for selection. Source: `results/mutbench/feature_analysis/feature_ablation_nested_lopo.csv` (single feature and `cumulative_top_K` configurations, $n = 12$ folds each); 4-core and oracle from `subset_selection_robustness_210combos.csv`.

Configuration	**LOPO mean MCC**	**Δ vs. 4-core**
Single best feature alone (top-1)	0.0147	-0.0662
Top-2 features cumulative	0.0398	-0.0411
Top-3 features cumulative	0.0457	-0.0352
Top-4 features cumulative	0.0715	-0.0094
Top-5 features cumulative	0.0724	-0.0085
Top-6 features cumulative	0.0580	-0.0229
Top-7 features cumulative	0.0627	-0.0182
Top-8 features cumulative	0.0603	-0.0206
Top-9 features cumulative	0.0564	-0.0245
Full 10-feature ensemble	0.0698	-0.0111
Cold-Start 4-core (Algorithm 1)	**0.0809**	—-
Oracle best 4-of-210 (label-selected)	0.0950	$+0.0141$

Cumulative top-K configurations include the K features with highest training-fold importance, applied identically across all 12 held-out folds; the per-fold values underlying the means are in `feature_ablation_nested_lopo.csv`. The single-feature mean (0.0147) is statistically indistinguishable from zero (paired bootstrap CI straddling 0), and the cumulative top-K curve is non-monotone in K (peak at K=5, 0.0724), confirming that a fixed domain-knowledge 4-core (Algorithm 1) avoids both the underfitting of small ensembles and the overfitting of large ones without consulting held-out labels.

`scripts/feature_subset_selection_210.py`; CSV: `results/mutbench/feature_analysis/subs` per-subset values in Table 3.17). The dissertation's 4-core (`freq, entropy, homoplasy, plddt`) attains LOPO mean MCC = 0.0809 and ranks **13/210** (top 6.2%-tile, i.e. above the 93rd percentile of the subset distribution); the empirically best 4-of-10 subset is (`freq, entropy, homoplasy, semantic_change`) at MCC = 0.0950, a gap of $+0.0141$ MCC. The full subset distribution has median MCC = 0.0525 and IQR [0.0377, 0.0700], so the 4-core sits well above the 75th percentile of arbitrary 4-of-10 subsets and slightly above the full 10-feature ensemble (0.0698). Homoplasy appears in all top-13 subsets and freq/entropy in the majority, supporting the choice of the canonical evolutionary panel; the one feature where the dissertation's choice (pLDDT) is dominated empirically is semantic-change, which contributes positively when paired with frequency and homoplasy under nested-LOPO. The headline interpretation is that the dissertation's 4-core is *near-optimal but not literally best*: it sits in the top 6.2% of all 210 subsets, with a $+0.0141$ MCC gap to the empirically best subset that is well within the paired-bootstrap CI for the nested-LOPO mean MCC ([+0.008, +0.166] for the 4-core; Table 3.14). The selection is therefore robust under the cold-start regime: any subset chosen by domain knowledge that includes homoplasy plus two of (freq, entropy) will land in the top decile, and the marginal gain from oracle-best subset selection ($+0.014$ MCC) is small relative to the cross-pathogen noise floor.

Full-lattice subset audit and per-feature Shapley. The full-lattice audit evaluated all $2^{10} - 1 = 1023$ non-empty subsets under the same nested-LOPO protocol. The Cold-Start 4-core ranks $87/1023$ (top 8.5%-tile; LOPO mean MCC 0.08088, exact), while the lattice top-1 7-feature subset reaches MCC 0.0968 and the 3-core (`freq, entropy, homoplasy`)

Table 3.17: Top-10 4-of-10 feature subsets by nested-LOPO mean MCC (paired protocol, training-pathogen-fit signs, within-fold z-scoring, uniform averaging, top-10% threshold). The dissertation's 4-feature core (`freq, entropy, homoplasy, plddt`) is row-shaded *Core* and ranks 13/210 overall (top 6.2%-tile). Full ranking in `subset_selection_robustness_210combos.csv`.

Rank	4-feature subset	LOPO mean MCC	Δ vs. full-10
1	freq, entropy, homoplasy, semantic_change	0.0950	+0.0252
2	freq, homoplasy, plddt, semantic_change	0.0926	+0.0229
3	freq, homoplasy, sasa, semantic_change	0.0876	+0.0179
4	freq, entropy, homoplasy, sasa	0.0870	+0.0172
5	entropy, homoplasy, sasa, dnds_proxy	0.0864	+0.0166
6	freq, entropy, homoplasy, dnds_proxy	0.0857	+0.0159
7	freq, entropy, rare_freq, homoplasy	0.0856	+0.0158
8	entropy, homoplasy, dnds_proxy, semantic_change	0.0856	+0.0158
9	freq, homoplasy, plddt, sasa	0.0847	+0.0150
10	freq, homoplasy, dnds_proxy, semantic_change	0.0839	+0.0142
13 (Core)	**freq, entropy, homoplasy, plddt**	**0.0809**	**+0.0111**
—	Full 10-feature ensemble (reference)	0.0698	—
—	Median of 210 subsets	0.0525	−0.0173

Best subset MCC − 4-core MCC = $+0.0141$ (within the paired bootstrap CI for the 4-core mean MCC, [+0.008, +0.166]; Table 3.14). Homoplasy appears in all top-13 subsets; `semantic_change` is the most frequently-paired non-core feature in the top-10 (5/10 subsets), reflecting its complementarity to frequency and homoplasy under nested-LOPO uniform weighting.

strictly dominates the 4-core (MCC 0.0866 vs. 0.0809, $\Delta = +0.0058$, lattice rank $27/1023$). Shapley values retain homoplasy ($+0.0294$), frequency ($+0.0220$), and entropy ($+0.0219$) as the only positive MCC contributors; pLDDT contributes -0.0015, and `esm2_llr` contributes -0.0144. The 4-core is therefore a historical fixed configuration, not an optimum; pLDDT is only an optional structural extension, and the 3-core is the stronger compact descriptive variant (see Table 3.15).

Per-pathogen rank reliability and decision-curve framing. A second audit bins positions into 10 score deciles and bootstraps top-vs-bottom Layer A prevalence differences. The 12-pathogen aggregate ranks the 3-core ahead of the 4-core and full-10 ensemble: $+0.084$ (3-core), $+0.063$ (4-core), $+0.043$ (full-10), versus $+0.011$ for random rank; mean Spearman trends are $+0.37$, $+0.28$, $+0.30$, and -0.04. Only $4/12$ pathogens (Norovirus, Dengue, H3N2, Influenza B) show a positive 4-core CI strictly excluding zero (HIV-1 is positive but its CI lower bound is 0.000, so it does not strictly exclude zero), and *Rabies inverts* (delta -0.123, CI $[-0.245, -0.003]$, Spearman $\rho = -0.76$). At 50 sites/pathogen, the decision curve yields 66 expected true positives for 4-core and 68 for 3-core over 12 pathogens, versus 41 for random allocation ($\sim 1.6\times$; `p2_decision_curve.csv`); Algorithm 1 is therefore a ranked screening-budget allocator, not a calibrated-probability model (ECE 0.03–0.04 at $\sim 5\%$ base rate; see Table 3.15).

Null-calibrated significance under four null models. A pre-registered null-calibration audit compared observed 4-core and 3-core metrics against $100{,}000$ permutations under four increasingly structured nulls per pathogen: iid labels, circular shift, 50-residue blocks, and strict local-burden preservation in ± 5 windows. The aggregated one-sided Stouffer p-value across 12 pathogens for 4-core MCC is $p = 1.00$ under all four nulls because $6/12$ pathogens have observed MCC at or below the null mean under the iid and circular-shift nulls (SARS-CoV-2, HIV-1, RSV, MERS, Zika, Rabies; verified from `p1_null_per_pathogen.csv`; the protein-block-shuffle and local-burden-preserving nulls give 4/12 and 7/12 respectively). Per-pathogen, $5/12$ pass the iid null at $p < 0.05$ (Norovirus $p = 1{\times}10^{-5}$, Influenza B $p = 7{\times}10^{-5}$, H3N2 $p = 0.0018$, HCV $p = 0.0092$, Dengue $p = 0.0055$), falling to $3/12$, $2/12$, and $1/12$ (Norovirus only) under the stricter nulls. Algorithm 1 is therefore an exploratory cold-start prioritization heuristic for a callable subset, not a globally calibrated detector (see Table 3.15).

Pre-registered callability and abstention rule (P5). The pre-registered P5 abstention rule operationalises the callable-subset framing of Paragraph 9 with D1–D4 and an inversion guardrail using training-fold-only information except held-out annotation density. Under defaults, $\mathbf{0/12}$ folds are callable for either the 3-core or 4-core method; D1 is rate-limiting (maximum quorum $5/11$; threshold $7/11$). The sensitivity sweep (D1 quorum $\in \{5, 6, 7, 8\}$ $\times$ D2 lower bound $\in \{1.05, 1.10, 1.20, 1.50\}$, 32 configs) preserves $\mathbf{0/12}$ callable folds at any quorum $\geq 6/11$; only $(5/11, 1.05)$ yields calls, and those are P1-negative (3-core: {SARS-CoV-2, RSV, Zika}, mean MCC -0.027; 4-core: {Rabies}, MCC -0.049). D3 also reproduces the P3 stability penalty for pLDDT ($3/12$ vs. $11/12$ median-Jaccard passes). Algorithm 1 is therefore formally abstained from deployment on the current 12-pathogen panel (see Table 3.15).

Biologically-realistic null calibration (P6). P6 tests whether the callable-subset signal is reducible to structural or evolutionary clustering using four explicit nulls: `sasa_strat` (5 SASA quintiles), `plddt_strat` (5 pLDDT quintiles), `sasa_freq_joint` (3×3 SASA×frequency), and `plddt_ent_joint` (3×3 pLDDT×entropy), each with $n_{\text{perm}} = 100{,}000$. All nulls were valid for **12/12** pathogens. For 4-core MCC, individual-significance counts were 5/12 for `sasa_strat` {H3N2, Norovirus, Dengue, Influenza B, HCV}, 5/12 for `plddt_strat` with the same set, 4/12 for `sasa_freq_joint` {H3N2, Norovirus, Dengue, Influenza B}, and 4/12 for `plddt_ent_joint` with the same set; 3-core counts were 5/3/4/3. Norovirus survives all four P6 nulls under both methods at $p = 0.00001$ and also passes P1 local-burden; Stouffer global $p = 1.0$ reflects signed-MCC cancellation. The "strongly defensible" branch fired, but P6 remains descriptive robustness under structural conditioning, not independent evidence or a deployability claim (27/96 3-core and 29/96 4-core P1+P6 cells significant; see Table 3.15).

Tier 2 features-only LOPO and expanded-target stability (W4). Wave 4 asked whether features-only Tier 2 expansion dissolves the panel-size limit at $n \approx 30$. The hscore harmonization audit failed on Dengue:dengue_e, HIV-1:hiv1_gp120, and Norovirus:norovirus_vp1, with **0/3** PASS at median Spearman $\rho \geq 0.70$ and top-10 Jaccard ≥ 0.50; Branch 5 fired, forcing the 2-core fallback `freq, entropy`. Labeled-only 12-fold LOPO gave mean 2-core MCC $= 0.077$ (sd 0.136), $8/12$ positive folds, and paired Wilcoxon $p = 0.368$ versus recomputed 4-core; D1 callability remained **0/12**. Expanded 28-target stability was $1/28$ stable (union quorum 16, `quorum_pass = False`), sign-fitting agreement was 81.6% for `freq` and 94.4% for `entropy`, and **16/16** unlabeled targets were feature-envelope compatible. Branch 2 is closest: feature expansion alone does not relax the panel-size limit, and Layer A curation remains rate-limiting (see Table 3.15).

Layer A′ structured-database sensitivity benchmark (W5). Wave 5 tested whether reviewed UniProtKB feature annotations can substitute for literature-curated Layer A labels. Of 16 feature-compatible Wave 4 targets, **10/16** passed refined Class A after coordinate reconciliation (length-delta tolerance $\leq$ 6 residues); Layer A$'$ prevalence ranged from 2.0% (lassa_gpc) to 14.5% (measles_virus_h). The 10-fold leave-one-target-out 2-core LOPO gave mean MCC $= -0.055$ (sd 0.068, median -0.057), **2/10** positive folds, and precision@20 $= 0.005$. Sign stability over 1,000 target-level bootstraps was 1.000 for both `freq` and `entropy`, but the sign was *negative*, opposite to the Wave 4 12-target literature-curated panel; the cross-panel Welch comparison was $t = -2.96$, two-sided $p = 0.0089$. Branch C fired: Layer A and Layer A$'$ are not interchangeable label provenances, the four flavivirus fusion-peptide-loop targets at residues 98–111 are not independent replications, and Layer A$'$ is not a panel extension (see Table 3.15).

Hierarchical Bayesian adaptive-weighting robustness check. The pre-registered Cycle-7 HBFWS audit tested whether hierarchical Bayesian shrinkage could outperform fixed EqualWeight-4core under 12-pathogen nested LOPO, with prior $P(\text{HBFWS} > \text{EqualWeight-4core}) \in [0.15, 0.25]$. HBFWS mean MCC was 0.058 versus EqualWeight-4core 0.078, paired $\Delta = -0.020$, paired Wilcoxon one-sided greater $p = 0.78$; family-stratified deltas were -0.014 for paired-family folds and -0.028 for singleton folds, with three singleton folds collapsing to $\Delta = 0$. The pre-registered hypothesis was rejected at $n = 11$ pathogens with 5/8 singleton families, consistent with LOPO 0/11, Friedman $p = 0.990$, and the 0.265 oracle-vs-generalised gap; Tier 1 remains the cold-start operationalisation (see Table 3.15).

Cycle 7B: comprehensive comparison of six additional adaptive-weighting paradigms. Six additional adaptive-weighting paradigms were evaluated against EqualWeight-4core under the same 12-pathogen nested-LOPO protocol; Table 3.18 retains the method-level results.

Cycle 7B gives seven adaptive-weighting failures at $n = 11$ pathogens (HBFWS, XG-

Table 3.18: Cycle 7B: six adaptive-weighting paradigms vs. EqualWeight-4core baseline (12-pathogen nested-LOPO; same folds as Table 3.14). All six fail to beat the baseline; paired Wilcoxon H_1: method > EqualWeight-4core, one-sided.

Method	**Mean MCC**	**Δ vs. EQ-4**	**Wilcoxon p**	**Paradigm**
EqualWeight-4core (baseline)	$+0.078$	—	—	Fixed weights
EqualWeight-10feat	$+0.083$	$+0.006$	0.29	Fixed weights
HBFWS (v171)	$+0.058$	-0.020	0.78	Bayesian shrinkage (ICTV)
XGBoost stacked	$+0.016$	-0.062	0.97	Gradient boosting
Random Forest stacked	$+0.019$	-0.059	0.91	Tree ensemble
Logistic Regression L1	$+0.056$	-0.022	0.70	Sparse linear
Algorithm Selection (1-NN)	$+0.066$	-0.012	0.69	Pathogen-meta classifier
HBFWS-phylo (3 groups)	$+0.063$	-0.015	0.84	Bayesian shrinkage (alt. group)
Rank Aggregation (top-3)	$+0.062$	-0.016	0.56	Training-rank ensemble

Boost, Random Forest, Logistic L1, Algorithm Selection, HBFWS-phylo, Rank Aggregation). None of the six new methods rejects H_0: method $\leq$ EqualWeight-4core (smallest $p = 0.56$); HBFWS-phylo, Algorithm Selection, and Rank Aggregation stay within 0.012–0.016 MCC, while XGBoost and Random Forest underperform by -0.06 MCC. Combined with LOPO 0/11, Friedman $p = 0.990$, and the 0.265 oracle-vs-generalised gap, the failure is most consistent with panel size; the single positive EV-A71 fold (homoplasy selected by 1-NN; held-out MCC $= +0.146$ vs. baseline $+0.013$) remains underpowered at $n = 11$ (see Table 3.15).

3.5.3 Ground Truth Heterogeneity

A potential confound in interpreting the pathogen-dependent performance patterns is that the Layer A ground truth definitions may themselves differ across pathogens. To assess this, each Layer A position was tagged with its biological definition type (immune_escape, functional, or convergent_evolution), and the composition was analyzed across the same 12-pathogen feature-analysis scope used in Section 3.5.2 (the core 11 pathogens plus Zika). Table 3.19 presents the results.

Table 3.19: Ground truth heterogeneity across 12 pathogens (11 main + Zika, matching Section 3.5.2). Best discriminative feature differs across pathogens despite homogeneous immune_escape composition. "Consistent" indicates feature stability when restricted to immune_escape-only. HIV-1 $n_{\mathrm{A}} = 26$ here is the curation-table tag count from `layer_a_tags.csv`; the benchmark feature matrix uses $n = 23$ Env gp120-mapped positions for Stage 2 MCC (Table 2.4; see Section 4.4.4).

Pathogen	n_{A}	**% immune**	**Best feature (all)**	**AUC**	**Consistent**
SARS-CoV-2	25	88%	esm2_llr	0.586	Yes
H3N2	25	100%	plddt	0.787	Yes
Norovirus	15	100%	homoplasy	0.944	Yes
HIV-1	26	100%	entropy	0.730	Yes
Dengue	24	71%	homoplasy	0.770	Yes
RSV	20	100%	plddt	0.561	Yes
Influenza B	17	100%	homoplasy	0.805	Yes
MERS	9	78%	esm2_llr	0.695	No
HCV	54	100%	dnds_proxy	0.680	Yes
Zika	27	93%	homoplasy	0.581	Yes
Rabies	39	97%	plddt	0.650	Yes
EV-A71	27	93%	esm2_llr	0.596	No

Despite homogeneous ground truth composition (immune_escape $\geq$ 71% for all 12 pathogens), 5 different best features emerge, and 10/12 pathogens retain the same best feature when restricting to immune_escape-only positions ("Consistent = Yes"). This argues against ground-truth composition being the sole explanation for the ANOVA interaction effect ($\omega^2 = 0.296$), while leaving room for the broader limitations discussed in Chapter 4: pathogen-specific benchmark behavior — driven by some combination of pathogen biology, curation protocol, auxiliary input quality, and their interaction — contributes to the observed method–pathogen dependence, without isolating any single component as causal.

3.5.4 Practical Validation: Search Space Reduction

The analysis results above demonstrate that multiple information types contribute to hotspot prediction, but the most informative feature differs by pathogen. This subsection evaluates whether integrating all 10 features reduces the experimental search space for hotspot validation.

The integration method uses equal-weight combination, instantiated in two complementary protocols. The *production* EqualWeight column underlying the headline scoring $\times$ pathogen ANOVA (`scripts/run_stage3_full_detectors.py`) applies pre-defined domain orientations (e.g., the inverse-pLDDT transform $\mathrm{pLDDT}_{\mathrm{max}} - \mathrm{pLDDT}$ rather than raw pLDDT) followed by uniform $1/10$ averaging over min–max normalised features, with no per-feature sign fit from labels at this stage. The *nested-LOPO* ablation protocol used in Section 3.5.2 (`scripts/feature_ablation_nested_lopo.py`; Table 3.14) additionally fits a directional sign $\hat{s}_f \in \{-1, +1\}$ on the eleven training pathogens excluding the held-out pathogen and applies within-fold z-scoring; the held-out fold combination $s_i = \sum_{f=1}^{10} (1/10) \cdot \mathrm{sign}(\hat{s}_f) \cdot z_{i,f}$ shown above corresponds to this nested-LOPO protocol. The two protocols agree to within ≤ 0.013 MCC mean (production in-sample: 0.083; nested-LOPO: 0.070), so the directional sign step is not load-bearing for the headline cross-pathogen claim. Positions exceeding a percentile threshold are called as hotspots. Equal weighting was chosen because, with only 11 reference pathogens, adaptive weighting strategies introduce more noise than signal (see below).

Leave-One-Pathogen-Out (LOPO) cross-validation was conducted across the same 12-pathogen feature-ablation evaluation set used in Section 3.5.2 (core 11 + Zika), comparing EqualWeight against frequency-only (FreqThresh) and random baselines. Under the EqualWeight top-10% threshold, mean MCC across the 12-pathogen set is 0.083, outperforming FreqThresh (0.075) and random (≈ 0.000); restricting to the core 11 pathogens shifts the baseline to 0.093 (consistent with the per-pathogen counts reported in Section 3.5.2, where 9 of 11 core pathogens carry positive EqualWeight MCC under the top-10% threshold and 8 of 11 do for FreqThresh).

Vaccine escape position validation. To assess practical relevance, the 20 scoring $\times$ 39 detector combinations were evaluated against experimentally confirmed vaccine/antibody escape positions for 3 pathogens: H3N2 (29 antigenic drift sites in HA sites A and B), SARS-CoV-2 (30 Spike escape sites in RBD, NTD, and S2), and HIV-1 (45 Env gp120 antibody escape sites). A total of 2,340 enrichment evaluations were performed (Table 3.20), with enrichment defined as

$$E = \frac{\text{observed overlap}}{\text{expected overlap}} = \frac{|D \cap A|}{|D|\,|A|/L},$$

where D is the set of detected hotspot positions, A is the set of annotated escape positions, and L is the protein length; significance uses Fisher's exact test on $\{|D \cap A|,\ |D \setminus A|,\ |A \setminus D|,\ L - |D \cup A|\}$. These three pathogens were selected only by availability of curated antibody-escape annotations; the remaining 8 lacked comparable residue-level escape databases.

Table 3.20: Vaccine escape enrichment audit (20 scoring $\times$ 39 detectors, Fisher's exact test). Enrichment is the observed/expected overlap ratio defined above; only the top scoring type per pathogen is shown for single-scoring rows.

Pathogen	Best scoring	Category	Detection	Enrichment	Overlap	p-value		Role
H3N2	FUBAR Bayes factor	Phylogenetic	Wavelet(t=2.0)	**9.36**×	9/29	< 0.0001	$***$	self-consistency
HIV-1	freq	Frequency	Wavelet(t=1.5)	**7.19**×	23/45	< 0.0001	$***$	external anchor
SARS-CoV-2	FUBAR pos. sel.	Phylogenetic	FreqThresh(p=85)	**7.63**×	2/30	0.026	$*$	exploratory
Multi-source integration — fixed top-5% protocol (`escape_validation_adapt.csv`; SARS-CoV-2 / H3N2):								
H3N2	EqualWeight	Composite	top-5% (fixed)	4.012×	6/29	**0.0023**	$**$	self-consistency
SARS-CoV-2	EqualWeight	Composite	top-5% (fixed)	2.67×	2/15	0.171	n.s.	exploratory
Multi-source integration — best-detector grid (`vaccine_escape_stage3.csv`; HIV-1 not yet in fixed-threshold pipeline):								
HIV-1	EqualWeight	Composite	CUSUM(d=0.5, h=5.0)	4.17×	—	0.0037	$**$	external anchor

$***$: $p < 0.001$; $**$: $p < 0.01$; $*$: $p < 0.05$; n.s.: not significant. Bonferroni uses 780 combinations per pathogen. Adjusted single-scoring tiers are H3N2 $p_{\text{adj}} \approx 2.5 \times 10^{-5}$ (unrounded 2.63×10^{-5}), HIV-1 $p_{\text{adj}} \approx 2.5 \times 10^{-16}$ (unrounded 2.47×10^{-16}), and SARS-CoV-2 $p_{\text{adj}} \rightarrow 1.00$; tier assignments are unchanged under $\alpha = 0.01$ or an 8,580-test cross-pathogen denominator. EqualWeight provenance: H3N2/SARS-CoV-2 use `scripts/validate_escape_adapt.py` ($\rightarrow$ `escape_validation_adapt.csv`); HIV-1 uses `scripts/run_vaccine_escape_stage3.py` ($\rightarrow$ `vaccine_escape_stage3.csv`) because HIV-1 was not yet integrated into the fixed top-5% pipeline. Layer A overlap is 20/29 for H3N2 (69%), 18/30 for SARS-CoV-2 (60%), and 8/45 for HIV-1 (18%); HIV-1 retains 37/45 Layer-A-disjoint positions (82%), novel-only enrichment 7.16–8.24×, worst-case Fisher $p = 5.0 \times 10^{-12}$, and novel-only $p_{\text{adj}} = 5.0 \times 10^{-12} \times 780 \approx 3.9 \times 10^{-9}$. Top-1 enrichments are upper envelopes; Bonferroni-survivor means are lower (H3N2 $\approx 4.7\times$ (67/780 survivors), HIV-1 $\approx 2.3\times$ across all 132/780 survivors (or $\approx 2.7\times$ when restricted to the freq-scoring subset)).

Key findings: The load-bearing external anchor is the single-scoring HIV-1 7.19$\times$ enrichment row (82% Layer A-disjoint escape set, novel-only $p_{\text{adj}} \approx 3.9 \times 10^{-9}$); the EqualWeight block is complementary because H3N2/SARS-CoV-2 and HIV-1 use different pipelines. All three pathogens are nominally significant ($p < 0.05$) with their optimal scoring–detection combination; after 780-combination Bonferroni correction, H3N2 and HIV-1 survive while SARS-CoV-2 remains exploratory. Category-level concordance matches the Layer A benchmark for H3N2 and HIV-1, although HIV-1 changes from Layer A *entropy* with Wavelet(t=2.0), MCC 0.275, to escape *freq* with Wavelet(t=1.5), enrichment 7.19$\times$; both are frequency-derived and correlated at $\rho \approx 0.97$ (Section 3.5).

For H3N2, FUBAR Bayes factor captures 9 of 29 antigenic drift sites with 9.36-fold enrichment, but 8 of those 9 are already in Layer A; removing them leaves 1 novel-only capture among 21 Layer-A-disjoint H3N2 escape positions (Fisher $p = 0.132$), so this is a self-consistency check rather than the HIV-1 external anchor. EqualWeight still gives a useful top-5% check at 4.012-fold enrichment ($p = 0.0023$), capturing antigenic site A position 137 and site B positions 155, 159, 160, 186, 188; frequency alone (FreqThresh top-5 $= \{137, 155, 159, 188\}$) misses positions 160 and 186.

Vaccine-escape validation was feasible for only 3 of 11 pathogens (H3N2, HIV-1, SARS-CoV-2): the remaining 8 were excluded on data-availability grounds prior to enrichment computation, falling into four categories—(i) annotation granularity too coarse for residue-level mapping (Dengue serotype-level, Influenza B HI-distance, Rabies antigenic-domain-level, Norovirus loop-level); (ii) sample size below the $n \geq 15$ threshold (MERS $<$10, EV-A71 $<$10); (iii) confound with the benchmark's Layer A set (RSV palivizumab/nirsevimab escape partially overlaps Site II); (iv) mechanism mismatch (HCV NS3/NS5A/NS5B drug-resistance is direct-acting antiviral resistance, not immune escape). The generalisability of the enrichment finding to the other 8 pathogens therefore remains to be established, and the three validated pathogens should be read as a data-availability convenience sample rather than a balanced subset.

Layer A / escape-list overlap (circularity audit). Because Layer A is partly constructed from immune-escape-tagged positions (Chapter 2, `layer_a_tags.csv` with an `immune_escape` evidence tag), H3N2 and SARS-CoV-2 escape enrichments should be read mainly as self-consistency checks; Table 3.20 gives the overlap counts and shows why HIV-1 is the least circular external validation.

Search space reduction. For a typical viral protein of 500 positions, the optimal scoring–detection combination narrows the candidate list to approximately 10–50 positions with 7–9-fold enrichment for escape sites in the reported vaccine-escape anchors. This transforms hotspot detection from an exhaustive search into a guided investigation, providing a ranked candidate list that directs limited experimental resources toward the most promising positions.

Chapter 4

Discussion, Limitations, and Future Directions

This chapter interprets the results from Chapter 3, summarises the contributions, lays out the dissertation's limitations and future research directions, and ends with concluding remarks. The multi-source information integration analysis (Section 3.5) identifies a compact 4-feature cold-start core (homoplasy, pLDDT, entropy, frequency) that is operationally equivalent to the full 10-feature ensemble under nested-LOPO ($\Delta = +0.011$ MCC, $p = 0.61$) and shows that integration enables practical search-space reduction; uniform full-feature integration is not uniformly superior to the best single scoring channel on every pathogen, and adaptive per-pathogen weighting is not yet learnable from the current $n = 11$ panel.

4.1 Methodological Validity of MutBench

The wide region-level BCa CIs ($n = 7$ regions; e.g., MutClust-Hybrid $[0.323, 0.635]$) reflect region-dependent recall sensitivity, but paired bootstrap comparisons cancel that variability and confirm method differences at $p < 0.001$. The synthetic–real gap (synthetic MutClust-Hybrid *HS* $= 0.778$ vs. sparse real GISAID 0.383, with 6.9–8.0$\times$ enrichment preserved across both) is a methodological feature: dual evaluation quantifies the distance between theoretical and practical detection.

4.1.1 Independence of Ground Truth and Detection Methods

The synthetic benchmark's ground truth is defined independently of the detection methods (DMS experimental data and convergent evolution literature), ensuring no *algorithmic* circularity between evaluation and detection. A more subtle concern is that Layer A positions (convergent evolution) could correlate with mutation frequency at the *per-position* level, which can make frequency-based scoring appear structurally strong. A per-pathogen audit of the point-biserial correlation between Layer A membership and per-position frequency finds a mean $|\rho| = 0.12$ across the 11 pathogens (range $[-0.06, +0.33]$). This per-position membership-vs-frequency $|\rho| = 0.12$ audit is conceptually distinct from the feature-feature Pearson correlation $\rho \approx 0.97$ between the *frequency* and *entropy* scoring inputs reported in Section 3.5 (specifically the feature–feature correlation analysis in Chapter 3 Section 3.5.1); the former measures circularity between ground truth and a scoring signal, while the latter measures redundancy among scoring inputs themselves. Per pathogen: HCV (+0.33), Influenza B (+0.26) and Norovirus (+0.23) show moderate positive correlations, while SARS-CoV-2 ($\rho \approx 0$), MERS (≈ 0) and EV-A71 (≈ 0) are essentially uncorrelated. The effect is therefore pathogen-specific rather than uniform, and is driven by ground-truth curation style: pathogens whose Layer A uses hypervariable/epochal-variant criteria (HCV, Influenza B, Norovirus) couple more strongly with frequency than pathogens whose Layer A uses phylogenetically-confirmed convergent positions (SARS-CoV-2, H3N2). At the individual-position level, D614G achieved the highest single frequency through a founder event (homoplasy count = 2), while position 476 has moderate frequency but the highest homoplasy count (53); such anecdotes confirm that frequency and convergent-evolution signals are not interchangeable. Readers should therefore interpret frequency scoring's strong benchmark performance as partly structural (frequency $\approx$ Layer A by construction) and partly genuine signal (homoplasy provides the 3-lineage discriminator that mere frequency cannot). *Implication for per-pathogen rankings*: HCV's frequency-best ranking and Norovirus's epochal-variant Layer A coupling with frequency scoring should be read as partly structural rather than purely mechanistic; the per-pathogen Stage 2 rankings tabulated in Chapter 3 (Table 3.5) carry this circularity discount implicitly for these three pathogens (HCV, Influenza B, Norovirus), and any interpretation that relies on their single-feature "best" identity should propagate this caveat. This pathogen-specific circularity is itself a downstream consequence of the Layer A curation heterogeneity discussed in Section 4.4.4. To provide a frequency-independent validation dimension, Layer C (DMS experimental fitness) is used where available (6 of 11 pathogens), and the vaccine escape enrichment analysis (Section 4.4.1) provides partial external cross-validation: HIV-1 (82% of its escape set disjoint from Layer A) is the cleanest external anchor, whereas H3N2's 69% Layer A/escape overlap means that the H3N2 enrichment re-measures benchmark performance more than it provides independent replication. The circularity audit in Chapter 3 Table 3.20 reports novel-only enrichment values that quantify this distinction.

Layer A curation-reproducibility limitations. Three structural limitations of the Layer A curation procedure are not measured by the position-level circularity audit above and remain disclosed as bounding caveats rather than resolved failure modes. *First*, the literature-curated position-level annotation table (`results/mutbench/layer_a_tags.csv`, 309 positions across 12 pathogens) was assembled as a single-team evidence synthesis from 1–3 an-

Table 4.1: Layer A provenance summary, one row per pathogen. "Dominant source" is the most-cited paper string in `layer_a_tags.csv`; tag mix shows the qualitative evidence type. The HCV and HIV-1 "unknown" dominant source labels reflect HVR-region and aggregated-anchor entries that do not parse as a single first-author+year string; the underlying CSV exposes the per-row evidence string for verification. Counts are n in `layer_a_tags.csv`; 3 HIV-1 positions outside the gp120 mapped window are not counted in the benchmark feature matrix ($n = 23$ used in analysis).

Pathogen	n	**Dominant source**	**Tag mix**
HCV	54	HVR1/HVR2 region (multi-source)	immune_escape(54)
Rabies	39	Aditham 2025	immune_escape(38), functic
Zika	27	Long 2019	immune_escape(25), conver
EV-A71	27	Foo 2007	immune_escape(25), functic
HIV-1	26 (23 mapped to gp120)	Moore–Williamson 2015 (multi-source)	immune_escape(26)
SARS-CoV-2	25	Cao 2023	immune_escape(22), functic
H3N2	25	Koel 2013	immune_escape(25)
Dengue	24	Twiddy 2002	immune_escape(17), conver
RSV	20	Mas 2018	immune_escape(20)
Influenza B	17	Ni 2013	immune_escape(17)
Norovirus	15	Lindesmith 2012	immune_escape(15)
MERS	9	Kim 2016	immune_escape(7), functior
Total	**308**	11 dominant sources	immune_escape $\sim$90%

chor publications per pathogen; no duplicate independent recuration subset, inter-annotator agreement statistic, or adjudicated disagreement log is available, so label reproducibility is not directly measured against an alternate curator. *Second*, several pathogens are dominated by one anchor source (e.g., HCV's 54 positions derive primarily from HVR1/HVR2 region definitions, H3N2's 25 positions from Koel 2013, HIV-1's 26 from Moore–Williamson 2015), so a source-selection sensitivity check that swapped one dominant source for an alternative same-topic publication could plausibly alter both Layer A membership and prevalence; this dissertation does not measure that source-selection sensitivity directly. *Third*, for entries derived from source-reported regions rather than residue-specific assays, the labels should be interpreted as region-supported position annotations rather than as residue-equivalent experimental evidence; the sensitivity weights reported in Chapter 3 Table 3.20

(HIV-1 82% Layer-A-disjoint novel-only audit) partially address this by quantifying how much of the practical-anchor result depends on positions with independent escape evidence, but a full IEDB-grade source-to-label audit trail (per-position columns: PMID, evidence type, original-vs-review flag, source coordinate, benchmark coordinate, extraction text pointer, assay context, inclusion rule, curator, adjudication status) is reserved as Priority-5 future work. The biological validity of detected hotspots was further supported by 3D structural analysis (Section 3.3), which showed 5.92-fold spatial contact enrichment, and by the consistent superiority of domain-specific methods (MutClust variants) over general-purpose alternatives (OPTICS, KDE) in Section 3.1.3. Min-max normalization was used for detector inputs and for the production EqualWeight composite column underlying the scoring $\times$ pathogen ANOVA; z-scoring was used only in the nested-LOPO EqualWeight ablation protocol described in Section 2.2.4. Rank transformations were not evaluated as they can introduce artifacts in sparse data distributions. With benchmark validity established, the temporal limitations of the underlying H-score metric warrant examination.

4.2 Limitations of H-score-Based Detection and Alternatives

H-score saturation in late-pandemic data limits reference-based metrics: in 2024–2025 SARS-CoV-2 Spike data, all 1,259 post-truncation positions have non-zero H-scores with $\sigma = 0.098$, eroding discriminatory power. TC-freq achieves the highest enrichment ($3.43\times$) on convergent-evolution ground truth, indicating that temporal-contrast scoring is a viable direction whose appropriate resolution remains a future challenge. The Jaccard similarity between DMS-escape and convergent-evolution positions is only 0.006 (Section 3.1.2), so the “best” method depends on the ground-truth definition—a Stage 1 finding that foreshadows the pathogen-dependent rankings revealed by the cross-pathogen benchmark.

Homoplasy-based scoring is one alternative motivated by the founder-bias problem ($\rho = -0.876$ between H-score and independent-origin count, Section 3.3): Fitch-parsimony homoplasy counts independent mutational origins, decoupling frequency from selective pressure (D614G receives a homoplasy count of 2 vs. 53 for convergent position 476). Cross-pathogen validation, however, is itself pathogen-dependent: HIV-1 attains AUC $= 0.706$ ($p = 0.0002$), H3N2 a weak trend (0.561, $p = 0.147$), and HCV fails (0.390) because parsimony saturates within HVR1. Homoplasy is therefore unreliable for pathogens with extreme intra-protein diversity, mirroring the main scoring $\times$ pathogen interaction finding ($\omega^2 = 0.296$).

4.3 Cross-Pathogen Generalization

4.3.1 Implications of Cross-Pathogen Generalization

The cross-pathogen benchmark demonstrated that no universally best-performing method exists (Chapter 3), with implications for real-time surveillance systems such as Nextstrain [17]: applying a previously optimized combination to a new pathogen may not generalize. The underlying cause is that pathogen-specific genomic characteristics create fundamentally different signal landscapes—for example, H3N2's uniformly nonzero H-scores neutralize threshold-based seed detection, while MERS's sparse genome favors weak-signal detectors.

A notable finding is the asymmetry between scoring and detection contributions: the detection method family main effect ($\omega^2 = 0.013$) is the smallest variance component, approximately 23 times smaller than the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$). This implies that *the choice of input information* matters far more than *the choice of detection* *algorithm*. Practically, this means that effort should be directed toward selecting the right information source for each pathogen rather than optimizing detection algorithm parameters.

The pattern is formalisable as Rice's algorithm-selection problem [68] with the No-Free-Lunch theorem [69] as theoretical scaffold: the empirical mapping f from pathogen features to optimal scoring is not yet learnable from $n = 11$ pathogens (LOPO 0/11, Friedman $p = 0.990$, 0.265 oracle-vs-generalised gap are jointly null-consistent), and operationalising f requires substantially larger panels (Section 4.6). The ESM-2 cross-paradigm validation (Section 3.4.4) is consistent: ESM-LLR tops SARS-CoV-2 but is dominated by homoplasy on Norovirus (AUC 0.763 vs. 0.944).

The pathogen-dependence finding is consistent with related work but addresses a different task: EVEREST [61] and ProteinGym [12] target per-variant fitness/escape prediction, while MutBench evaluates per-region hotspot detection. The contribution here is the quantified scoring $\times$ pathogen interaction ($\omega^2 = 0.296$ / 0.103 at 20-type / 6-category granularity) and the HIV-1 vaccine-escape cross-check ($7.19\times$, 82% Layer-A-disjoint) as the strongest external anchor.

4.3.2 Upper Bound of Detection Performance

Mean oracle MCC across 11 pathogens is 0.341 (range 0.196–0.660; Table 3.5), which appears low but is explained by three structural factors—position-level exact matching, extreme class imbalance (0.7%–15.9% positive rates), and ground-truth stringency—and is unambiguously non-random (113 standard deviations above the random baseline 0.001, comparable to TFBS prediction F1 $=$ 0.3–0.5 [122]). Region-overlap analysis (Table 4.2) shows mean oracle MCC rises from 0.341 to 0.472 under ± 5 position windows (Influenza B reaches 0.894); HCV and Norovirus show decreased MCC under expansion because their oracle detections are already precise.

Table 4.2: Region-overlap MCC under three evaluation modes. *Exact* column is the per-pathogen Stage 2 best MCC reproduced from Table 3.11 (source: `stage3_full_results.csv`); ±5 and ±10 window columns are window-tolerance re-evaluations of each pathogen's Stage 2 best detector under the corresponding position tolerance. The legacy `region_overlap_mcc.csv` archive evaluates a different (Stage 1-style H-score) scoring set and is not the direct source for this table; values for EV-A71 and Rabies were computed identically to the upper block. Window columns: ±5, ±10 position tolerance.

Pathogen	**Exact**	**±5 window**	**±10 window**
Reproducible from `region_overlap_mcc.csv` (9 pathogens):			
HCV	0.660	0.588	0.519
H3N2	0.534	0.639	0.548
Norovirus	0.510	0.335	0.275
Influenza B	0.392	0.894	0.924
HIV-1	0.275	0.406	0.343
Dengue	0.274	0.441	0.508
SARS-CoV-2	0.211	0.480	0.497
MERS	0.196	0.470	0.593
RSV	0.196	0.394	0.318
Subset mean (9)	0.361	0.516	0.503
Oracle-derived (no archived fixed-combination CSV row):			
EV-A71	0.260	0.275	0.199
Rabies	0.248	0.265	0.266
Mean (all 11)	**0.341**	**0.472**	**0.454**

Despite these ceiling effects, the benchmark results have practical implications for public health surveillance.

4.3.3 Recall Analysis: Which Positions Are Missed?

Mean oracle recall across 11 pathogens is 0.552 (Table 3.11). Three observations bound the interpretation: (i) missed positions are not random—low-recall positions are typically Layer A entries documented in only the minimum 3 lineages or governed by epistatic selection (e.g., 8 of the 25 SARS-CoV-2 Layer A positions are NTD supersite residues where immune evasion operates through conformational change, not direct frequency elevation);

(ii) multi-source integration recovers positions missed by a single-feature view (the H3N2 EqualWeight ensemble picks up antigenic site B positions 160 and 186 that frequency-only Top-5 misses, Section 3.5.4); and (iii) region-level coverage is substantially higher than position-level recall—under ± 10 position windows mean MCC rises from 0.341 to 0.454 (Influenza B 0.924), so most "missed" positions are detected within 1–2 residues of the true site, sufficient to guide experimental prioritisation. The remaining risk is concentrated in EV-A71 (recall 0.185) and Rabies (0.308), whose short post-truncation alignments (261 / 491 AA) limit statistical power; users applying MutBench to such pathogens should treat detected positions as a prioritised subset rather than an exhaustive list and complement with DMS or epitope mapping.

4.4 Practical Implications and Fundamental Limitations

Returning to the three problems identified in Chapter 1: the evaluation framework, comprising the three-layer ground truth and the hotspot-score metric (Contribution 1), is designed for standardised use, pending external benchmarking by independent teams (with the saturation caveat in Section 4.2); pathogen-dependence is quantified by the interaction effect ($\omega^2 = 0.296$ at the evaluation level, $\omega^2 = 0.234$ at the cell level) and is corroborated by LOPO 0/11 read as the operational failure mode that illustrates why the interaction matters, not as standalone proof (Contribution 2); cross-validation against vaccine-escape positions supports retrospective practical relevance, anchored by HIV-1 with 82% Layer-A-disjoint novel-only enrichment (Contribution 3).

4.4.1 Vaccine Escape Position Validation

Vaccine escape validation (Table 3.20) showed enrichment values of 9.36$\times$ for H3N2, 7.19$\times$ for HIV-1, and 7.63$\times$ for SARS-CoV-2 (nominal Fisher's exact p-values). Under Bonferroni correction across 780 combinations, HIV-1 ($p_{\text{adj}} = 2.5 \times 10^{-16}$) and H3N2 ($p_{\text{adj}} = 2.6 \times 10^{-5}$) survive while SARS-CoV-2 does not. The circularity audit in Chapter 3 further distinguishes HIV-1 (82% of escape set disjoint from Layer A; external validation) from H3N2 (69% Layer A overlap; self-consistency check). The 10-feature EqualWeight integration (top-5% threshold; `escape_validation_adapt.csv`) further achieves statistically significant H3N2 escape enrichment (4.012$\times$, $p = 0.0023$) in the integration-vs-single comparison reported in that file (note: single-scoring H3N2 FUBAR survives Bonferroni in Table 3.20; the "single-feature approaches do not" statement here applies specifically to the EqualWeight/FreqThresh/BestSingle top-5 integration block, not to the full 780-combination Stage 2 sweep). The Stage 2 single-feature 7.19$\times$ HIV-1 enrichment and the Stage 3 EqualWeight integration 4.012$\times$ H3N2 enrichment are not directly comparable as "larger = better" numbers: the former is the per-pathogen winner of a 780-combination exhaustive sweep (winner's-curse-prone, top-of-search selection), while the latter is a fixed 10-feature uniform-weight ensemble at a pre-specified top-5% threshold (winner's-curse-free, no per-pathogen tuning). The smaller integration number is therefore *more reliable* as an externally-comparable estimate, not weaker. The primary practical value lies in *reducing the experimental search space* rather than achieving high absolute prediction accuracy (see Section 3.5.4 for detailed discussion).

While LOPO cross-validation yields 0/11 generalization, MutBench provides practical value through three mechanisms: (1) the family-level scoring recommendation table (Table 4.3 in this chapter) provides coarse-grained starting hypotheses based on virus family characteristics; (2) the 10-feature EqualWeight ensemble (under uniform-weight LOPO top-10% evaluation the 4-feature core retains 97.6% of the 10-feature ensemble's mean MCC—a relative-to-ensemble retention figure within the same protocol, not an absolute floor relative to an oracle; no paired-sample significance test was performed in this within-pathogen demonstration, and the 4-core ranks 13/210 (top 6.2%-tile) under the exhaustive 4-of-10 nested-LOPO subset audit in Table 3.17) achieves 4.012-fold H3N2 vaccine escape enrichment in the integration comparison without requiring pathogen-specific optimization; and (3) once scoring features have been computed for a pathogen, the 20-scoring $\times$ 39-detector evaluation grid can be rerun in approximately 25 seconds on a single workstation, making exhaustive detector comparison operationally feasible. The earlier Stage 1 single-scoring escape analysis is therefore retained only as historical development context; the dissertation-level evidence uses the full 20-scoring benchmark in Table 3.20.

4.4.2 Cross-Validation via Vaccine Escape Enrichment

A critical question for any benchmark is whether its rankings reflect genuine biological relevance or merely circular evaluation artifacts. The vaccine escape enrichment analysis across three pathogens (2,340 evaluations) provides *partial* cross-validation of the Layer A benchmark results, with the strength of external validation differing substantially by pathogen.

Three-tier evidence rule. External value is judged by Bonferroni survival and Layer A independence. HIV-1 passes both criteria (18% Layer A overlap, $p_{\text{adj}} = 2.5 \times 10^{-16}$) and remains the primary external anchor; the direct novel-only test on 37 Layer A-disjoint positions gives 7.16–$8.24\times$ enrichment, with the conservative worst case still significant after 780-fold correction. H3N2 survives Bonferroni correction but has 69% Layer A overlap, so it is treated as a self-consistency check; its novel-only enrichment is not significant ($p = 0.132$). SARS-CoV-2 does not survive correction and is exploratory. Thus the benchmark is not algorithmically circular, but the practical-value claim rests mainly on HIV-1.

Concurrent-benchmark task-orthogonality. ViroGym and EVEREST overlap with MutBench on some pathogens, but they optimize per-variant fitness Spearman whereas MutBench evaluates per-region binary hotspot calls. A fair head-to-head would require task-aligned binarization of fitness predictions, so it remains future work rather than a missing leaderboard. This is especially relevant for pathogens lacking DMS Layer C, where Layer A plus HIV-1-like external escape validation remains the current evidential bridge.

4.4.3 Practical Workflow for New Pathogens

For a novel RNA virus, MutBench's methodology operationalises through four steps. **(1) Sequence collection**: ≥ 200 protein-coding sequences from NCBI GenBank or GISAID with temporal/geographic diversity, aligned with MAFFT [84] ($\sim$5 min for $1{,}000$ sequences). **(2) Feature computation** in order of increasing cost: frequency/entropy from the MSA (seconds), homoplasy via Fitch parsimony on an inferred tree (2–5 min), ESM-2 masked-marginal scores (3–5 min/protein, GPU), and structural features pLDDT/SASA from AlphaFold/ESMFold (omitted as a reduced-feature baseline when no reliable structural model exists). **(3) Scoring category selection**: Table 4.3 provides empirically grounded family-level starting priors; users should treat the recommendation as a first-pass prior rather than a fixed rule. **(4) Detection and evaluation**: KDE or Wavelet-based detection as robust defaults; with pre-computed features the 39-detector grid evaluates in under 3 s per pathogen (the full $8{,}580$-evaluation benchmark in $\sim$25 s). Total per-pathogen runtime is 15–20 min on a single GPU; the 4-feature core (homoplasy, pLDDT, entropy, frequency; nested-LOPO paired delta $+0.011$, $p = 0.61$, Table 3.14) is the cold-start option when labels are unavailable, eliminating PLM inference and bringing runtime under 10 min on CPU (AlphaFold2 itself remains GPU-bound). Live-deployment integration with Nextstrain [17]/GISAID [64] APIs remains future work.

Table 4.3: Starting scoring-category priors by virus family, based on cross-pathogen benchmark results

Virus Family	Starting Scoring Prior	Rationale
Orthomyxoviridae	Phylogenetic (FUBAR)	Strong diversifying selection signal
Coronaviridae	AI-based (PLM)	Shallow phylogeny; PLM captures broader context
Caliciviridae	MSA-derived (homoplasy)	Epochal evolution with convergent patterns
Retroviridae	MSA-derived (entropy)	High diversity; entropy captures allelic variation
Flaviviridae	Frequency-based	Hypervariable regions dominate signal
Pneumoviridae	AI-based (ESM-2)	Limited phylogenetic signal

The two operating regimes are complementary, not a single universal algorithm: Contribution 2 gives the *informed-use ceiling* (pathogen-specific selection explains the $\omega^2 = 0.296$ interaction and the 0.265 oracle-vs-generalised gap), and the 4-feature core gives the *cold-start floor* (statistically indistinguishable from the 10-feature ensemble under nested-LOPO at $p = 0.61$).

4.4.4 Defense Limitations Matrix

The residual risks concentrate in three examiner-relevant areas. First, Layer A is intentionally heterogeneous across pathogens (Table 2.4); the interaction claim is therefore a claim about relative method rankings under real curation heterogeneity, not directly comparable absolute MCC across viruses. Second, the retrospective signal does not yet transfer cleanly to forward time: the 73-window backtest has grand mean AUROC 0.539 and the predeclared callability rule later refuses all 12 folds. Third, the outputs are dual-use because they prioritize immune-escape-relevant regions, so release is limited to computed region-level scores and summaries, with no raw GISAID FASTA redistribution and dedicated IBC/P3CO/Select Agent review for future higher-risk extensions.

Table 4.4 consolidates the detailed limitations formerly separated across Sections 4.4.4–4.4.4; the rightmost column preserves the audit anchor for each issue.

Table 4.4: Defense limitations matrix for MutBench

Limitation	**Implication**	**Mitigation in this dissertation**	**Where addressed**
Phylogenetic and sampling bias	Founder effects can mimic selection; GISAID coverage is geographically and temporally uneven ($>80\%$ high-income-country sequences).	Homoplasy proof-of-concept separates convergent from founder mutations (AUC $= 1.000$); real GISAID H-score vs. homoplasy remains distinct ($\rho = -0.107$, $p = 1.36 \times 10^{-4}$).	Sections 3.3–3.3
Temporal-window dependence	Frequency AUC shifts from 0.337 (2020-H1) to 0.472 (2021-H1), and entropy from 0.126 to 1.241.	Treated as a time-stratification caveat; prospective claims require time-forward validation.	Section 3.2.2

Limitation	Implication	Mitigation in this dissertation	Where addressed
PLM leakage and proxy collinearity	UniRef-trained ESM-2 contains the target proteins; Tranception proxy and ESM-2 LLR are near-equivalent for 11 of 12 pathogens ($\rho = 0.643$–0.836, all $p < 10^{-50}$).	PLM claims are framed as region-level detection signals, not model-family wins; HIV-1 is the lone dissociating case ($\rho = 0.043$, n.s.; Spearman 0.158).	Table 2.9
Detector-family Simpson's paradox and scoring redundancy	Family main effect is small ($\omega^2 = 0.013$), but family $\times$ pathogen is medium (0.082); 20 scoring types partly share inputs.	Claims compare modeled main effects only; 6-category aggregation gives a collinearity-robust lower bound ($\omega^2_{\text{category}\times\text{pathogen}} = 0.103$).	Section 3.5
Winner's curse in per-pathogen top picks	Each pathogen searches 780 combinations; H3N2 top $9.36\times$ exceeds the Bonferroni-survivor mean ($\approx 4.7\times$).	Interaction effect ($\omega^2 = 0.296$) and vaccine-escape validation are emphasized over top-1 absolutes; 9 scoring types from 4 of 6 categories reduce the chance-uniqueness interpretation.	Table 3.20

Limitation	Implication	Mitigation in this dissertation	Where addressed
Small pathogen panel for adaptive inference	$n = 11$ is below Bolker's $\geq$20–30 stable-random-effects regime; Friedman $p = 0.990$ and LOPO 0/11 are uninformative for a deployable selector.	Fixed-effect analysis is bounded to these pathogens; five robustness frames retain lower bounds above $\omega^2 > 0.14$; expansion to 20–30+ pathogens is future work.	Sections 2.2.5, 3.4.4
Independent reproduction	8,580 evaluations come from one team, one codebase, one snapshot, seed $= 42$.	Repository, archived CSVs, and provenance manifest expose headline values ($\omega^2 = 0.296/0.234$, oracle MCC 0.341, HIV-1 $7.19\times$, $p_{\text{adj}} = 2.5 \times 10^{-16}$) for external rerun.	Section 4.7
Technical covariates and input heterogeneity	Oracle MCC correlates with protein length ($\rho = -0.73$, $p = 0.011$) and Layer A positive rate ($\rho = +0.64$, $p = 0.035$); tree, structure, and PLM inputs vary by pathogen.	ANCOVA with $\log(\text{length})$ and Layer A rate amplifies rather than attenuates the interaction ($0.234 \rightarrow 0.264$, $p < 10^{-167}$); most detectors internally normalize scale.	Section 2.2.5

Limitation	**Implication**	**Mitigation in this dissertation**	**Where addressed**
Layer A heterogeneity	HCV, Norovirus, SARS-CoV-2, H3N2, and immune-escape-led pathogens use different label logics, so absolute MCC is not cross-pathogen comparable.	Subset refits remain medium or larger: $\omega^2 = 0.199$ ($n = 10$), 0.187 ($n = 9$), 0.137 ($n = 4$); 6-category aggregation gives 0.103.	Tables 2.4, 4.1
Sub-lineage pooling	Influenza B, SARS-CoV-2, HIV-1, and HCV pool biologically distinct sub-lineages or genotypes.	Winners are interpreted as best on pooled MSAs; per-sub-lineage reruns are reserved as sensitivity analyses.	Paragraph 4.4.4
Scope and mutation regime	Current evidence covers RNA-virus surface glycoprotein substitutions, not DNA viruses, indels, recombination, frameshifts, non-coding regions, or other pathogen systems.	HBV polymerase pilot ($n = 708$, $L = 882$) shows a 2.6–13$\times$ density gap and nonsignificant 14-position entropy test ($p = 0.354$ global, $p = 0.136$ RT-domain).	Paragraph 4.4.4

Limitation	Implication	Mitigation in this dissertation	Where addressed
Prospective validation gap	Retrospective MCC (0.341 exact / 0.454 at ± 10) does not imply deployment.	Sliding-window backtest reports EV-A71 0.77, Rabies 0.79, Influenza B 0.70, grand mean 0.539, enrichment $1.57\times$; later abstention rule gives 0/12 callable.	Section 3.2.2
Dual-use risk	Region priorities can aid vaccine surveillance and, in principle, escape-variant design.	No raw GISAID FASTA or variant-level engineering recipes are released; future P3CO/Select Agent or non-RNA pandemic-potential extensions require dedicated institutional biosafety review.	Section 4.4.4
MAFFT `--auto` artifact	Alignment mode (FFT-NS-2 / FFT-NS-i / L-INS-i / G-INS-i) is confounded with pathogen.	Disclosed as an input-quality caveat; expected effect is small but unquantified until a single-mode sensitivity rerun.	Chapter 2

The dominant residual risks are therefore bounded scope, label-provenance heterogeneity, and insufficient pathogen count for deployable adaptive learning rather than hidden evidence for a universal predictor.

4.5 Summary of Research Contributions

This dissertation presents three contributions. **First**, to our knowledge, MutBench is the broadest *region-level, single-position* hotspot-detection benchmark to date for RNA viruses, combining 10 biological information types (20 scoring formulas) $\times$ 14 detection families (39 variants) $\times$ 11 RNA viruses = 8,580 evaluations, a three-layer ground truth (Layer A: a pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types, with immune-escape positions dominating; Layer B: purifying-selection negatives; Layer C: DMS experimental fitness where available), and the hotspot-score composite metric within a two-stage main experimental design plus a Stage 3 information-integration layer. **Second**, the benchmark shows that the optimal biological information source is pathogen-dependent: the scoring $\times$ pathogen interaction is the largest modeled effect ($\omega^2 = 0.296$ at 20-type granularity, 95% cluster-bootstrap CI [0.201, 0.346]; 0.103 at 6-category granularity), the 11 pathogens have 9 distinct optimal information types, and the oracle-vs-generalized MCC gap is 0.265. Friedman $\chi^2 = 7.69$, $p = 0.990$ (cluster-robust main statistic) shows no universally best combination across pathogens, consistent with the interaction effect; LOPO 0/11 is null-consistent under permutation and is reported as a corroborative consistency check rather than independent evidence. **Third**, the framework yields practical search-space reduction validated on independently-curated vaccine-escape positions. For a typical viral protein the optimal scoring–detection combination narrows the candidate list from $\sim$1,000 to approximately 10–50 positions (top-decile budget; 11–32 detected positions in the three reported escape-enrichment anchors) with 7–9$\times$ enrichment for escape sites. External validation is anchored by HIV-1: the escape set is largely Layer A-disjoint (82%) and yields full-set 7.19$\times$ enrichment ($p_{\text{adj}} = 2.5 \times 10^{-16}$) plus a novel-only 7.16–8.24$\times$

Bonferroni-significant subset (full numerics and circularity audit in Chapter 3 Table 3.20); H3N2 ($9.36\times$) is a self-consistency check (69% Layer A overlap) and SARS-CoV-2 ($7.63\times$) is exploratory after multiple-comparison correction. The H3N2 EqualWeight integration case yields $4.012\times$ enrichment ($p = 0.0023$). A multi-source feature-ablation robustness check (nested-LOPO; 4-feature core paired delta $+0.011$, 95% CI $[-0.018, +0.042]$, $p = 0.61$; Section 3.5.2, Table 3.14) confirms the compact 4-feature core (homoplasy, pLDDT, entropy, frequency) provides a practical cold-start recipe statistically indistinguishable from the full 10-feature EqualWeight ensemble. An exploratory adaptive evidence-fusion redesign was developed alongside the dissertation but is not adopted as a contribution; it remains future work pending a larger pathogen panel.

Falsification-resistant audits and the boundary of Algorithm 1's claim. Three pre-registered audits (Sections 9, 9, 9) further bound the deployable scope of Algorithm 1 on the current 12-pathogen panel. *First*, the full 1023-subset lattice with formal Shapley decomposition shows the dissertation's fixed 4-core (`freq, entropy, homoplasy, plddt`) ranks $87/1023$ (top 8.5%-tile, lower than the rank reported on the restricted 4-of-10 lattice in Paragraph 9) and is strictly dominated by the 3-feature variant (`freq, entropy, homoplasy`) (MCC 0.0866 vs. 0.0809, $\Delta = +0.0058$, lattice rank $27/1023$); pLDDT contributes a slightly negative Shapley value (-0.0015) and the ESM-2 LLR contributes the only doubly-negative Shapley (-0.0144 on MCC, -0.158 on enrichment). The Cold-Start 4-core is therefore correctly characterised as a historical fixed configuration, not as a non-arbitrary optimum, and pLDDT is recommended as an optional structural extension. *Second*, per-pathogen rank reliability (1,000-resample bootstrap of top-vs-bottom decile prevalence) shows a positive CI strictly excluding zero in only $4/12$ pathogens for the 4-core (Norovirus, Dengue, H3N2, Influenza B; HIV-1's CI lower bound is 0.000 so it is positive but does not strictly exclude zero) and the score *inverts* on Rabies (delta -0.123, CI $[-0.245, -0.003]$, Spearman $\rho = -0.76$ across deciles), so the operational claim is reduced to a screening-budget allocator with documented per-pathogen heterogeneity rather than a calibrated probability. *Third*, 100,000-permutation null calibration under four nulls (iid, circular shift, protein-block shuffle, local-burden-preserving) yields Stouffer one-sided $p = 1.00$ globally for the 4-core MCC across the 12-pathogen panel; per-pathogen iid significance is $5/12$ but degrades to $1/12$ (Norovirus only) under the strictest local-burden-preserving null. Most "above-null" signal is therefore explained by local clustering of Layer A epitopes rather than by exact-site discrimination. These three audits collectively reframe Algorithm 1 as an exploratory cold-start prioritization heuristic with prospectively definable callable subsets, not a globally calibrated detector. The five iid-significant pathogens form a descriptive callable set. Wave 2 executed the abstention rule on the current panel ($\mathbf{0/12}$ callable, Paragraph 9) and Wave 3 added structurally explicit local nulls (Paragraph 9): under SASA-, pLDDT-, and joint structural-evolutionary stratification at $n_{\text{perm}} = 100{,}000$, four pathogens (H3N2, Norovirus, Dengue, Influenza B) remain individually significant under the 4-core MCC across all four P6 nulls, and Norovirus additionally survives the strictest P1 local-burden-preserving null, so the descriptive signal on this callable subset is not reducible to simple surface or flexibility clustering.

Wave 4 expanded-feature feasibility and callability failure. The expanded-panel work remained the natural prospective test. Wave 4 (Paragraph 9) tested the features-only expansion on the `experiments/waves` branch: hscore did not harmonize (0/3 PASS), so the 2-core `freq, entropy` fallback was used; labeled 12-fold LOPO was bounded positive but indistinguishable from the recomputed 4-core (mean MCC 0.077, $8/12$ positive, paired Wilcoxon $p = 0.368$), D1 callability remained $0/12$, expanded 28-target sign stability was $1/28$, and the $16/16$ unlabeled-target envelope establishes feature-space feasibility for Layer A curation without relaxing the panel-size limit. Wave 4 therefore left open whether any structured per-position label set — not specifically literature-curated Layer A — would preserve the same directional signal at the new targets.

Wave 5 Layer A$'$ negative provenance-sensitivity result. Wave 5 (Paragraph 9) implemented this directly as a UniProt-feature-derived Layer A$'$ sensitivity benchmark on the **10** targets that passed coordinate and feature-density filters; mean LOPO MCC $= -0.055$ with $2/10$ positive folds and a stable but *negative* `freq/entropy` sign opposite to the 12-panel positive sign, firing the predeclared Branch C. The negative result shows that Layer A and Layer A$'$ are not interchangeable label provenances; the dissertation therefore retains the literature-curated 12-pathogen panel as the pre-existing primary benchmark but does not treat the Layer A$'$ failure as validation of Layer A. Layer A$'$ is reported as a bounded sensitivity check on label-provenance equivalence rather than a panel extension or a deployment claim.

Recommended use today. Within the boundaries established above, the recommended near-term use of MutBench is *retrospective prioritization* only: as a screening-budget allocator on the 11-pathogen panel and as an HIV-1-anchored search-space-reduction tool, with abstention status reported alongside any recommended position list. Prospective detector performance — including any deployment claim or callability assertion — is not supported until (a) the predeclared callability rule (Paragraph 9) calls at least a quorum-majority of held-out folds on an externally validated expanded panel, and (b) adaptive methods exceed EqualWeight under paired nested-LOPO. Until both conditions hold, users should treat Algorithm 1 as an exploratory cold-start heuristic, not as a calibrated detector or family-conditional classifier.

4.6 Future Research Directions

Table 4.5 separates completed negative gates from engineering-feasible extensions and true future validation work.

Table 4.5: Prioritized future research roadmap with current status

Priority	Status	Research direction	Gate / expected impact
1	Done negative	Adaptive method selection	EqualWeight not beaten under nested-LOPO; current panel remains non-deployable
2	Engineering feasible	Expanded feature pipeline	Automated features scale; external labels are the bottleneck
3	Future	20–30+ pathogen validation	Curated labels, time-forward validation, and paired nested-LOPO/callability gates
4	Future	Scope extension and reproduction	DNA viruses, multi-protein panels, indels, independent reruns, Zenodo provenance

Adaptive method selection and prospective callability. Adaptive selection remains a direction rather than a deployable result: the adaptive-weight oracle exceeds EqualWeight in principle (MCC 0.160 vs. 0.083), but the seven tested adaptive paradigms fail under nested-LOPO, and the predeclared abstention rule calls $0/12$ folds. The next test is therefore not another weighting model on the same panel, but a 20–30+ pathogen expansion with curated external labels, time-forward validation, and the same paired nested-LOPO/callability gates; until those gates pass, Algorithm 1 remains a retrospective prioritization heuristic rather than a prospective classifier.

Expanded-panel engineering and label provenance. The deadline scaling pilot established that the automated feature path is engineering-bound rather than science-bound: Layer A annotation was deferred, so the pilot is a sequence-and-features extension only, not a benchmark-validated panel expansion; NCBI provenance and per-target accessions are archived in `results/mutbench/pilot_accession_manifest.csv`. Wave 4 and Wave 5 already report the decisive bounded results in Paragraphs 9 and 9: feature expansion is feasible, but label-provenance equivalence failed, so the bottleneck is curated external labels rather than detector arithmetic.

Integration of Phylogenetic Correction. Homoplasy-based scoring captures signals distinct from frequency (simulation $\rho = -0.876$; real GISAID $\rho = -0.107$) and should be integrated through TreeTime [31] or scalable alternatives such as UShER [34]. This would mainly affect founder-effect-prone pathogens such as SARS-CoV-2, and should be paired with temporal/geographic stratification.

Additional Research Directions. Scope extensions are DNA viruses, multi-protein panels, indels/recombination, virus-tuned PLM features, real-time surveillance integration via Nextstrain pipelines, and non-coding regions; current evidence does not support claims outside RNA-virus surface-glycoprotein substitutions. The HBV polymerase pilot reported in Section 4.4.4 provides empirical sanity-check evidence for the rate-based boundary, and HIV-1 Env remains the only retroviral target evaluated.

Reproduction and Release. The archived CSVs, provenance manifest, and Zenodo tag (Section 4.7) close the reproducibility gap as a defense-time deliverable, instantiating Priority 4 of Table 4.5.

4.7 Concluding Remarks

MutBench shows that viral hotspot detection is driven more by the fit between pathogen and biological information source than by detector choice alone. Within the single-position scoring regime and modeled ANOVA terms, the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$) is the largest identified factor, and nine distinct optimal information types appear across

11 pathogens. The practical validation is strongest for HIV-1 escape enrichment (7.19×, 82% Layer A-disjoint), while H3N2 is mainly a self-consistency check and SARS-CoV-2 remains exploratory. The 4-feature core (homoplasy, pLDDT, entropy, frequency) provides a compact starting recipe for new pathogens, but adaptive deployment remains bounded: the dissertation is consistent with a feasible direction for evidence fusion, not a universal adaptive predictor.

Table 4.6: Defense map: dissertation claims with their evidence tier, scope boundary, and committee-facing oral wording. The table is the verbal-defense reference; full numeric evidence and caveats are in the cited paragraphs.

Claim	**Evidence tier**	**Boundary / not yet**	**Allowed oral wording**
C1. Standardised three-layer benchmark on 11 RNA-virus surface glycoproteins (Paragraph 4.5)	Proven (artefact)	Not externally re-benchmarked; literature-curated Layer A heterogeneity (Section 4.4.4)	"Reproducible benchmark with three-layer ground truth and 8,580 evaluations; awaiting external re-benchmark."
C2. Scoring × pathogen interaction $\omega^2 = 0.296$ is the dominant modeled component (Paragraph 9)	Proven (within-panel, robust under counterfactual null)	Cell-level $\omega^2 = 0.234$; small-cluster regime; counterfactual sanity, not external validation	"On this 11-pathogen panel, choosing the information source matters more than choosing the detector; this is robust under five inferential frames and a counterfactual null."
C3. HIV-1 search-space reduction with 7.19× enrichment (Section 4.4.1)	Retrospective	3/11 pathogens externally validated; no prospective time-forward; $0/12$ callable under abstention rule (Paragraph 9)	"Retrospective practical anchor on HIV-1 with 82% Layer A-disjoint novel-only enrichment; not a prospective deployment claim."
Wave 5 Layer A′ Branch C (Paragraph 9)	Negative (sensitivity)	UniProt label provenance is not interchangeable with literature-curated Layer A	"Wave 5 is a label-provenance sensitivity check; the negative result establishes non-equivalence rather than validating Layer A."
Adaptive method selection (Cycle 7B + HBFWS, Section 3.5.2)	Negative (seven paradigms)	11-pathogen panel below the Bolker 20–30 threshold for stable random effects	"Seven adaptive paradigms fail to beat EqualWeight at $n = 11$; the panel-size constraint is the principal lever, not algorithm choice."

Take-home message: choosing the right biological information source for the pathogen explains more performance variation than the detection-algorithm main effect alone, and MutBench quantifies that asymmetry.

Dual-use considerations. MutBench outputs are dual-use because they prioritize immune-escape-relevant regions on viral surface glycoproteins. The release mitigations and IBC/P3CO/Select Agent boundaries are specified in Section 4.4.4: no raw GISAID FASTA, no variant-level engineering recipes, and dedicated biosafety review for higher-risk extensions.

Table 4.7: Code, data, and numerical artifact availability

Item	Availability / constraint
Repository and release	Source code, scripts, ground truth definitions, and configuration files are archived at `https://github.com/kksuhj/mutbench` under tag `dissertation-v1`; commit hash is recorded in `provenance/commit.txt`, and a Zenodo DOI will be minted at final submission.
Licenses	MutBench implementation code is MIT; dissertation text and figures are CC-BY-NC 4.0; upstream tools and resources (MAFFT, IQ-TREE, HyPhy, ESM-2, Tranception, AlphaFold-DB, GenBank, Bloom-lab DMS datasets) retain their own licenses as summarized in Chapter 2, Section 2.2.7.
Redistributed artifacts	The release contains computed scores, detection outputs, summary statistics, and provenance manifests only; upstream FASTA bundles, GISAID raw sequences, and external model weights are not redistributed.
Reproducibility and provenance	With precomputed features, the full 8,580-evaluation detector grid runs in approximately 25 seconds on a single workstation using Python 3.10, NumPy/SciPy/Pandas, MAFFT, IQ-TREE, HyPhy, ESM-2, and seed 42.

Code and data availability. The CSV-row-to-table-cell provenance manifest is bundled with the examiners' archive and mirrored with the Zenodo deposit; the load-bearing numerical files are `stage3_full_results.csv`, `vaccine_escape_stage3.csv`, and `feature_analysis/feature_ablation_nested_lopo_summary.csv`.

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바이러스 변이 핫스팟 탐지를 위한 생물학적 정보원의 체계적 벤치마킹

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바이러스 게놈의 변이 핫스팟 탐지는 변이주 감시와 백신 설계에 필수적이나, 기존 접근법은 변이 빈도와 Shannon 엔트로피에 거의 전적으로 의존해 왔으며, 계통 분석 선택 신호, 단백질 언어모델 예측, 구조적 특성 등 풍부한 보완적 정보원은 체계적으로 탐색되지 않았다. 어떤 정보 유형이 어떤 병원체에 가장 효과적인지를 비교한 체계적 연구는 존재하지 않았다.

본 연구에서는 이 문제를 해결하기 위해 MutBench를 제안한다. MutBench는 어떤 요인이 핫스팟 탐지 성능에 영향을 미치는지 분석하기 위한 실험 프레임워크로, 10가지 생물학적 정보 유형(6개 카테고리의 20개 점수화 공식)을 5개 범주(임계값 기반, 신호처리, 밀도/클러스터링, 통계, 적응형 윈도우)에 걸친 14개 탐지 방법군(39개 매개변수 변형)과 결합하여 11개 RNA 바이러스에 걸쳐 총 8,580회 평가를 수행하며, 세 가지 핵심 구성요소를 포함한다: (1) 같은 바이러스 종 내에서 서로 다른 계통에서 독립적으로 반복 출현한 수렴 진화 위치를 양성 선택 기준(Layer A), 정화 선택 하의

보존 구조 영역을 음성 기준(Layer B), Deep Mutational Scanning(DMS) 실험적 적합도 데이터를 기능적 검증(Layer C)으로 통합하는 3층 ground truth 체계, (2) hotspot-score = (recall + precision + stability)/3으로 정의되는 복합 평가 지표, (3) Stage 1에서 SARS-CoV-2에 대한 심층 분석, Stage 2에서 대규모 교차 병원체 비교를 수행하는 2단계 실험 설계이다.

Stage 1에서는 MutClust-Hybrid가 최고 hotspot-score(0.778)를 달성하여, 도메인 특화 사전지식과 데이터 기반 클러스터링의 결합이 효과적임을 확인하였다. Stage 2의 핵심 발견은 최적의 생물학적 정보원이 병원체마다 근본적으로 다르다는 것이다: 점수화 방법과 병원체 간 상호작용이 전체 성능 분산의 29.6%를 설명하여($\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$), 가장 큰 분산 원천으로 나타났다. Friedman 검정($\chi^2 = 7.69$, $p = 0.990$)에서 상위 20개 조합 간 유의한 차이가 없음을 확인하였으며, LOPO 교차검증에서 11개 병원체 모두 서로 다른 최적 정보원을 요구하였다(일반화 성공률 0/11). 또한 탐지된 핫스팟이 실험적으로 확인된 백신 escape 위치와 최대 9.36배 enrichment($p < 0.0001$)를 보이며, 벤치마크에서 최적으로 식별된 scoring 유형이 3개 병원체(2,340회 평가)에서 백신 escape 포착률도 최대화함을 확인하여 공중보건 감시에 대한 실용적 가치를 입증하였다.

6개 정보 카테고리(빈도 기반, MSA 유래, 계통 분석, 구조 기반, AI 기반, 복합)에 걸친 20개 점수화 유형 분석에서 11개 병원체에 걸쳐 9가지 서로 다른 최적 정보 유형이 관찰되었다: FUBAR 계통선택 점수(H3N2), 호모플래시(Norovirus), 단백질 언어모델(SARS-CoV-2, RSV), 의미적 변화(EV-A71), 안정성 예측(Rabies), 빈도 기반(HCV, Influenza B, MERS). 다중 정보 통합이 실험적 탐색 범위를 축소하며(H3N2 백신 escape enrichment 4.012배, $p = 0.0023$), 4개 핵심

특징(homoplasy, pLDDT, entropy, frequency)이 전체 앙상블 성능의 98%를 달성하였다.

주요어: 변이 핫스팟 탐지, MutBench, 생물학적 정보 유형, 바이러스 게놈, 다중 정보원 분석, 병원체 의존적 최적성, SARS-CoV-2, Deep Mutational Scanning